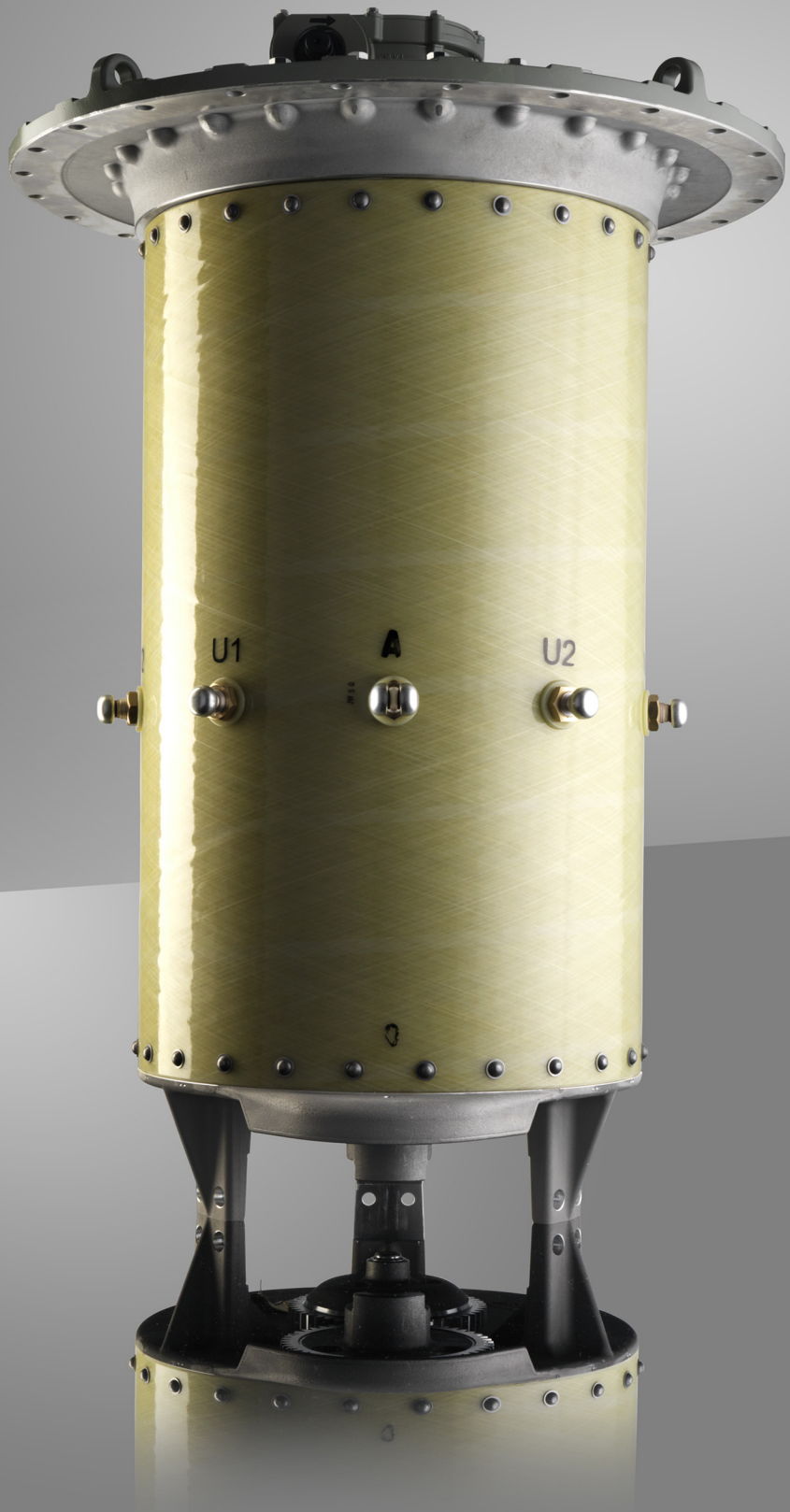
On-load tap-changer VACUTAP® VM

Technical Data



© All rights reserved by Maschinenfabrik Reinhausen

Copying and distribution of this document and the utilization and communication of its contents are strictly prohibited unless expressly authorized.

Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or ornamental design registration.

The product may have been modified after this document went to press.

We expressly reserve the right to make changes to the technical data, the design or the scope of delivery.

The information provided and the arrangements agreed during processing of the relevant quotations and orders are generally binding.

The original operating instructions were drawn up in German.

Table of contents

1 General / Technical Data 11

1.1 On-load tap-changer designations 11

1.2 Technical data for VACUTAP® VM 12

1.2.1 Technical data for VACUTAP® VM I 12
1.2.2 Technical data for VACUTAP® VM II 13
1.2.3 Technical data for VACUTAP® VM III 14
1.2.4 Step capacity diagram VM 15

1.3 Rated insulation level 17

1.4 Rated withstand voltages 17

1.4.1 Rated withstand voltages of the internal on-load tap-changer insulation (with the exception of VACUTAP® VM 300) 19
1.4.2 Rated withstand voltages of the internal on-load tap-changer insulation for VACUTAP® VM 300 20
1.4.3 VM rated withstand voltages — multiple coarse change-over selector 22

2 Special designs 23

2.1 Parallel bridges for parallel connections 23

2.2 On-load tap-changer VACUTAP® VM 650 Y...VM I 1503 with multiple coarse change-over selector (up to maximum of 5 coarse tap connections) 24

2.3 Two-pole on-load tap-changer VACUTAP® VM II 302/352/502/602 24

2.4 On-load tap-changer VACUTAP® VM III 350 Y...VM I 1503 for linear voltage adjustment, up to a maximum of 34 operating positions 24

3 Appendix 25

3.1 Dimensional drawings/connection diagrams 25

3.1.1 VACUTAP® VM III 300 Y-0/W/G (898038) 25
3.1.2 VACUTAP® VM II 302-0/W/G (898039) 26
3.1.3 VACUTAP® VM I 301-0/W/G (898040) 27
3.1.4 VACUTAP® VM III 350/500/650 Y-0/W/G (746219) 28
3.1.5 VACUTAP® VM II 352/502/652-0/W/G (746220) 29

3.1.6 VACUTAP® VM II 351/501/651-0/W/G (746221) ... 30
3.1.7 VACUTAP® VM I 802/1002-0/W/G (746222) ... 31
3.1.8 VACUTAP® VM I 1203/1503-0/W/G (746223) ... 32
3.1.9 VACUTAP® VM III K-0/W/G (746224) ... 33
3.1.10 VACUTAP® VM III 650 Y with multiple coarse change-over selector (746226) 34
3.1.11 VACUTAP® VM I 601 with multiple coarse change-over selector (746227) ... 35
3.1.12 VACUTAP® VM I 802/1002 with multiple coarse change-over selector (746228) ... 36
3.1.13 VACUTAP® VM I 1203/1503 with multiple coarse change-over selector (746229) ... 37
3.1.14 VACUTAP® VM III 300 tie-in resistors without potential switch (898695) ... 38
3.1.15 VACUTAP® VM III 300 tie-in resistors without potential switch (898694) ... 39
3.1.16 VACUTAP® VM III 300 tie-in resistors without potential switch (898693) ... 40
3.1.17 VACUTAP® VM III Y tie-in resistors with/without potential switch (898692) ... 41
3.1.18 VACUTAP® VM II tie-in resistors with/without potential switch (898691) ... 42
3.1.19 VACUTAP® VM I tie-in resistors with/without potential switch (898690) ... 43
3.1.20 VACUTAP® VM I 351...1503, tie-in resistor cylinder with potential switch without tie-in resistors (898804) ... 44
3.1.21 VACUTAP® VM 300 — installation drawing of centrical drive (765192) ... 45
3.1.22 VACUTAP® VM — installation drawing of centrical drive (746230) ... 46
3.1.23 VACUTAP® VM — on-load tap-changer head, centrical drive (893899) ... 48
3.1.24 VACUTAP® VM — variants of on-load tap-changer head (720026) ... 49
3.1.25 VACUTAP® VM — swivel range of gear unit (720027) ... 50
3.1.26 VACUTAP® VM — screenings on oil compartment contacts (730336) ... 51
3.1.27 VACUTAP® VM 300 — tap selector cross-sections (898041) ... 52
3.1.28 VACUTAP® VM 300 — arrangement of contacts on tap selector (blanks, 891114) ... 53
3.1.29 VACUTAP® VM — arrangement of contacts on tap selector, tap selector division 10...22 (898013) ... 54
3.1.30 VACUTAP® VM — installation position of tap selector connection contacts (890477) ... 55
3.1.31 VACUTAP® VM — connecting leads 3W, 1G, 3G (723590) ... 56
3.1.32 VACUTAP® VM — screenings on fine tap selector and change-over selector (730335) ... 57
3.1.33 VACUTAP® VM 802/803/1203/1503 — bridges to parallel connection of tap selector connection contacts (899598) ... 58
3.1.34 VACUTAP® VM — horizontal drive shaft, centrical drive (893896) ... 59
3.1.35 VACUTAP® VM III 350/500/650 Y — arrangement of tap selectors (891107) ... 60
3.1.36 VACUTAP® VM I 351/501/651 — arrangement of tap selectors (891108) ... 61
3.1.37 VACUTAP® VM I 1503 — arrangement of tap selectors (891109) ... 62
3.1.38 VACUTAP® VM I 802/1002 — arrangement of tap selectors (891110) ... 63
3.1.39 VACUTAP® VM, supporting flange, special design for bell-type tank installation for Um up to 300 kV (896762) ... 64

 © Maschinenfabrik Reinhausen 2010

3.1.40 VACUTAP® VM — flange for pressure relief valve (895168) 65
3.1.41 VACUTAP® VM — tap selector base with additional screening, reversing change-over selector design (893934) 66
3.1.42 VACUTAP® VM — tap selector base with additional screening, coarse tap connection design (893935) 67
3.1.43 VACUTAP® VM — tap selector base with through hole D20 and D13 (725649) 68

3.2 VACUTAP® VM 300 — overview of on-load tap-changer designs (765835) 69

3.3 VACUTAP® VM — overview of on-load tap-changer designs (899740) 70

3.4 Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616) 71

3.5 Connection diagram (contacts designated in accordance with MR standard) 74

3.5.1 VACUTAP® VM III 300 Y, basic connection diagram 10 19 1 W (2414642) 74
3.5.2 VACUTAP® VM III 300 Y, basic connection diagram 10 19 3 W (2414644) 75
3.5.3 VACUTAP® VM III 300 Y, basic connection diagram 14 27 1 G (2414648) 76
3.5.4 VACUTAP® VM III 300 Y, basic connection diagram 14 27 3 G (2414649) 77
3.5.5 VACUTAP® VM III 350/500/650 Y, basic connection diagram 10 19 1 W (2414658) 78
3.5.6 VACUTAP® VM III 350/500/650 Y, basic connection diagram 10 19 3 W (2414670) 79
3.5.7 VACUTAP® VM I 802/1002, basic connection diagram 14 27 1 G (2414631) ... 80
3.5.8 VACUTAP® VM I 1203/1503, basic connection diagram 18 35 1 W (2414636) 81
3.5.9 VACUTAP® VM I 802/1002, basic connection diagram 16 79 1 G, multiple coarse change-over selector design (2407535) 82

4 MR worldwide 83

List of figures

Figure 1 Step capacities (rated step voltage U_i, rated through-current I_u) 15

Figure 2 Step capacities (rated step voltage U_i, rated through-current I_u) 16

Figure 3 Rated withstand voltages 18

Figure 4 On-load tap-changer combination VM I 351/VM II 352 (a — VM I 351, b — VM II 352) for delta connection 23

List of tables

Table 1 Technical data for VACUTAP® VM I 12

Table 2 Technical data for VACUTAP® VM II 13

Table 3 Technical data for VACUTAP® VM III 14

Table 4 Rated insulation level 17

Table 5 Rated withstand voltages of the internal on-load tap-changer insulation (with the exception of VACUTAP® VM 300) 19

Table 6 Rated withstand voltages of the internal on-load tap-changer insulation for VACUTAP® VM 300 20

Table 7 Available connections (also available as 3 W, 3 G) 21

Table 8 On-load tap-changer VACUTAP® VM with multiple coarse change-over selector, rated withstand voltages of the internal on-load tap-changer insulation 22

1 General / Technical Data

1.1 On-load tap-changer designations

Example: VM III 650 Y - 72.5 / C - 10 19 1W R

Basic connection diagram

Type	VACUTAP® VM	**VM**
Number of phases	1 phase	I
	2 phases	II
	3 phases	**III**
I_{um} (in A)	VM I, VM II, VM III	300
	VM I, VM II, VM III	350
	VM I, VM II, VM III	500
	VM I, VM II, VM III	**650**
	VM I	802
	VM I	1002
	VM I	1203
	VM I	1503
Number of parallel main switching contacts	1 (3-phase Y)	**0**
	1 (1-phase)	1
	2 (1-phase)	2
	3 (1-phase)	3
Applications	For use with neutral point only	**Y**
U_m (in kV)	VM	**72.5**
	VM	123
	VM	170
	VM	245
	VM	300
Tap selector size		B
		C
		D
	not with multiple coarse change-over selector	DE
Number of maximum operating positions without change-over selector	10	**10**
	12	12
	14	14
	16	16
	18	18
	22	22
Number of maximum operating positions with change-over selector (reversing change-over selector or coarse tap selector)[1)]	10	**19**
	12	23
	14	27
	16	31
	18	35
Number of maximum operating positions with multiple coarse change-over selector	10	59
	12	71
	14	83
	16	95
	18	107
Mid-positions	0 mid-positions (without change-over selector)	0
	1 mid-position	**1**
	3 mid-positions	3
Change-over selector	Reversing change-over selector	**W**
	Coarse tap connection	G
Tie-in measures	Tie-in resistor	**R**
	Potential switch	S
	Potential switch with tie-in resistor	P

1) 300 amp variants with a maximum of 27 operating positions available

1.2 Technical data for VACUTAP® VM

1.2.1 Technical data for VACUTAP® VM I

On-load tap-changer VM I	301	302	351	501	651	802	803	1002	1203	1503
Number of phases and application	1	1	1	1	1	1	1	1	1	1
Maximum rated through current I_{um} (in A)	300	300	350	500	650	800	800	1000	1200	1500
Rated short-time current (in kA)	4	4	4.2	5	6.5	8	8	10	12	15
Rated duration of short-circuits (in s)	3	3	3	3	3	3	3	3	3	3
Rated peak withstand current (in kA)	10	10	10.5	12.5	16.25	20	20	25	30	37.5
Maximum rated step voltage U_{im} (in V)[1)]	3300	3300	3300	3300	3300	3300	3300	3300	3300	3300
Step capacity (PStN, in kVA)	990	990	1155	1625	1625	2640	2640	3300	3500	3500
Rated frequency (in Hz)	50...60									
Operating positions	Without change-over selector: maximum 18 With change-over selector: maximum 35[2)] With multiple coarse change-over selector: maximum 107[2)]									
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the diverter switch oil compartment are vacuum-proof.									
Temperature range	The on-load tap-changer VACUTAP® VM can be operated in the rated load range at surrounding oil temperatures of between -25 ° and +105 °C and with overload up to +115 °C in accordance with IEC 60214-1. For details of operation under Arctic conditions, please refer to the General Technical Data TD 61.									
Dimensions	Weight, displacement volume and oil content of the diverter switch oil compartment are shown in the relevant dimension drawings.									

Table 1 Technical data for VACUTAP® VM I

1) The maximum rated step voltage may be exceeded by 10 % due to overexcitation of the transformer if the step capacity is limited to its rated value.

2) 300 amp variants with a maximum of 27 operating positions available

 © Maschinenfabrik Reinhausen 2010

1.2.2 Technical data for VACUTAP® VM II

On-load tap-changer VM II	302	352	502	652
Number of phases and application	2	2	2	2
Maximum rated through current I_{um} (in A)	300	350	500	650
Rated short-time current (in kA)	4	4.2	5	6.5
Rated duration of short-circuits (in s)	3	3	3	3
Rated peak withstand current (in kA)	10	10.5	12.5	16.25
Maximum rated step voltage U_{im} (in V)[1)]	3300	3300	3300	3300
Step capacity (PStN, in kVA)	990	1155	1625	1625
Rated frequency (in Hz)	50...60			
Operating positions	Without change-over selector: maximum 18 With change-over selector: maximum 35[2)] With multiple coarse change-over selector: maximum 107[2)]			
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the diverter switch oil compartment are vacuum-proof.			
Temperature range	The on-load tap-changer VACUTAP® VM can be operated in the rated load range at surrounding oil temperatures of between -25 ° and +105 °C and with overload up to +115 °C in accordance with IEC 60214-1. For details of operation under Arctic conditions, please refer to the General Technical Data for TD 61.			
Dimensions	Weight, displacement volume and oil content of the diverter switch oil compartment are shown in the relevant dimension drawings.			

Table 2 Technical data for VACUTAP® VM II

1) The maximum rated step voltage may be exceeded by 10 % due to overexcitation of the transformer if the step capacity is limited to its rated value.

2) 300 amp variants with a maximum of 27 operating positions available

1.2.3 Technical data for VACUTAP® VM III

On-load tap-changer VM III	300 Y	350 Y	500 Y	650 Y
Number of phases and application	3	3	3	3
Maximum rated through current I_{um} (in A)	300	350	500	650
Rated short-time current (in kA)	4	4.2	5	6.5
Rated duration of short-circuits (in s)	3	3	3	3
Rated peak withstand current (in kA)	10	10.5	12.5	16.25
Maximum rated step voltage U_{im} (in V)[1)]	3300	3300	3300	3300
Step capacity (PStN, in kVA)	990	1155	1625	1625
Rated frequency (in Hz)	50...60			
Operating positions	Without change-over selector: maximum 18 With change-over selector: maximum 35[2)] With multiple coarse change-over selector: maximum 107[2)]			
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the diverter switch oil compartment are vacuum-proof.			
Temperature range	The on-load tap-changer VACUTAP® VM can be operated in the rated load range at surrounding oil temperatures of between -25 ° and +105 °C and with overload up to +115 °C in accordance with IEC 60214-1. For details of operation under Arctic conditions, please refer to the General Technical Data for TD 61.			
Dimensions	Weight, displacement volume and oil content of the diverter switch oil compartment are shown in the relevant dimension drawings.			

Table 3 Technical data for VACUTAP® VM III

1) The maximum rated step voltage may be exceeded by 10 % due to overexcitation of the transformer if the step capacity is limited to its rated value.

2) 300 amp variants with a maximum of 27 operating positions available

 © Maschinenfabrik Reinhausen 2010

1.2.4 Step capacity diagram VM

1.2.4.1 Step capacity diagram for network application

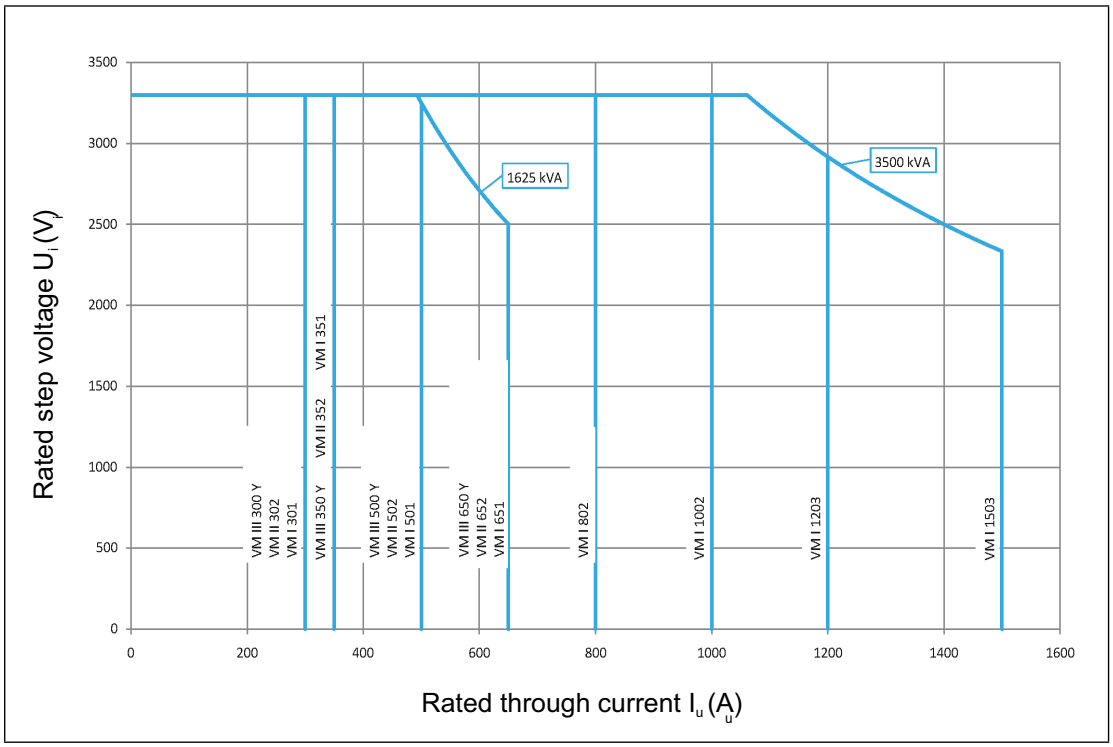


Figure 1 Step capacities (rated step voltage U_i, rated through-current I_u)

1.2.4.2 Step capacity diagram for autoclave application

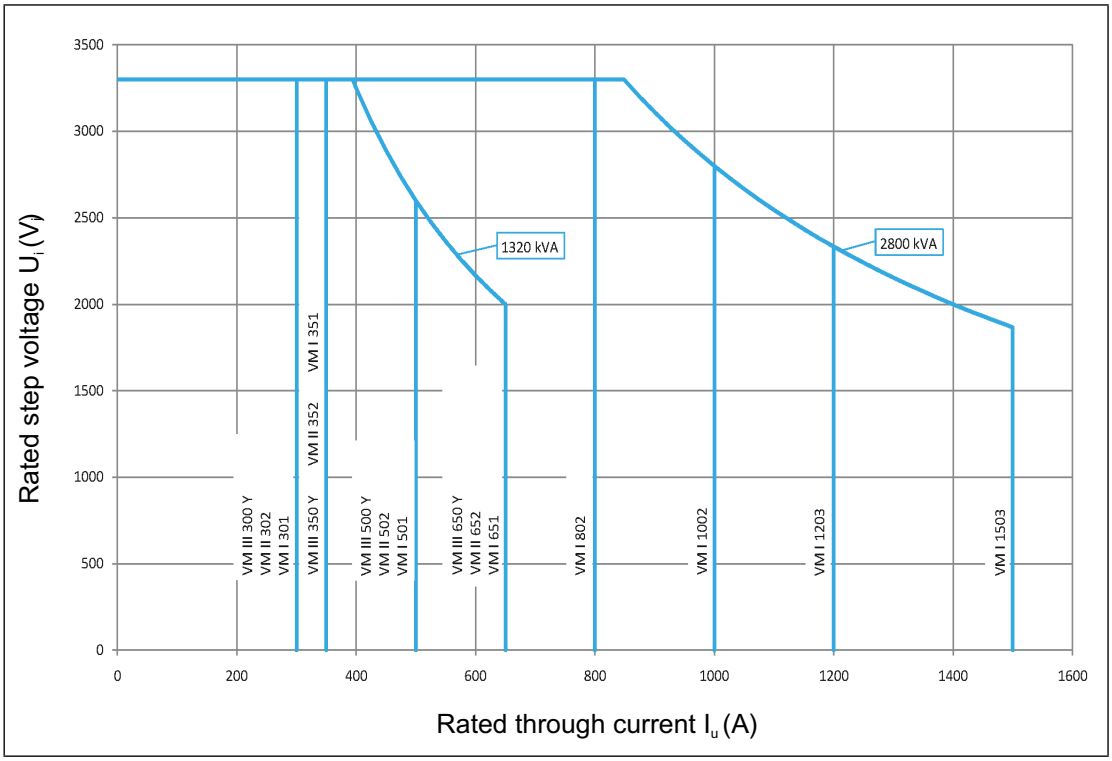


Figure 2 Step capacities (rated step voltage U_i, rated through-current I_u)

 © Maschinenfabrik Reinhausen 2010

1.3 Rated insulation level

Rated insulation level	For all on-load tap-changer variants				
Highest voltage for equipment U_m (in kV)[1]	72.5	123	170	245	300[2)]
Highest operating voltage Ub (phase-phase) on on-load tap-changer (kV)	55	79	145	170	245
Rated lightning impulse withstand voltage (in kV, 1.2/50 µs)	350	550	750	1050	1050
Rated switching impulse voltage (in kV)				850	850
Rated short-duration power frequency withstand voltage (in kV, 50 Hz, 1 min.)	140	230	325	460	460

Table 4 Rated insulation level

1) In accordance with IEC 60214-1, chapter 3.57: highest effective value for phase-to-phase voltage in a three-phase system for which an on-load tap-changer is designed with respect to its insulation.

2) 300 A variants with a maximum of 245 KV available.

1.4 Rated withstand voltages

Figure 3 shows diagrams of the voltage stress present on the tap winding of the three primary basic connections of three-pole on-load tap-changers and single-pole on-load tap-changers.

When selecting the on-load tap-changer, you must check that the highest stress on the tap selector does not exceed the related rated withstand voltages.

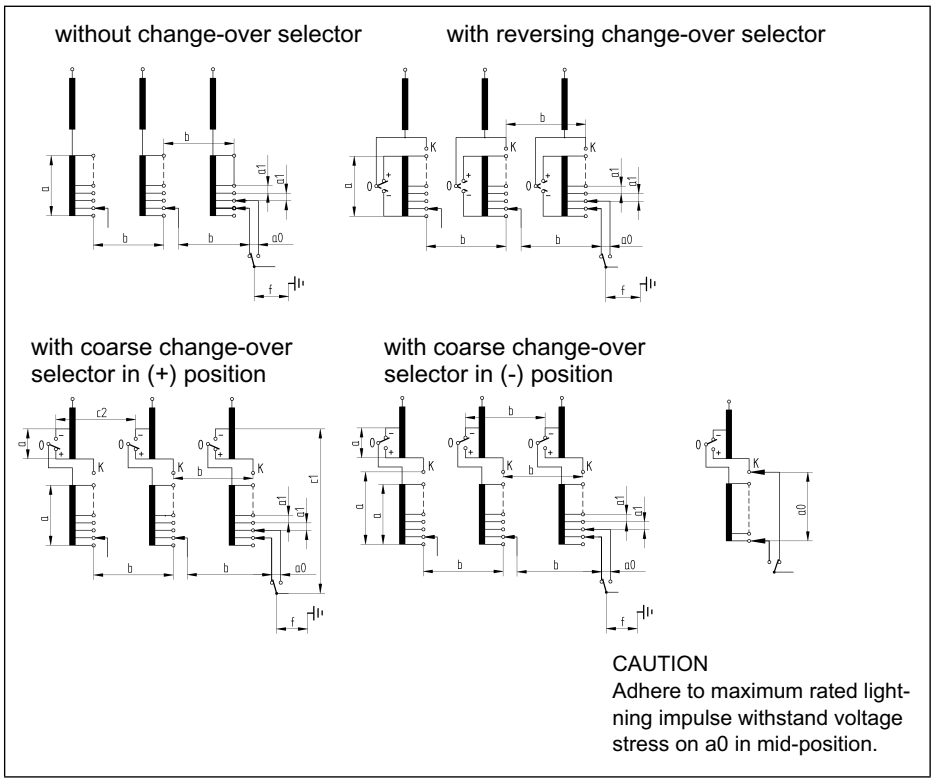


Figure 3 Rated withstand voltages

a0 = Between selected and preselected tap on the diverter switch and tap selector.

a1 = Between tap selector contacts of the winding of one tap position (connected or not connected)

a = Between beginning and end of a tapped winding and, with respect to a coarse winding, between beginning and end of a coarse winding.
Note for coarse tap selector connection (-) position of the change-over selector:
When stressed with impulse voltage, the permissible withstand voltage "a" must be adhered to between the end of a coarse tap winding connected with the K fine tap selector contact and the fine tap selector contact at the end of the tapped winding of the same phase.

b = Between the fine tap selector contacts of different phases and between change-over selector contacts of different phases, which are connected with the beginning/end of a tapped winding or with a fine tap selector contact.

f = Between diverter switch output terminal and ground

Additionally for coarse tapping arrangement in (+) position of the change-over selector:

c1 = From one (-) change-over selector contact to take-off lead of the same phase.

c2 = Between (-) change-over selector contacts of different phases.

 © Maschinenfabrik Reinhausen 2010

1.4.1 Rated withstand voltages of the internal on-load tap-changer insulation (with the exception of VACUTAP® VM 300)

The tap selector size (ID letters B, C, D, DE, E) characterizes the internal insulation of the tap selector, whose rated withstand voltages must be matched to the requirements of the transformer winding.

Insulation distances	Tap selector size B		Tap selector size C		Tap selector size D		Tap selector size DE	
	kV 1.2/ 50 µs	kV 50 Hz 1 min	kV 1.2/50 µs	kV 50 Hz 1 min	kV 1.2/50 µs	kV 50 Hz 1 min	kV 1.2/50 µs	kV 50 Hz 1 min
a0	Power-frequency voltage test (50 Hz/1 min): 20 kV 50 Hz 1 min; Response voltage with 1 mA leakage current: 35 kV 1.2/50 µs; 3 kA residual voltage (= lightning impulse protection level): 56 kV 1.2/50 µs with ZnO							
a1	150	30	150	30	150	30	150	30
a	265	50	350	82	490	105	550	120
b[1)]	265	50	350	82	490	146	550	160
c1	485	143	545	178	590	208	660	230
c2[1)]	495	150	550	182	590	225	660	250

Table 5 Rated withstand voltages of the internal on-load tap-changer insulation (with the exception of VACUTAP® VM 300)

1) Insulation distance omitted for single-pole on-load tap-changers

The admissible maximum operating voltage on the individual tap selector distances corresponds to half the value of the above mentioned rated short-duration power frequency withstand voltages.

1.4.2 Rated withstand voltages of the internal on-load tap-changer insulation for VACUTAP® VM 300

Insulation distances	VM III 300 Y		VM I 301	
	kV 1.2/ 50 µs	kV, 50 Hz, 1 min	kV 1.2/ 50 µs	kV, 50 Hz, 1 min
a0	Power-frequency voltage test (50 Hz/1 min): 20 kV 50 Hz 1 min; Response voltage with 1 mA leakage current: 35 kV 1.2/50 µs; 3 kA residual voltage (= lightning impulse protection level): 56 kV 1.2/50 µs with ZnO			
a	300	70	300	70
b	300	70	-	-
c1	400	120	400	120
c2	400	120	-	-

Table 6 Rated withstand voltages of the internal on-load tap-changer insulation for VACUTAP® VM 300

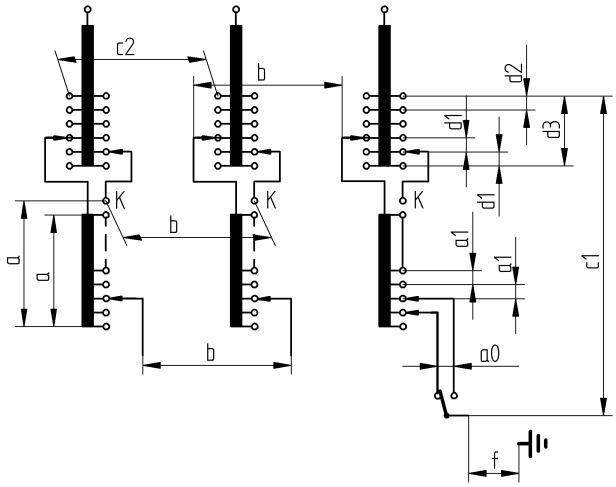
The admissible maximum operating voltage on the individual tap selector distances corresponds to half the value of the above mentioned rated short-duration power frequency withstand voltages.

 © Maschinenfabrik Reinhausen 2010

without change-over selector		with reversing change-over selector		with coarse change-over selector	
Connection	**Tap selector size**	**Connection**	**Tap selector size**	**Connection**	**Tap selector size**
10050	B/C/D/DE	10071W	B/C/D/DE	10071G	B/C/D/DE
10060	B/C/D/DE	10081W	B/C/D/DE	10081G	B/C/D/DE
10070	B/C/D/DE	10091W	B/C/D/DE	10091G	B/C/D/DE
10080	B/C/D/DE	12101W	B/C/D/DE	12101G	B/C/D/DE
10090	B/C/D/DE	12111W	B/C	12111G	B/C
10100	B/C/D/DE	14111W	D/DE	14111G	D/DE
12110	B/C/D/DE	14121W	B/C	14121G	B/C
12120	B/C/D/DE	14131W	B/C	14131G	B/C
14130	B/C/D/DE	16121W	D/DE	16121G	D/DE
14140	B/C/D/DE	16131W	D/DE	16131G	D/DE
16150	B/C/D/DE	16141W	B/C/D/DE	16141G	B/C/D/DE
16160	B/C/D/DE	16151W	B/C	16151G	B/C
18170	B/C/D/DE	18151W	D/DE	18151G	D/DE
18180	B/C/D/DE	18161W	B/C	18161G	B/C
22190	B/C/D/DE	18171W	B/C	18171G	B/C
22200	B/C/D/DE	10191W	B/C/D/DE	10191G	B/C/D/DE
22210	B/C	12231W	B/C/D/DE	12231G	B/C/D/DE
22220	B/C	14271W	B/C/D/DE	14271G	B/C/D/DE
		16311W	B/C/D/DE	16311G	B/C/D/DE
		18351W	B/C/D/DE	18351G	B/C/D/DE

Table 7 Available connections (also available as 3 W, 3 G)

1.4.3 VM rated withstand voltages — multiple coarse change-over selector



a0 = Between selected and preselected tap on the diverter switch and tap selector

a1 = Between tap selector contacts of the winding of one tap position (connected or not connected)

a = Between beginning and end of a tapped winding and also between the connected K contact and any points of the tapped winding of the same phase

b = Between the tap selector contacts of different phases and between the connected K contact of one phase and any points of the tapped winding of another phase

c1 = Between any coarse tappings of one phase to the diverter switch terminal of the same phase

c2 = Between identically-named, unconnected coarse tappings of different phases

d1 = Between connected and adjacent coarse tap contacts in one phase

d2 = Between unconnected, adjacent coarse tap contacts in one phase

d3 = Between beginning and end of all coarse tap connections of one phase

f = Between diverter switch output terminal and ground

Insulation distances	Tap selector size B		Tap selector size C		Tap selector size D	
	kV 1.2/ 50 µs	kV 50 Hz, 1 min	kV 1.2/50 µs	kV 50 Hz, 1 min	kV 1.2/50 µs	kV 50 Hz, 1 min
a0	Power-frequency voltage test (50 Hz/60 s): 20 kV 50 Hz 1 min; Response voltage with 1 mA leakage current: 35 kV 1.2/50 µs; 3 kA residual voltage (= lightning impulse protection level): 56 kV 1.2/50 µs with ZnO					
a1	150	30	150	30	150	30
a	265	50	350	82	450	105
b[1)]	265	50	350	82	450	146
c1	455	127	525	165	590	210
c2[1)]	455	127	525	165	590	215
d1	265	50	350	82	450	105
d2	350	82	450	105	450	105
d3	350	82	450	105	490	120

Table 8 On-load tap-changer VACUTAP® VM with multiple coarse change-over selector, rated withstand voltages of the internal on-load tap-changer insulation

1) Insulation distance omitted for single-pole on-load tap-changers

The admissible maximum operating voltage on the individual tap selector distances corresponds to half the value of the above mentioned rated short-duration power frequency withstand voltages.

 © Maschinenfabrik Reinhausen 2010

2 Special designs

2.1 Parallel bridges for parallel connections

Current division on the connection contacts of 2 tap selector planes only for on-load tap-changers VACUTAP® VM I 802 (803)/1002 and of 3 tap selector planes only for on-load tap-changers VACUTAP® VM I 1203/1503, see .**Chapter 3.1.33**

Parallel bridges on the tap selector terminals are mandatory if the tap winding has been wound in two or more branches and each of these branch taps is connected to the terminals of the tap selector.

This measure reliably prevents the following:

a) Introduction of circulating currents into the current paths of tap selector and diverter switch

b) Commutating arc on movable tap selector contact bridges

c) Overvoltage between adjacent tap selector connection contacts connected in parallel

On-load tap-changer combination for delta connection

The on-load tap-changers can also be used with the single-phase on-load tap-changer VM I 351 as a two column on-load tap-changer combination VM I 351/ VM II 352 for adjusting the voltage of transformer windings in a delta connection (analogous to VM I 501/VM II 502 and VM I 651/VM II 652).

The tap windings should correspond to those in **Figure 4**.

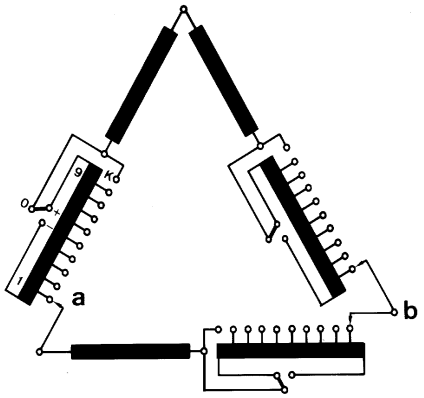


Figure 4 On-load tap-changer combination VM I 351/VM II 352 (a — VM I 351, b — VM II 352) for delta connection

2.2 On-load tap-changer VACUTAP® VM 650 Y...VM I 1503 with multiple coarse change-over selector (up to maximum of 5 coarse tap connections)

Extremely fine voltage adjustment requires a great number of operating positions which sometimes can only be achieved with a multiple coarse tapping arrangement.

For instance, 107 operating positions can be obtained by using a 5-tap coarse winding and a tapped winding with 18 taps.

The multiple coarse change-over selector is attached to both sides of the tap selector.

The on-load tap-changers are available (on request) for U_m = 72.5 up to max. 300 kV and for 2 to 5 coarse taps (tap selector sizes B and C) or 2 and 3 coarse tap connections (tap selector size D).

2.3 Two-pole on-load tap-changer VACUTAP® VM II 302/352/502/602

On-load tap-changer VM II 302/352/502/602 can be supplied as a two-pole on-load tap-changer for single-phase center point tap-change operations with the same technical data as on-load tap-changers VM III 350 Y, VM III 500 Y and/or VM III 650 Y (see dimensional drawings, **Chapter 3**).

2.4 On-load tap-changer VACUTAP® VM III 350 Y...VM I 1503 for linear voltage adjustment, up to a maximum of 34 operating positions

The high number of operating positions without change-over selector is achieved by doubling the tap selector contact planes compared with the standard design (requires a larger installation length).

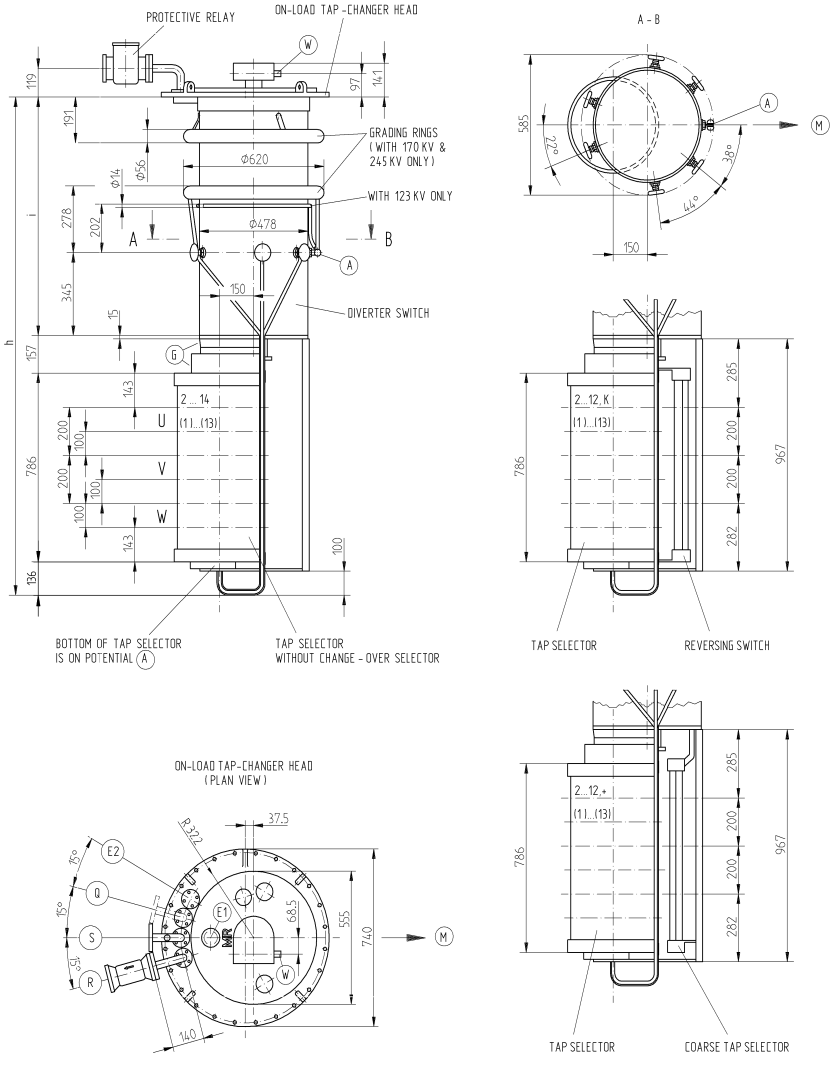
The on-load tap-changers are available for U_m = 72.5...300 kV and several tap selector sizes (rated withstand voltages and dimensional drawings on request).

 © Maschinenfabrik Reinhausen 2010

3 Appendix

3.1 Dimensional drawings/connection diagrams

3.1.1 VACUTAP® VM III 300 Y-0/W/G (898038)



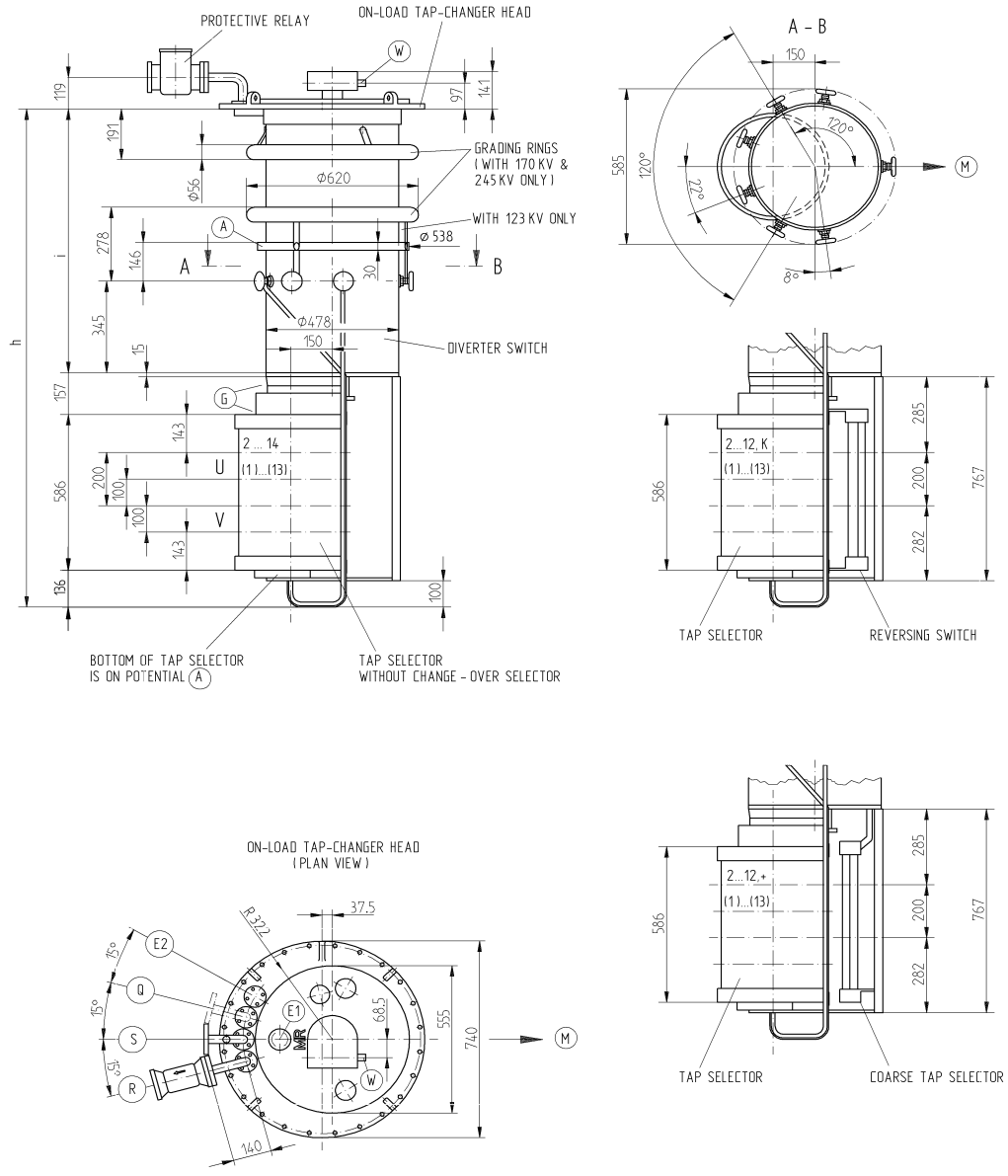
(A) ON-LOAD TAP-CHANGER CURRENT TAKE-OFF TERMINAL (NEUTRAL)
(E1) BLEEDING OF ON-LOAD TAP-CHANGER HEAD
(E2) BLEEDING OF TRANSFORMER TANK
(G) TAP SELECTOR GEAR (ON POTENTIAL (A))
(Q) CONNECTING FLANGE FOR OIL RETURN (WITH FILTER UNIT ONLY)
(R) CONNECTING FLANGE FOR PROTECTIVE RELAY
(S) CONNECTING FLANGE FOR SUCTION PIPE
(W) DRIVE SHAFT
(M) DRIVE SIDE OF TAP SELECTOR

FOR INHERENT DRAWINGS REFER TO 898 026.

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE CONNECTION DIAGRAM OF THE TAP CHANGER

TAP SELECTOR SIZE		B			
U_M IN KV		72.5	123	170	245
DIMENSIONS IN MM	h	1942	2072	2202	2302
	i	863	993	1123	1223
OIL VOLUME DM^3		130	150	170	190
DISPLACEMENT DM^3		190	220	240	260
WEIGHT KG		220			

3.1.2 VACUTAP® VM II 302-0/W/G (898039)



(A) ON-LOAD TAP-CHANGER CURRENT TAKE-OFF TERMINAL
(E1) BLEEDING OF ON-LOAD TAP-CHANGER HEAD
(E2) BLEEDING OF TRANSFORMER TANK
(G) TAP SELECTOR GEAR (ON POTENTIAL (A))
(Q) CONNECTING FLANGE FOR OIL RETURN (WITH FILTER UNIT ONLY)
(R) CONNECTING FLANGE FOR PROTECTIVE RELAY
(S) CONNECTING FLANGE FOR SUCTION PIPE
(W) DRIVE SHAFT
(M) DRIVE SIDE OF TAP SELECTOR

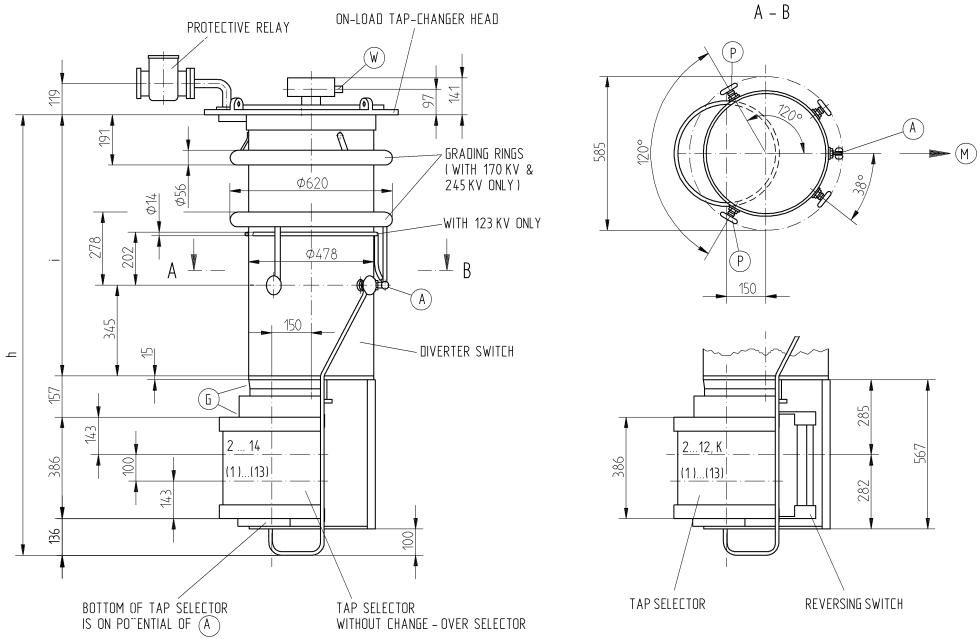
FOR INHERENT DRAWINGS REFER TO 898 026 :

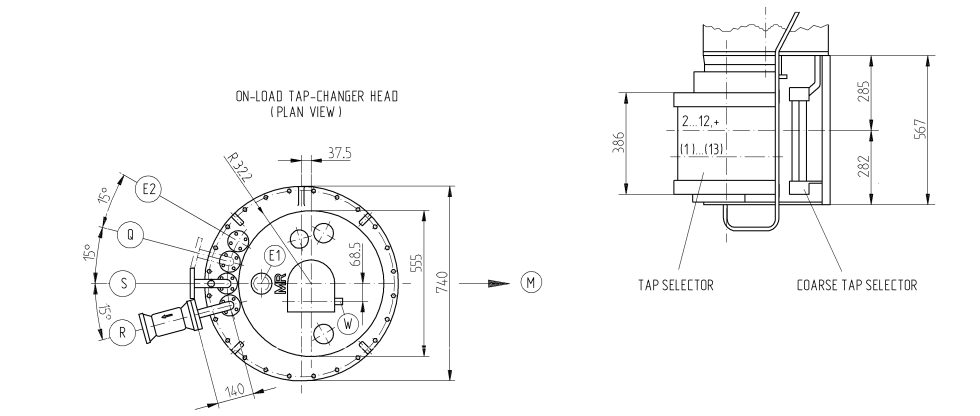
FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE CONNECTION DIAGRAM OF THE TAP CHANGER

TAP SELECTOR SIZE		B			
U_M IN KV		72,5	123	170	245
DIMENSIONS IN MM	h	1742	1872	2002	2102
	i	863	993	1123	1223
OIL VOLUME DM^3		130	150	170	190
DISPLACEMENT DM^3		180	210	230	250
WEIGHT KG		200			

 © Maschinenfabrik Reinhausen 2010

3.1.3 VACUTAP® VM I 301-0/W/G (898040)





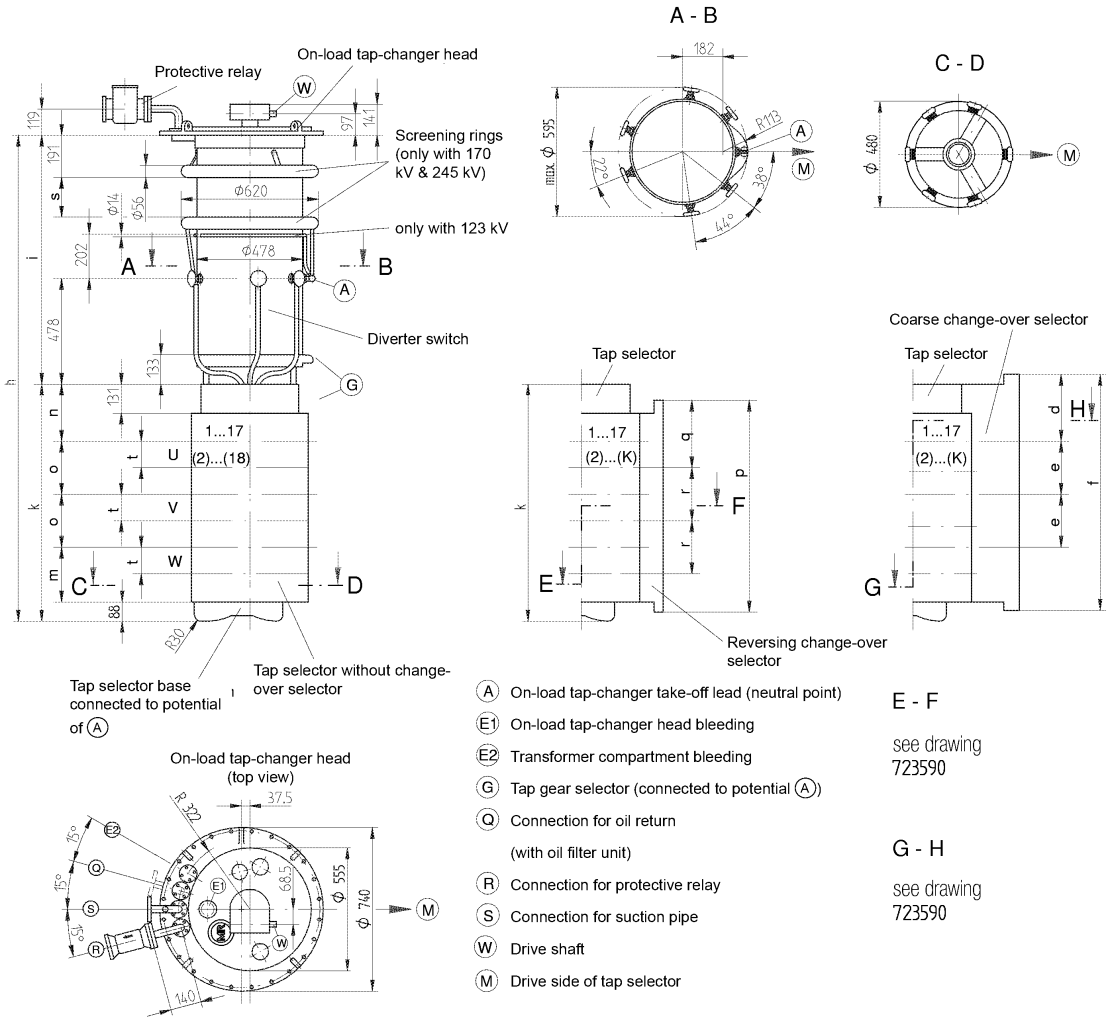
(A) ON-LOAD TAP-CHANGER CURRENT TAKE-OFF TERMINAL
(E1) BLEEDING OF ON-LOAD TAP-CHANGER HEAD
(E2) BLEEDING OF TRANSFORMER TANK
(G) TAP SELECTOR GEAR (ON POTENTIAL (A))
(Q) CONNECTING FLANGE FOR OIL RETURN (WITH FILTER UNIT ONLY)
(R) CONNECTING FLANGE FOR PROTECTIVE RELAY
(S) CONNECTING FLANGE FOR SUCTION PIPE
(W) DRIVE SHAFT
(M) DRIVE SIDE OF TAP SELECTOR
(P) IS ON POTENTIAL OF (A)

FOR INHERENT DRAWINGS REFER TO 898 026 :

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE CONNECTION DIAGRAM OF THE TAP CHANGER

TAP SELECTOR SIZE		B			
U_M IN KV		72,5	123	170	245
DIMENSIONS IN MM	h	1542	1672	1802	1902
	i	863	993	1123	1223
OIL VOLUME DM³		130	150	170	190
DISPLACEMENT DM³		160	190	210	230
WEIGHT KG		200			

3.1.4 VACUTAP® VM III 350/500/650 Y-0/W/G (746219)



(A) On-load tap-changer take-off lead (neutral point)
(E1) On-load tap-changer head bleeding
(E2) Transformer compartment bleeding
(G) Tap gear selector (connected to potential (A))
(Q) Connection for oil return (with oil filter unit)
(R) Connection for protective relay
(S) Connection for suction pipe
(W) Drive shaft
(M) Drive side of tap selector

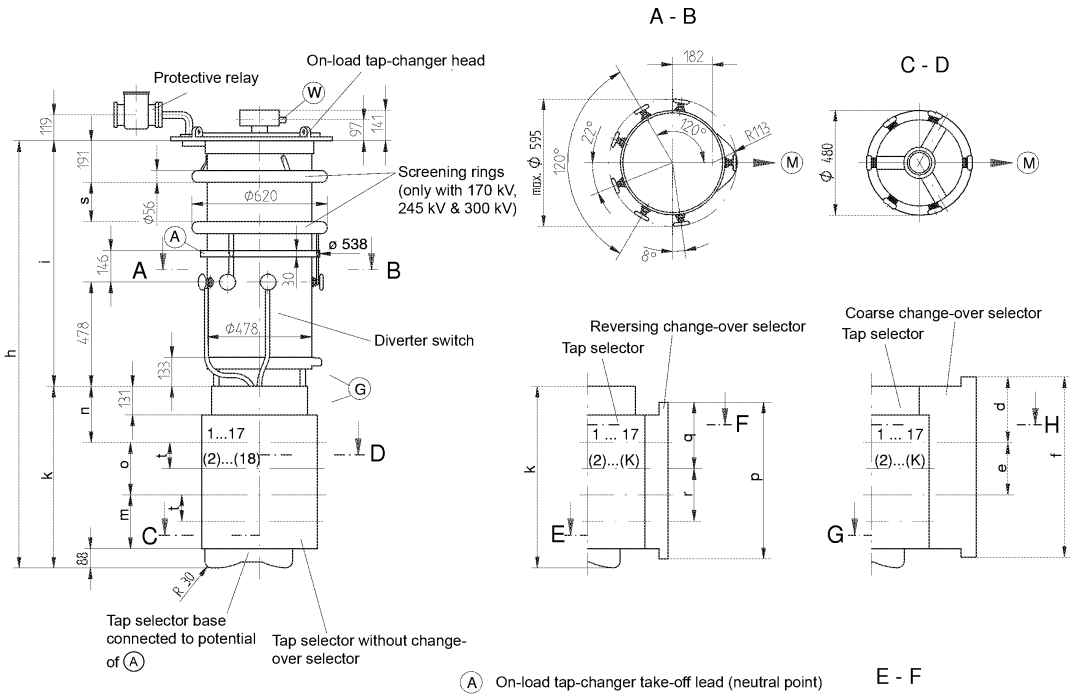
For associated drawings, see: 898012

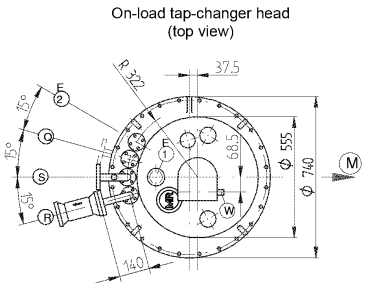
The connection diagram is binding for the designation of connection terminals and phases

Tap selector size		B				C				D/DE			
U_m [kV]		72,5	123	170	245	72,5	123	170	245	72,5	123	170	245
Dimensions [mm]	h	1894	2024	2154	2254	2069	2199	2329	2429	2524	2654	2784	2884
	i	996	1126	1256	1356	996	1126	1256	1356	996	1126	1256	1356
	s	-		267	367	-		267	367	-		267	367
	k	898				1073				1528			
	n	233				258				323			
	o	190				240				370			
	m	197				247				377			
	t	95				120				185			
	r	190				240				370			
	q	255				305				435			
	p	783				958				1413			
	d	276.5				301.5				366.5			
	e	190				240				370			
	f	892				1067				1522			
Oil content [dm³]		130	150	170	190	130	150	170	190	130	150	170	190
Displacement [dm³]		199	224	244	264	199	224	244	264	207	232	252	272
Max. weight [kg]		350	355	360	365	360	365	370	375	375	380	385	390

 © Maschinenfabrik Reinhausen 2010

3.1.5 VACUTAP® VM II 352/502/652-0/W/G (746220)





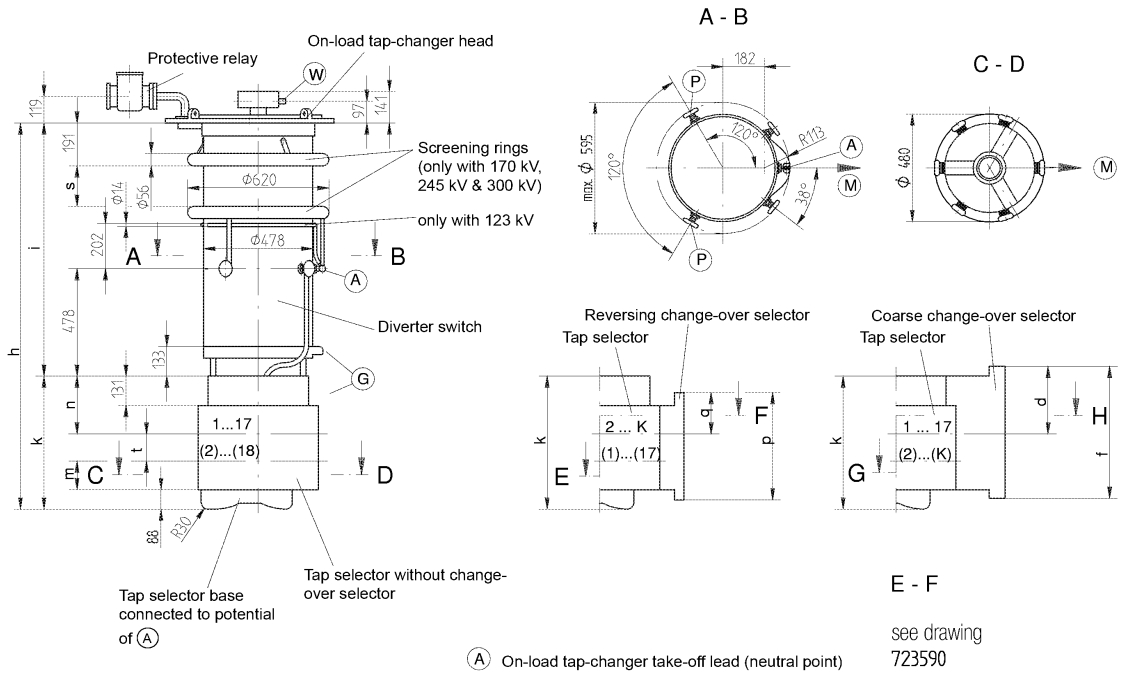
A On-load tap-changer take-off lead (neutral point)
E1 On-load tap-changer head bleeding
E2 Transformer compartment bleeding
G Tap gear selector (connected to potential A)
Q Connection for oil return
(with oil filter unit)
R Connection for protective relay
S Connection for suction pipe
W Drive shaft
M Drive side of tap selector

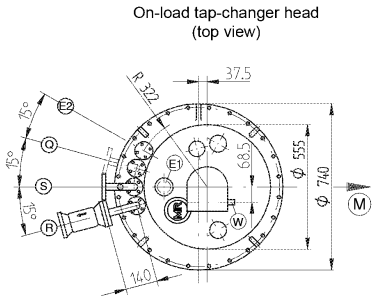
For associated drawings,
see: 898012

The connection diagram is binding for the designation of
connection terminals and phases

Tap selector size		B					C					D/DE				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1704	1834	1964	2064	2216	1829	1959	2089	2189	2341	2154	2284	2414	2514	2666
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	-		267	367	520	-		267	367	520	-		267	367	520
	k	708					833					1158				
	n	233					258					323				
	o	190					240					370				
	m	197					247					377				
	t	95					120					185				
	r	190					240					370				
	q	255					305					435				
	p	593					718					1043				
	e	190					240					370				
	d	276.5					301.5					366.5				
	f	702					827					1152				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		196	221	241	261	281	196	221	241	261	281	199	224	244	264	284
Max. weight [kg]		310	315	320	325	330	320	325	330	335	340	330	335	340	345	350

3.1.6 VACUTAP® VM II 351/501/651-0/W/G (746221)





A On-load tap-changer take-off lead (neutral point)
E1 On-load tap-changer head bleeding
E2 Transformer compartment bleeding
G Tap gear selector (connected to potential A)
Q Connection for oil return (with oil filter unit)
R Connection for protective relay
S Connection for suction pipe
W Drive shaft
M Drive side of tap selector
P connected to potential of A

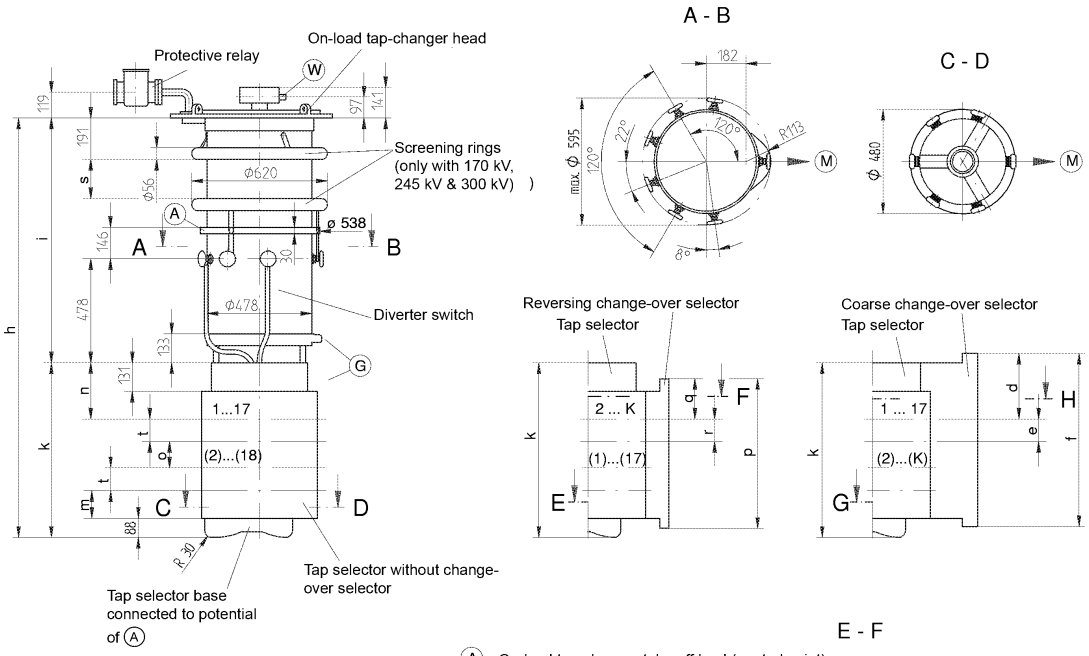
For associated drawings, see: 898012

The connection diagram is binding for the designation of connection terminals and phases

Tap selector size		B					C					D/DE				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1514	1644	1774	1874	2026	1589	1719	1849	1949	2101	1784	1914	2044	2144	2296
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	-		267	367	520	-		267	367	520	-		267	367	520
	k	518					593					788				
	n	233					258					323				
	m	102					127					192				
	t	95					120					185				
	q	160					185					250				
	p	403					478					673				
	d	276.5					301.5					366.5				
	f	512					587					782				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		193	218	238	258	278	193	218	238	258	278	195	220	240	260	280
Max. weight [kg]		280	285	290	295	300	290	295	300	305	310	300	305	310	315	320

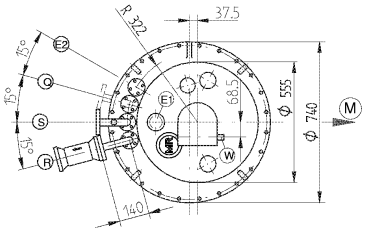
 © Maschinenfabrik Reinhausen 2010

3.1.7 VACUTAP® VM I 802/1002-0/W/G (746222)



A On-load tap-changer take-off lead (neutral point)
E1 On-load tap-changer head bleeding
E2 Transformer compartment bleeding
G Tap gear selector (connected to potential A)
Q Connection for oil return
(with oil filter unit)
R Connection for protective relay
S Connection for suction pipe
W Drive shaft
M Drive side of tap selector

On-load tap-changer head
(top view)

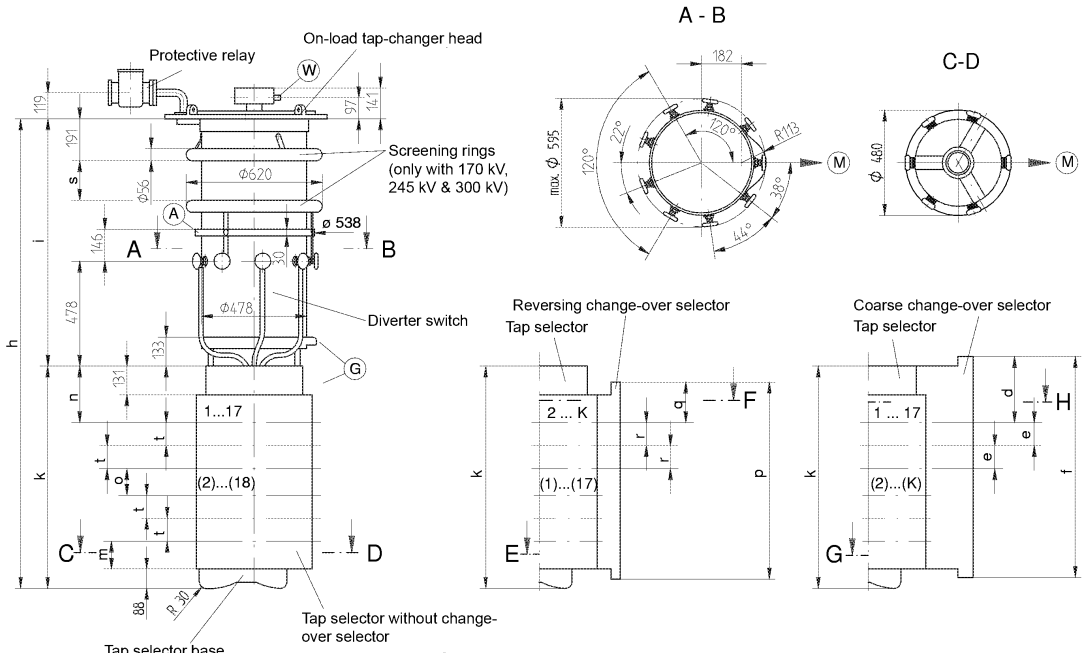


For associated drawings,
see: 898012

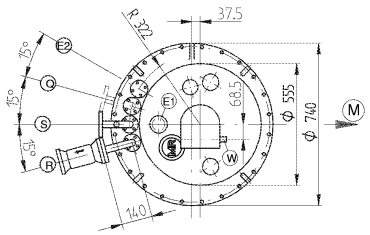
The connection diagram is binding for the designation of
connection terminals and phases

Tap selector size		B					C					D/DE				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1724	1854	1984	2084	2236	1799	1929	2059	2159	2311	1994	2124	2254	2354	2506
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	-		267	367	520	-		267	367	520	-		267	367	520
	k	728					803					998				
	n	233					258					323				
	o	95					120					185				
	m	102					127					192				
	t	105					105					105				
	r	105					105					105				
	q	160					185					250				
	p	613					688					883				
	e	105					105					105				
	d	276.5					301.5					366.5				
	f	722					797					992				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		196	221	241	261	281	196	221	241	261	281	199	224	244	264	284
Max. weight [kg]		310	315	320	325	330	320	325	330	335	340	330	335	340	345	350

3.1.8 VACUTAP® VM I 1203/1503-0/W/G (746223)



On-load tap-changer head (top view)



(A) On-load tap-changer take-off lead (neutral point)
(E1) On-load tap-changer head bleeding
(E2) Transformer compartment bleeding
(G) Tap gear selector (connected to potential (A))
(Q) Connection for oil return
(with oil filter unit)
(R) Connection for protective relay
(S) Connection for suction pipe
(W) Drive shaft
(M) Drive side of tap selector

For associated drawings, see:
898012

The connection diagram is binding for the designation of connection terminals and phases

Tap selector size		B					C					D/DE				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1934	2064	2194	2294	2446	2009	2139	2269	2369	2521	2204	2334	2464	2564	2716
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	-		267	367	520	-		267	367	520	-		267	367	520
	k	938					1013					1208				
	n	233					258					323				
	o	95					120					185				
	m	102					127					192				
	t	105					105					105				
	r	105					105					105				
	q	160					185					250				
	p	823					898					1093				
	e	105					105					105				
	d	276.5					301.5					366.5				
	f	932					1007					1202				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		200	225	245	265	285	200	225	245	265	285	204	229	249	269	289
Max. weight [kg]		350	355	360	365	370	360	365	370	375	380	375	380	385	390	395

 © Maschinenfabrik Reinhausen 2010

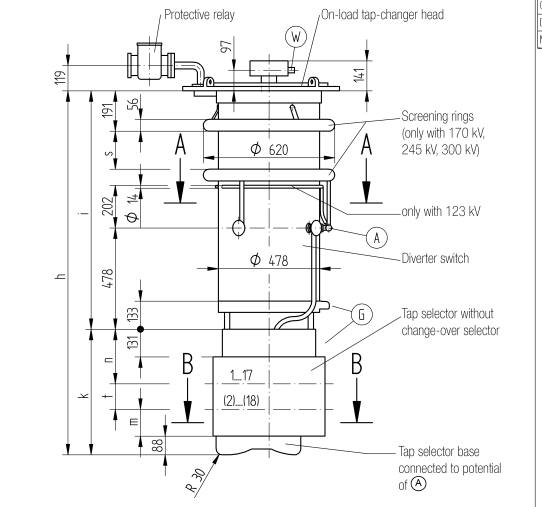
3.1.9 VACUTAP® VM III K-0/W/G (746224)

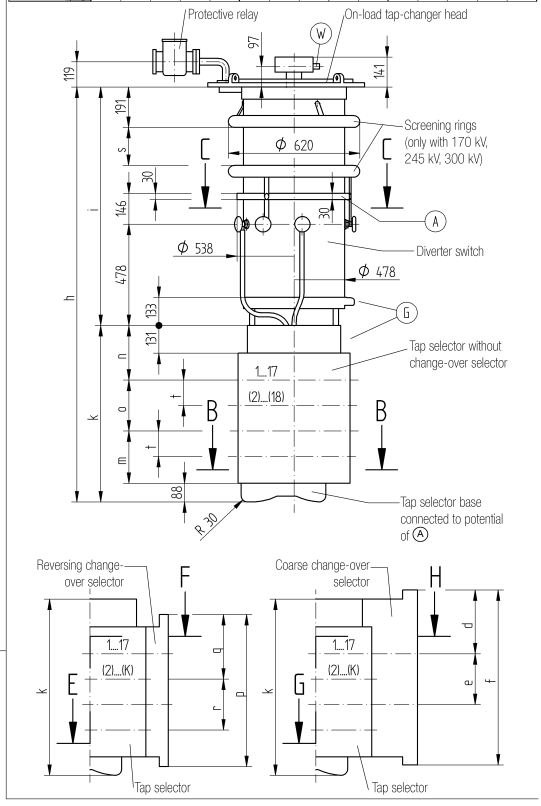
VMI 351 / 501 / 651 – 0 /W / G

Tap selector size		B					C					D/DE				
Um in kV		72.5	123	170	245	300	72.5	123	170	245	300	72.5	123	170	245	300
Dimensions in mm	h	1514	1644	1774	1874	2026	1589	1719	1849	1949	2101	1784	1914	2044	2144	2296
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	–		267	367	520	–		267	367	520	–		267	367	520
	k	518					593					788				
	n	233					258					323				
	m	102					127					192				
	t	95					120					185				
	q	160					185					250				
	p	403					478					673				
	d	276,5					301,5					366,5				
	f	512					587					782				
Oil content	dm³	130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement	dm³	193	218	238	258	278	193	218	238	258	278	195	220	240	260	280
Max. weight	kg	350	355	360	365	370	360	365	370	375	380	375	380	385	390	395

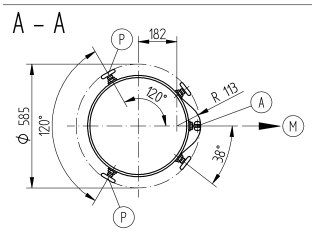
VMII 352 / 502 / 652 – 0 /W / G

Tap selector size		B					C					D/DE				
Um in kV		72.5	123	170	245	300	72.5	123	170	245	300	72.5	123	170	245	300
Dimensions in mm	h	1704	1834	1964	2064	2216	1829	1959	2089	2189	2341	2154	2284	2414	2514	2666
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	–		267	367	520	–		267	367	520	–		267	367	520
	k	708					833					1158				
	n	233					258					323				
	o	190					240					370				
	m	197					247					377				
	t	95					120					185				
	r	190					240					370				
	q	255					305					435				
	p	593					718					1043				
	e	190					240					370				
	d	276,5					301,5					366,5				
	f	702					827					1152				
Oil content	dm³	130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement	dm³	196	221	241	261	281	196	221	241	261	281	199	224	244	264	284
Max. weight	kg	310	315	320	325	330	320	325	330	335	340	330	335	340	345	350

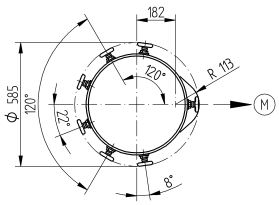


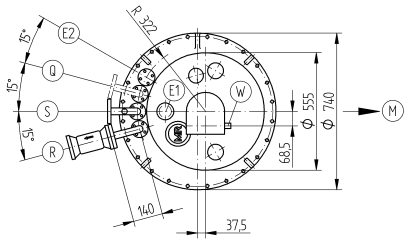


On-load tap-changer head (view from above)



C – C





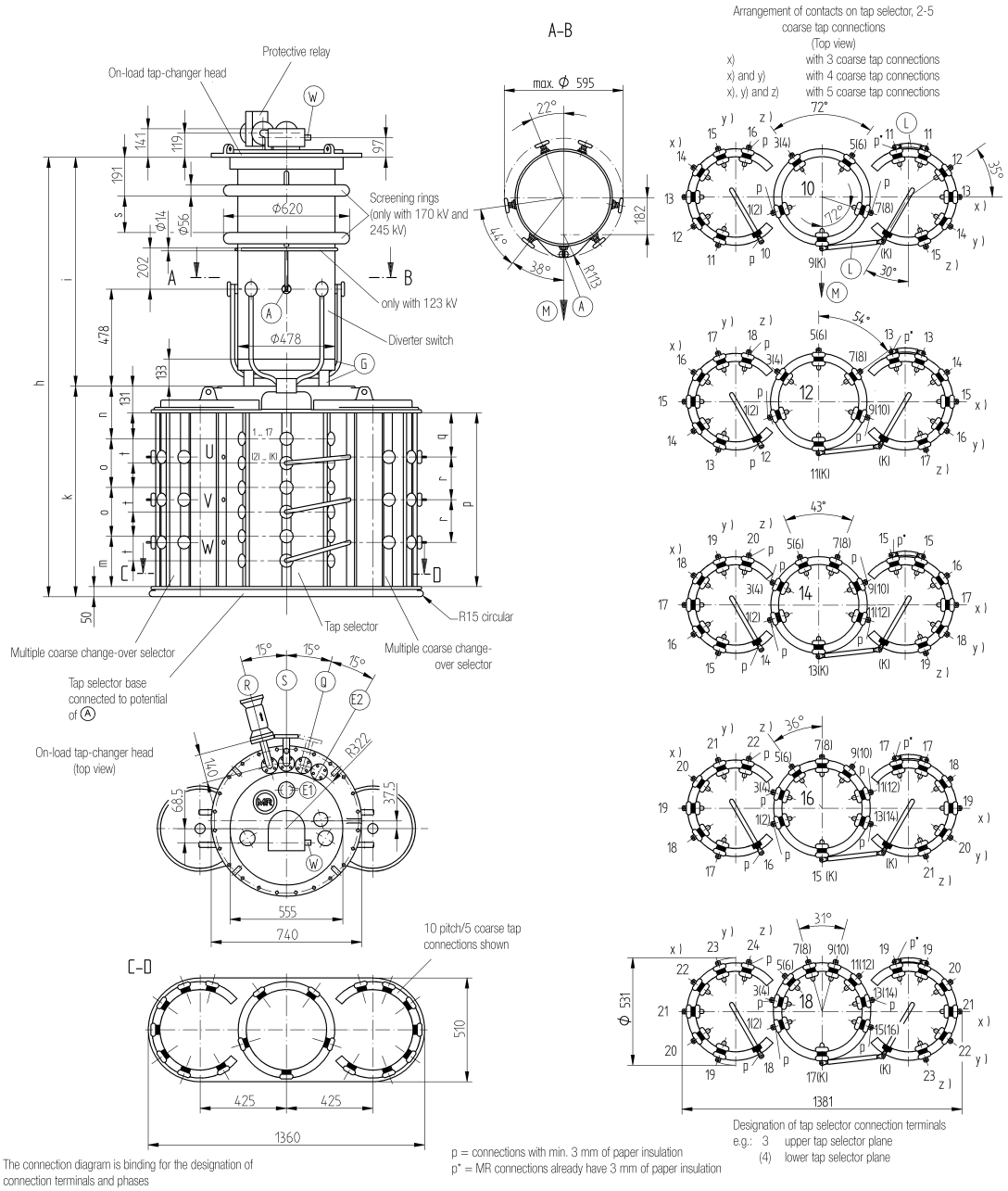
E – F / G – H see drawing 723590

(A) On-load tap-changer take-off lead
(E1) On-load tap-changer head bleeding
(E2) Transformer compartment bleeding
(G) Tap selector gear (connected to potential (A))
(L) Connecting leads (only with 10 19 3W to 18 35 3W)
(M) Drive side of the tap selector
(Q) Connection for oil return (with oil filter unit)
(R) Connection for protective relay
(S) Connection for suction pipe
(W) Drive shaft
(P) connected to potential of (A)

For associated drawings 898012

The connection diagram is binding for the designation of connection terminals and phases

3.1.10 VACUTAP® VM III 650 Y with multiple coarse change-over selector (746226)

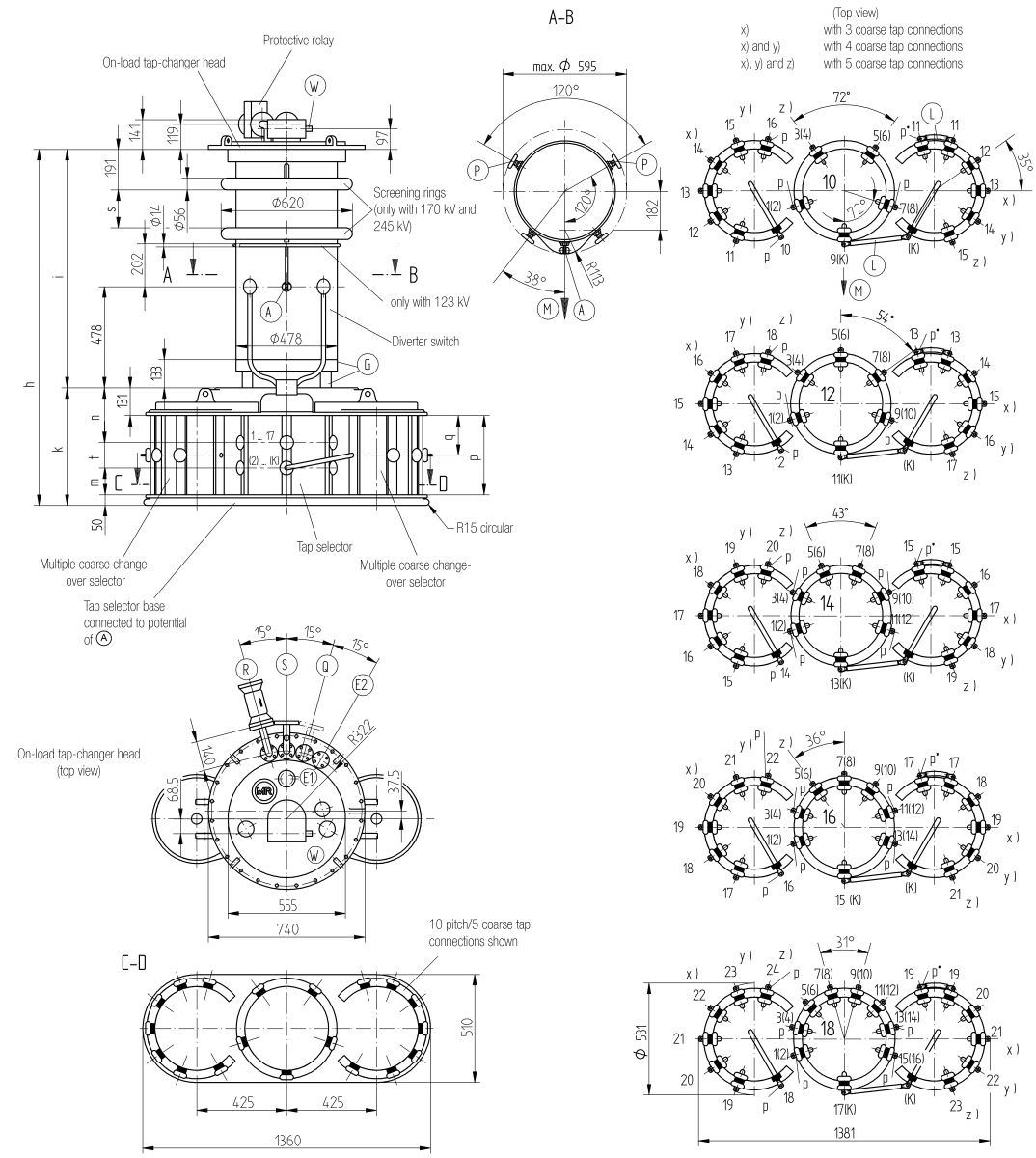


- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap selector gear (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector

Tap selector size		B				C				D			
U_m [kV]		72,5	123	170	245	72,5	123	170	245	72,5	123	170	245
Dimensions [mm]	h	1856	1986	2116	2216	2031	2161	2291	2391	2486	2616	2746	2846
	i	996	1126	1256	1356	996	1126	1256	1356	996	1126	1256	1356
	s	–		267	367	–		267	367	–		267	367
	k	860				1035				1490			
	n	233				258				323			
	o	190				240				370			
	m	197				247				377			
	t	95				120				185			
	r	166,5				210				327			
	q	173				217				327,5			
	p	679				854				1309			
Oil content [dm³]		130	150	170	190	130	150	170	190	130	150	170	190
Displacement [dm³]		321	346	366	386	321	346	366	386	333	353	375	395
Max. weight [kg]		460	465	470	475	470	475	480	485	485	490	495	500

© Maschinenfabrik Reinhausen 2010

3.1.11 VACUTAP® VM I 601 with multiple coarse change-over selector (746227)



The connection diagram is binding for the designation of connection terminals and phases

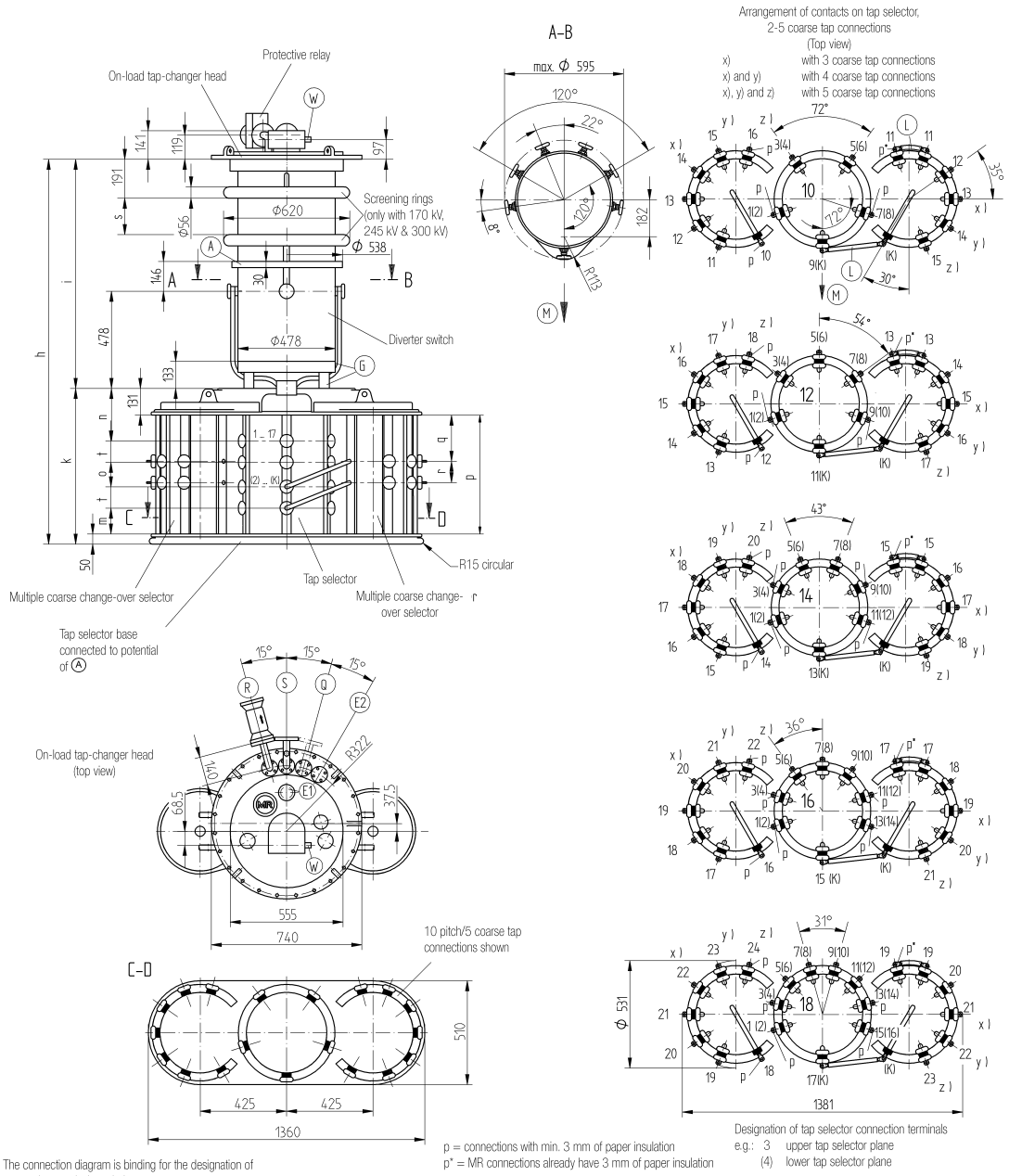
Ⓐ On-load tap-changer take-off lead
Ⓔ1 On-load tap-changer head bleeding
Ⓔ2 Transformer compartment bleeding
Ⓖ Tap selector gear (connected to potential Ⓐ)
Ⓠ Connection for oil return (with oil filter unit)
Ⓡ Connection for protective relay
Ⓢ Connection for suction pipe
Ⓦ Drive shaft
Ⓛ Connecting lead
Ⓜ Drive side of the tap selector
Ⓟ connected to potential Ⓐ

p = connections with min. 3 mm of paper insulation
p* = MR connections already have 3 mm of paper insulation

Designation of tap selector connection terminals
e.g.: 3 upper tap selector plane
(4) lower tap selector plane

Tap selector size		B					C					D				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1476	1606	1736	1836	1988	1551	1681	1811	1911	2063	1746	1876	2006	2106	2258
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	–		267	367	520	–		267	367	520	–		267	367	520
	k	480					555					750				
	n	233					258					323				
	m	102					127					192				
	t	95					120					185				
	q	149,5					187					284,5				
	p	299					374					569				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		198	223	243	263	283	198	223	243	263	283	203	223	248	268	288
Max. weight [kg]		370	375	380	385	390	380	385	390	395	400	390	395	400	405	410

3.1.12 VACUTAP® VM I 802/1002 with multiple coarse change-over selector (746228)



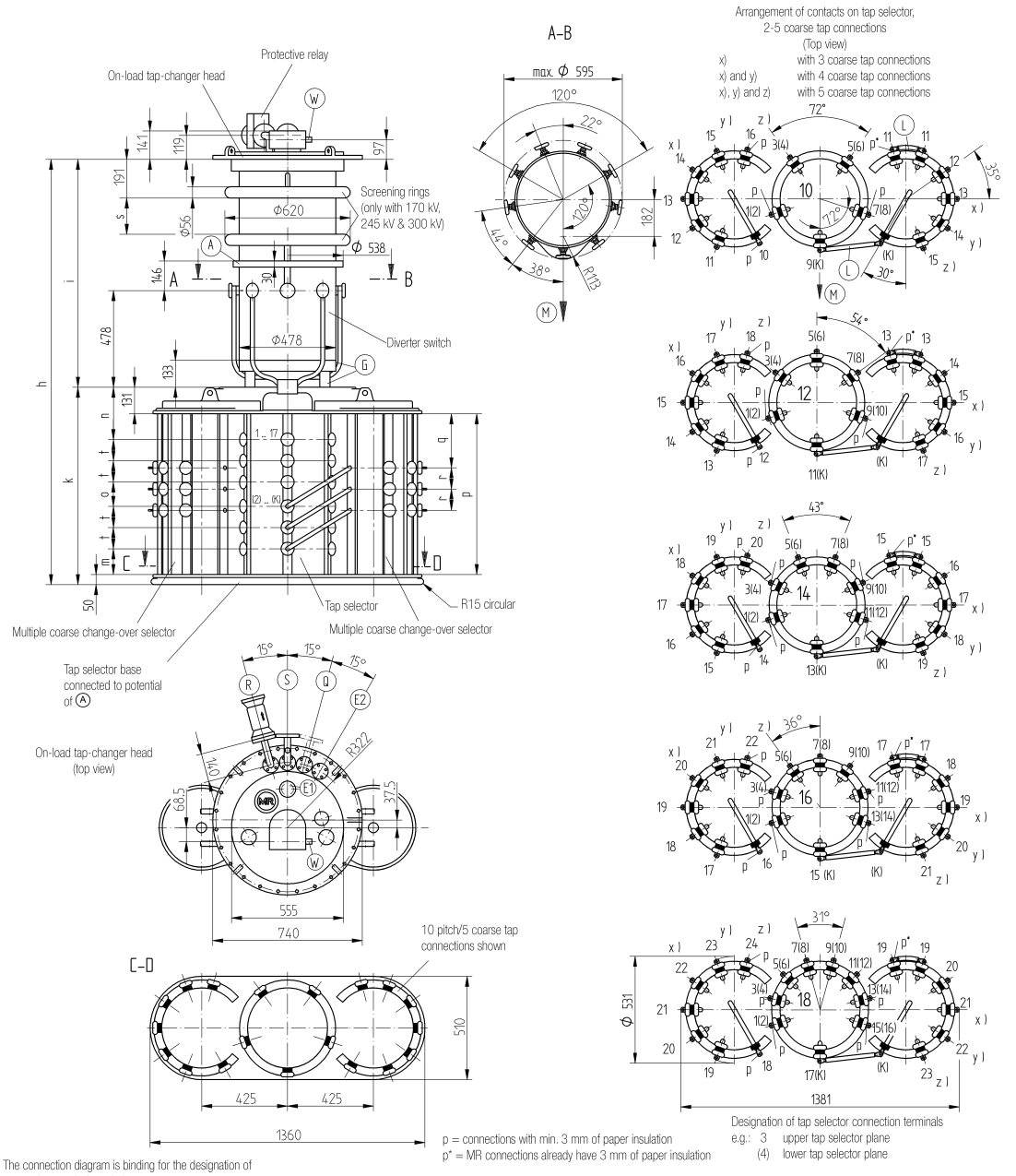
The connection diagram is binding for the designation of connection terminals and phases

- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap selector gear (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector

Tap selector size		B					C					D				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1686	1816	1946	2046	2198	1761	1891	2021	2121	2273	1956	2086	2216	2316	2468
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	–		267	367	520	–		267	367	520	–		267	367	520
	k	690					765					960				
	n	233					258					323				
	o	95					120					185				
	m	102					127					192				
	t	105					105					105				
	r	105					105					105				
	q	189,5					227					324,5				
	p	509					584					779				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		200	225	245	265	285	200	225	245	265	285	208	233	253	273	293
Max. weight [kg]		410	415	420	425	430	420	425	430	435	440	430	435	440	445	450

 © Maschinenfabrik Reinhausen 2010

3.1.13 VACUTAP® VM I 1203/1503 with multiple coarse change-over selector (746229)



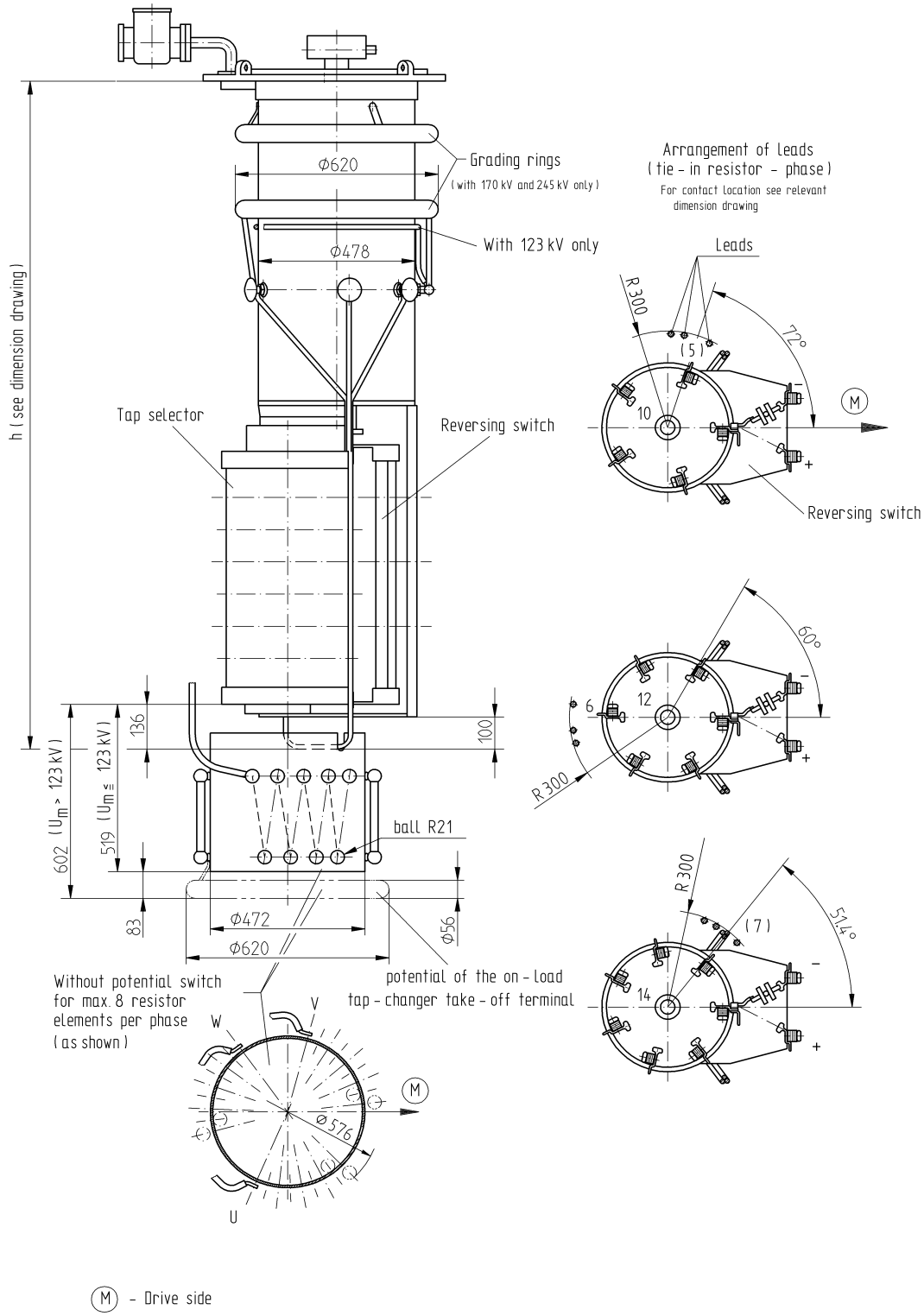
p = connections with min. 3 mm of paper insulation
p* = MR connections already have 3 mm of paper insulation

The connection diagram is binding for the designation of connection terminals and phases

- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap selector gear (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector

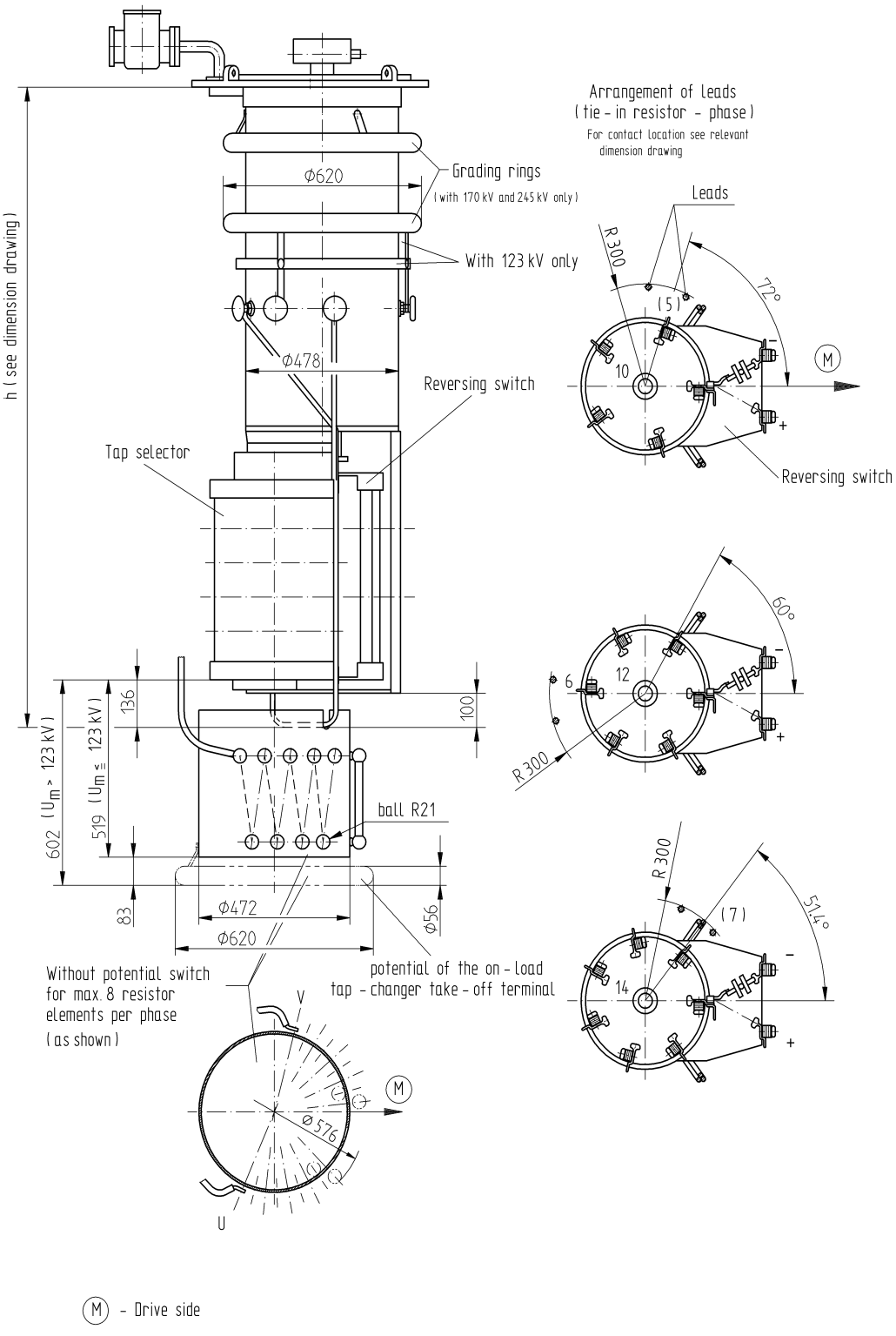
Tap selector size		B					C					D				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1896	2026	2156	2256	2408	1971	2101	2231	2331	2483	2166	2296	2426	2526	2678
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	–		267	367	520	–		267	367	520	–		267	367	520
	k	900					975					1170				
	n	233					258					323				
	o	95					120					185				
	m	102					127					192				
	t	105					105					105				
	r	105					105					105				
	q	229,5					267					364,5				
	p	719					794					989				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		320	345	365	385	405	320	345	365	385	405	331	351	373	393	413
Max. weight [kg]		460	465	470	475	480	470	475	480	485	490	485	490	495	500	505

3.1.14 VACUTAP® VM III 300 tie-in resistors without potential switch (898695)

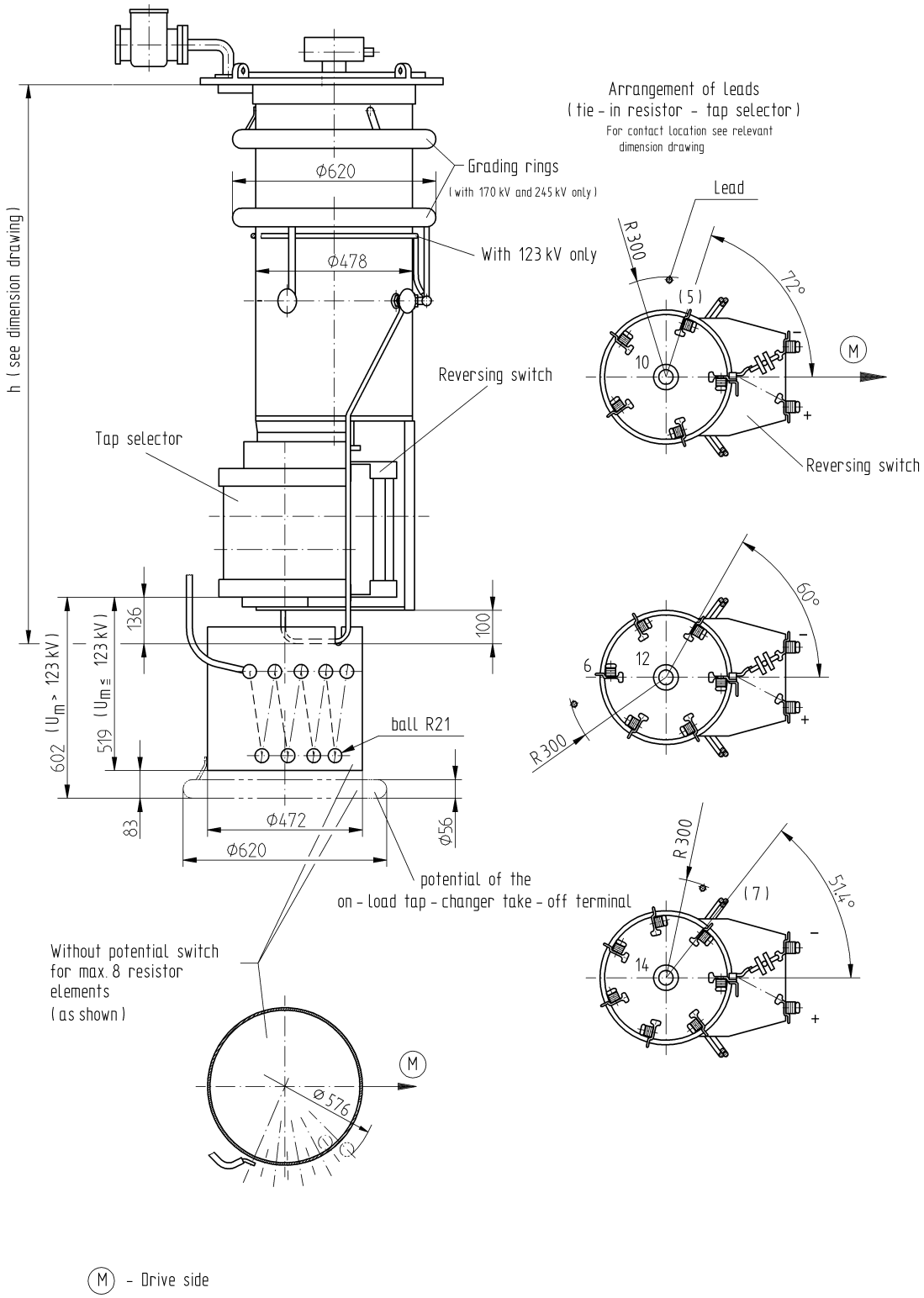


 © Maschinenfabrik Reinhausen 2010

3.1.15 VACUTAP® VM III 300 tie-in resistors without potential switch (898694)

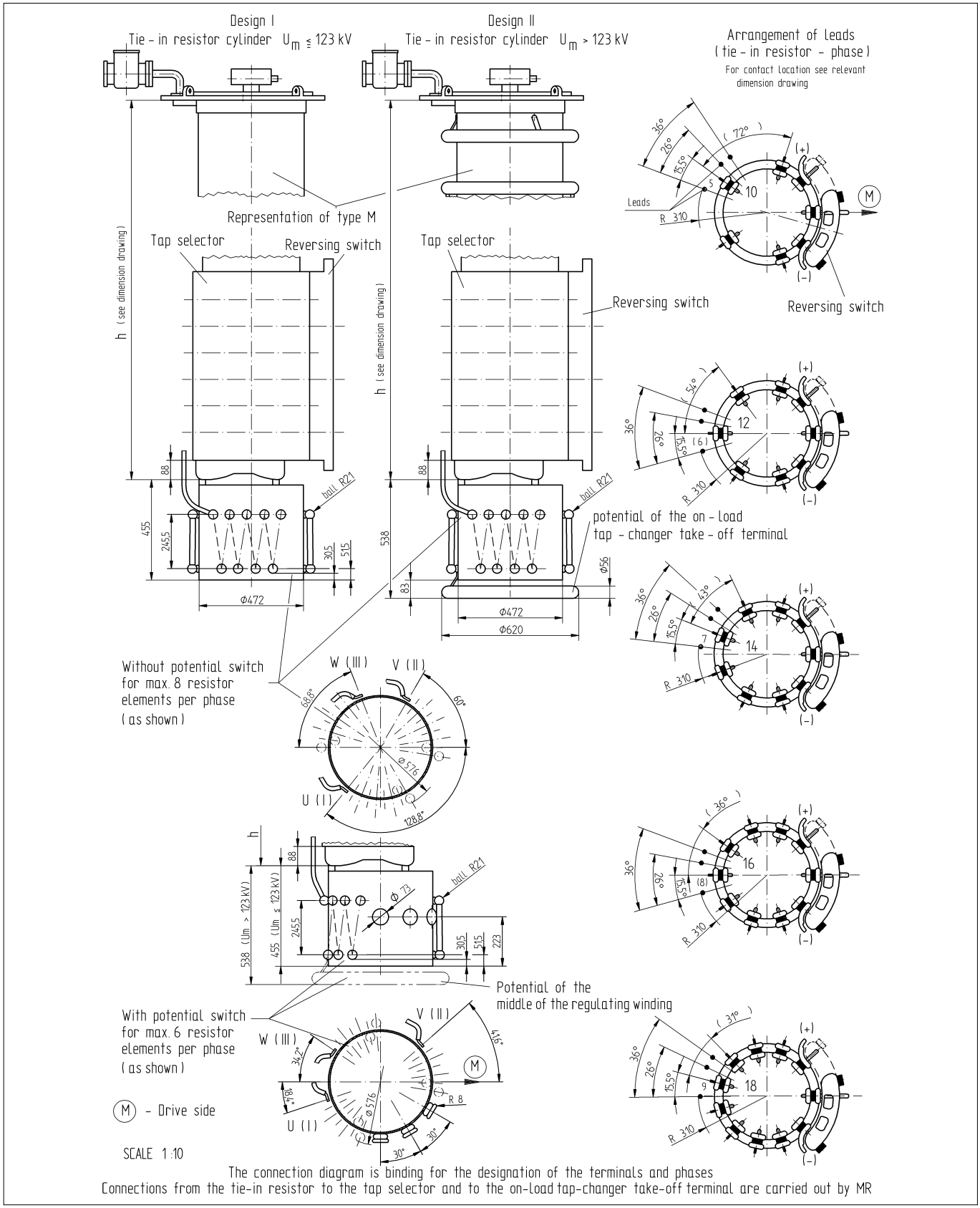


3.1.16 VACUTAP® VM III 300 tie-in resistors without potential switch (898693)

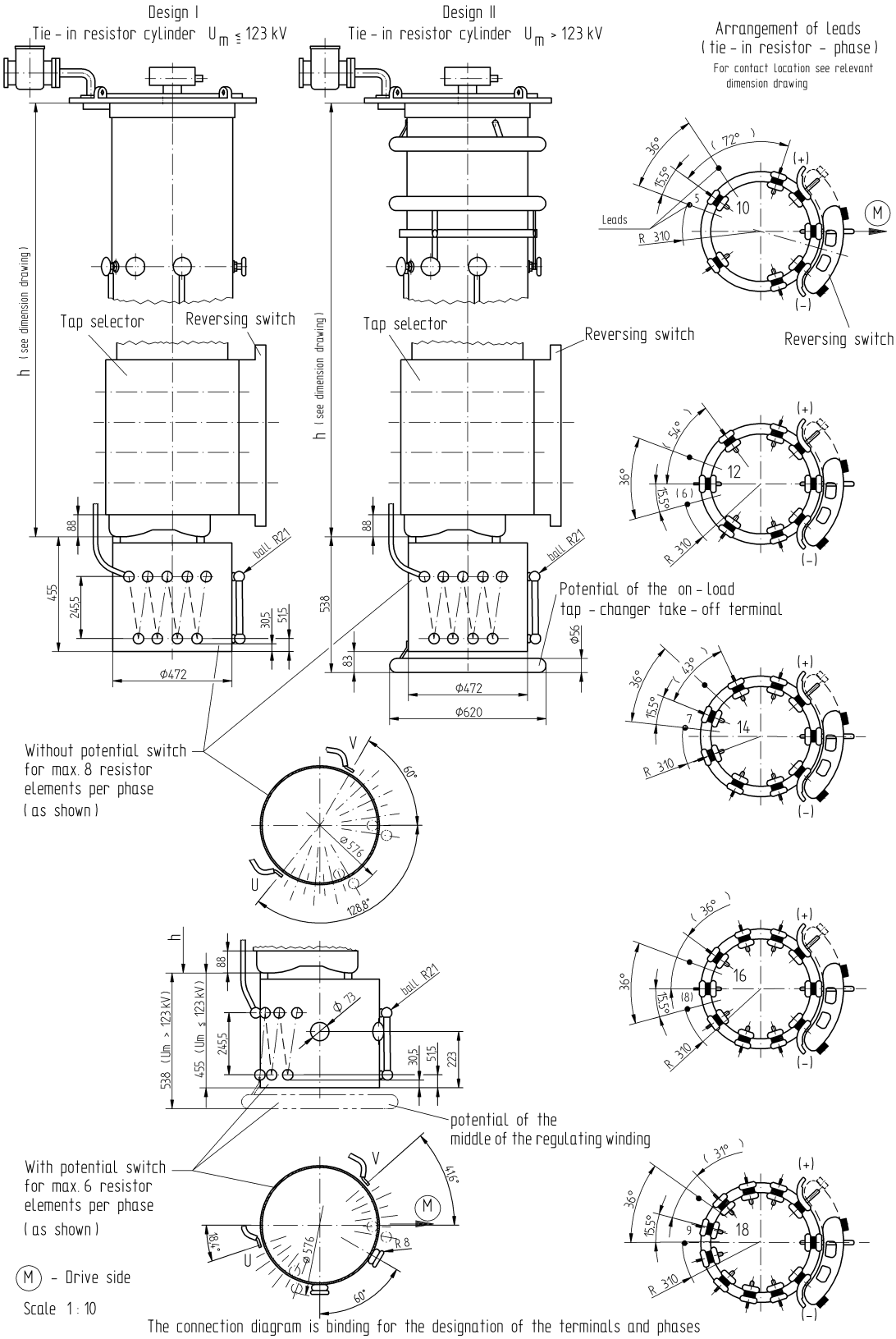


 © Maschinenfabrik Reinhausen 2010

3.1.17 VACUTAP® VM III Y tie-in resistors with/without potential switch (898692)

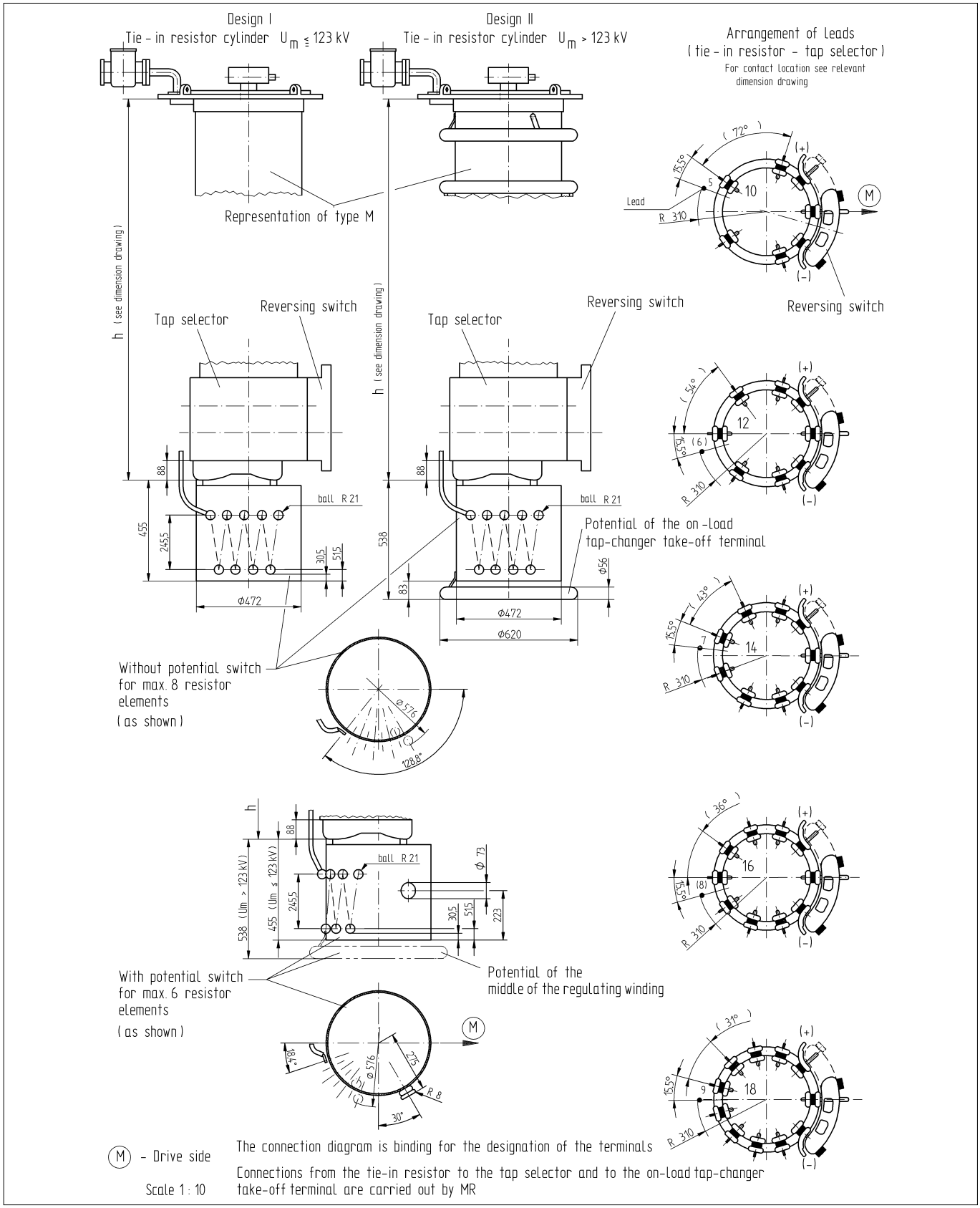


3.1.18 VACUTAP® VM II tie-in resistors with/without potential switch (898691)

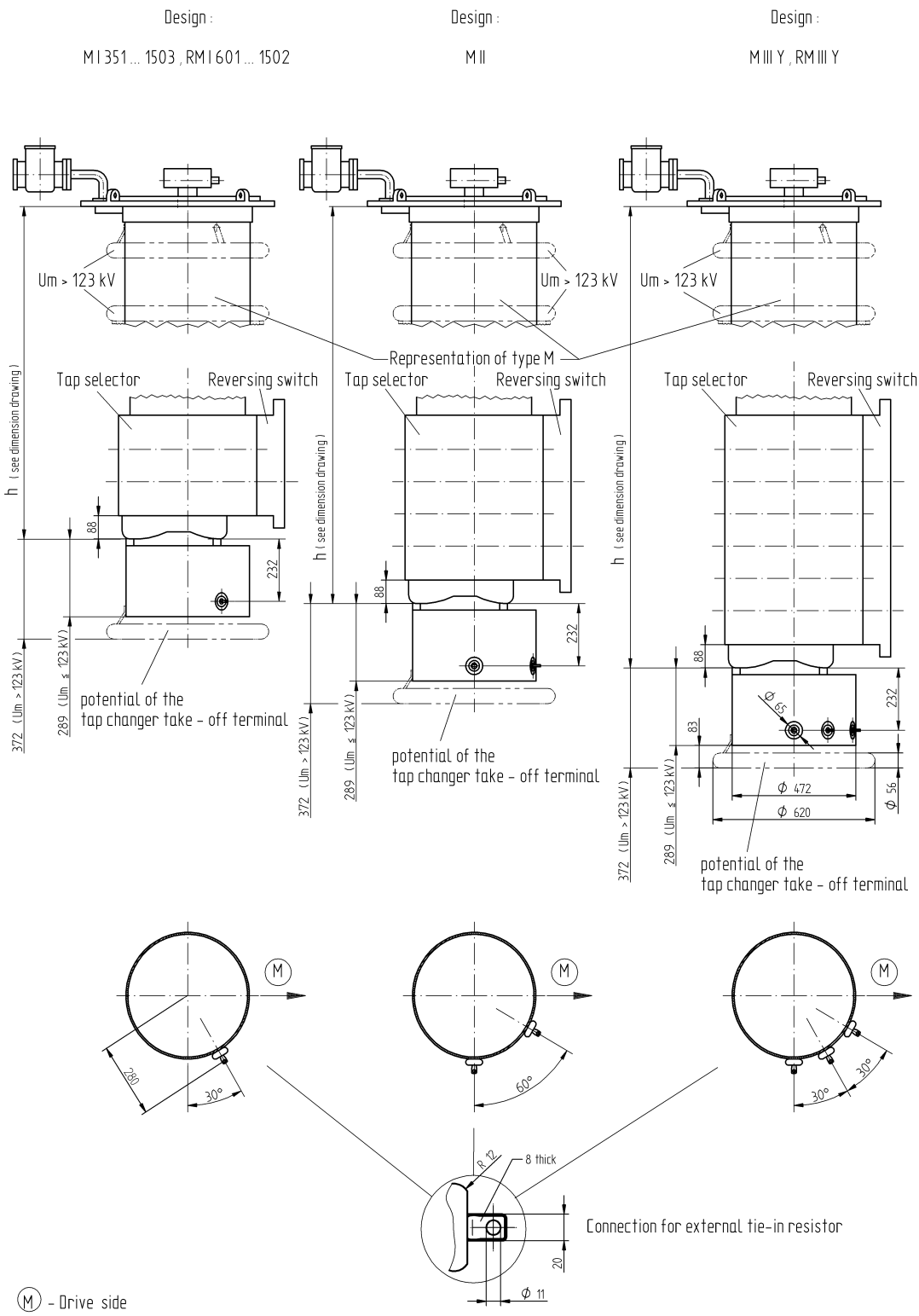


 © Maschinenfabrik Reinhausen 2010

3.1.19 VACUTAP® VM I tie-in resistors with/without potential switch (898690)

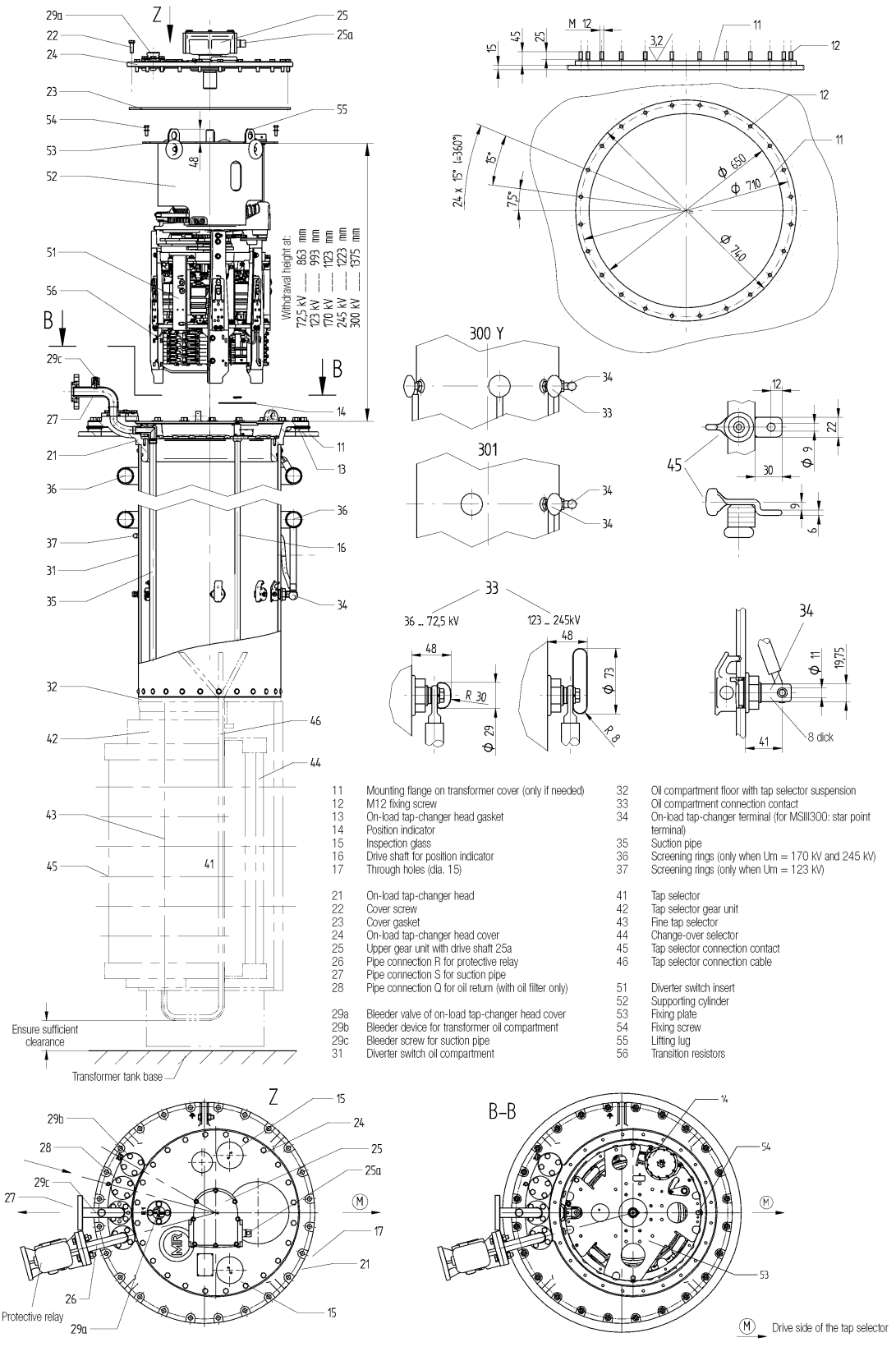


3.1.20 VACUTAP® VM I 351...1503, tie-in resistor cylinder with potential switch without tie-in resistors (898804)



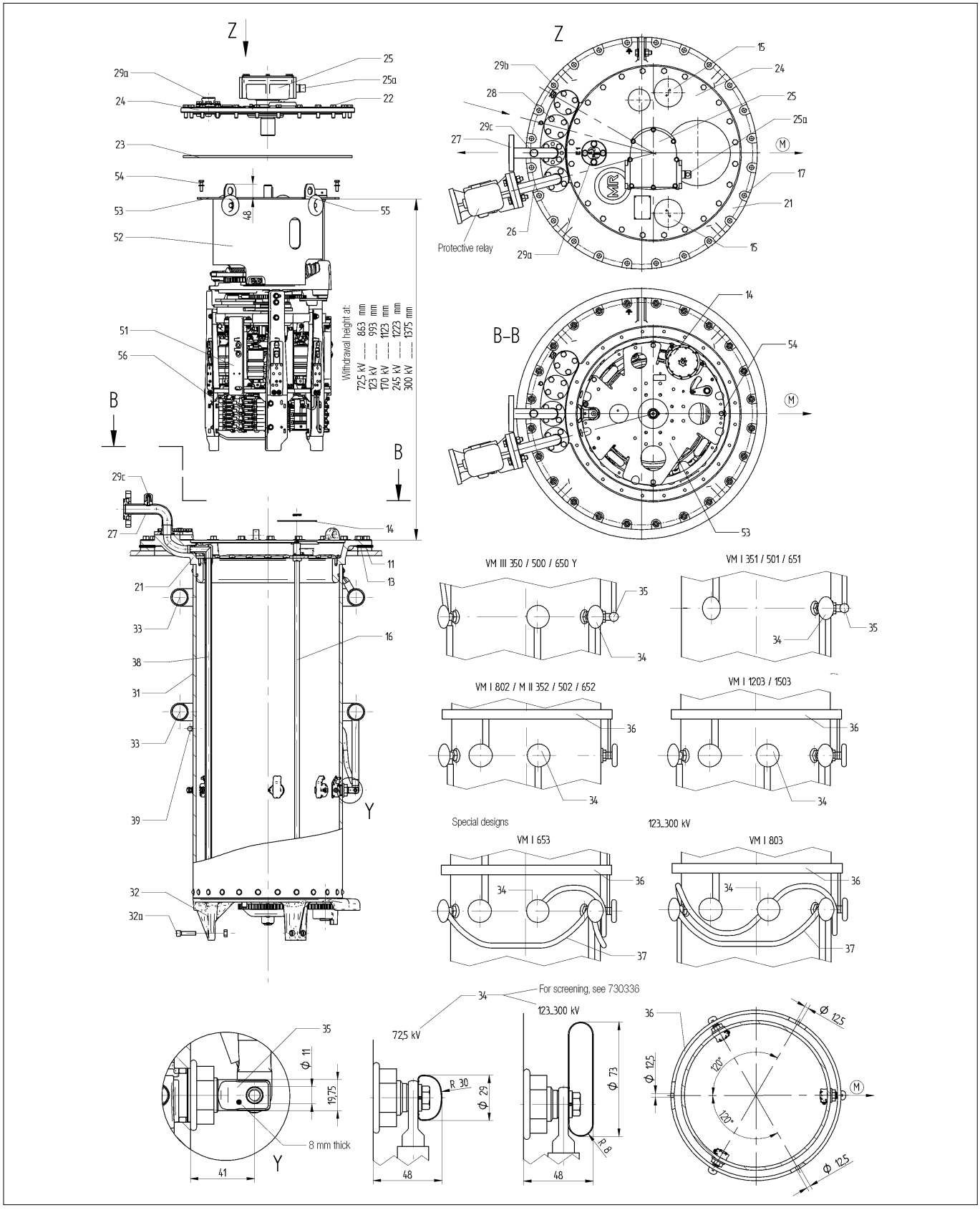
 © Maschinenfabrik Reinhausen 2010

3.1.21 VACUTAP® VM 300 — installation drawing of centrical drive (765192)



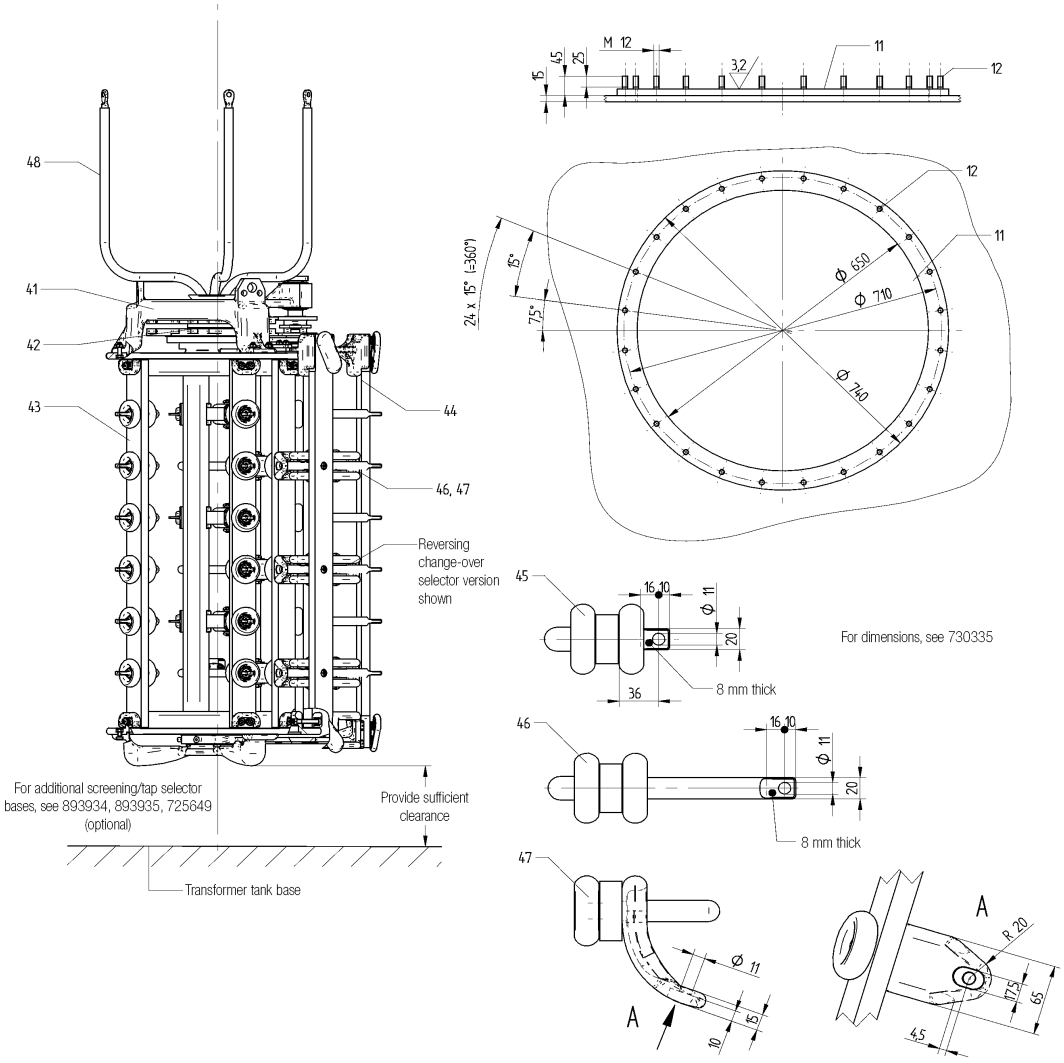
11	Mounting flange on transformer cover (only if needed)
12	M12 fixing screw
13	On-load tap-changer head gasket
14	Position indicator
15	Inspection glass
16	Drive shaft for position indicator
17	Through holes (dia. 15)
21	On-load tap-changer head
22	Cover screw
23	Cover gasket
24	On-load tap-changer head cover
25	Upper gear unit with drive shaft 25a
26	Pipe connection R for protective relay
27	Pipe connection S for suction pipe
28	Pipe connection Q for oil return (with oil filter only)
29a	Bleeder valve of on-load tap-changer head cover
29b	Bleeder device for transformer oil compartment
29c	Bleeder screw for suction pipe
31	Diverter switch oil compartment
32	Oil compartment floor with tap selector suspension
33	Oil compartment connection contact
34	On-load tap-changer terminal (for MSIII300: star point terminal)
35	Suction pipe
36	Screening rings (only when Um = 170 kV and 245 kV)
37	Screening rings (only when Um = 123 kV)
41	Tap selector
42	Tap selector gear unit
43	Fine tap selector
44	Change-over selector
45	Tap selector connection contact
46	Tap selector connection cable
51	Diverter switch insert
52	Supporting cylinder
53	Fixing plate
54	Fixing screw
55	Lifting lug
56	Transition resistors

3.1.22 VACUTAP® VM — installation drawing of centrical drive (746230)



 © Maschinenfabrik Reinhausen 2010

VACUTAP® VM — installation drawing of centrical drive (746230)



11 Mounting flange on transformer cover
12 Fixing bolt M12
13 On-load tap-changer head gasket
14 Remove position indicator before withdrawal of diverter switch
15 Inspection window
16 Drive shaft for position indicator
17 Through hole 15 mm DIA

21 On-load tap-changer head
22 Cover bolt
23 Cover gasket
24 On-load tap-changer head cover
25 Centric gear unit with drive shaft 25a
26 Pipe connection R for protective relay
27 Pipe connection S for suction pipe
28 Pipe connection Q for oil return with oil filter only
29a Air-vent valve of on-load tap-changer head cover
29b Bleeding facility for transformer oil compartment
29c Vent screw for suction pipe

(M) Drive side of the tap selector

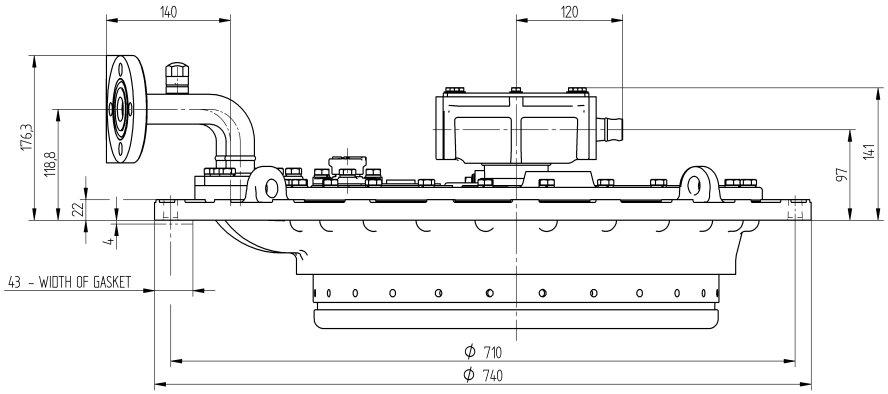
31 Diverter switch oil compartment
32 Oil compartment base with tap selector support screw 32a
33 Screening rings (with 170 kV, 245 kV, 300 kV only)
34 Oil compartment connection terminal
35 Connection terminal (VM III 350/500/650: neutral point VM I 351/501/651 on-load tap-changer take-off lead)
36 Connecting ring for on-load tap-changer take-off lead (only with VM I 802/803/1203/1503)
37 Connecting lead (only with VM I 653/803)
38 Suction pipe
39 Screening ring (only with 123 kV)

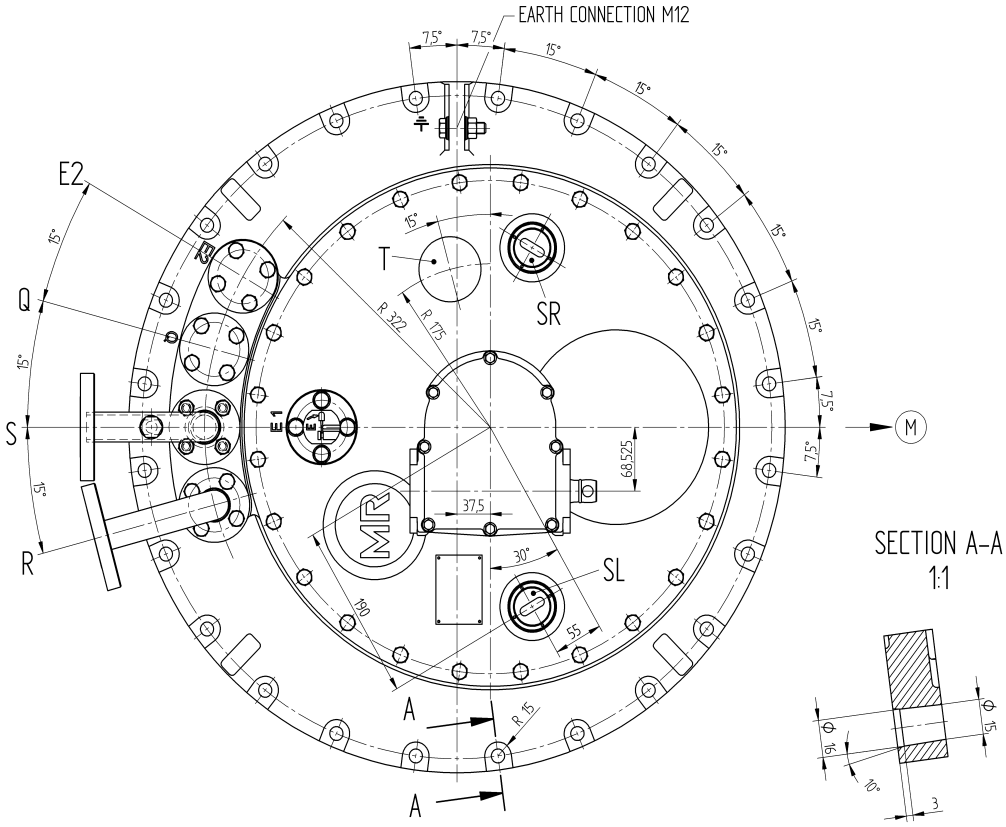
41 Tap selector suspension
42 Tap selector gear
43 Fine tap selector
44 Change-over selector
45 Tap selector contacts (see corresponding dimension drawing)
46 Change-over selector connection terminal "K"and/or "0" **
47 Change-over selector connection terminal "+" and/or "-" **
48 Tap selector connecting lead

51 Diverter switch insert
52 Supporting cylinder
53 Base plate
54 Fixing bolt
55 Lifting lug through hole 25 mm DIA
56 Transition resistors

** not with multiple coarse change-over selector

3.1.23 VACUTAP® VM — on-load tap-changer head, centrical drive (893899)





SECTION A-A
1:1

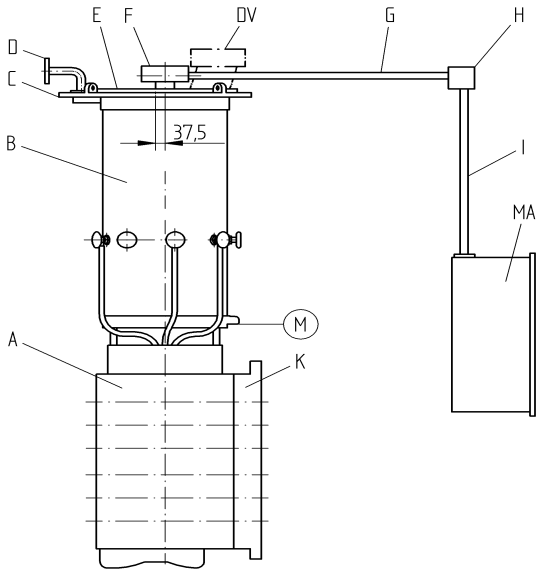
E1 = BLEEDING FACILITY FOR ON-LOAD TAP-CHANGER HEAD
E2 = BLEEDING FACILITY FOR SPACE UNDER THE HEAD OUTSIDE
THE TAP-CHANGER OIL COMPARTMENT (SAME PIPE CONNECTION AS R, S, Q OR BLEEDER SCREW CAN BE USED)
Q = CONNECTION FOR OIL RETURN PIPE (ONLY FOROIL FILTER) - CONNECTIONS ORIENTABLE THROUGH 360°
S = CONNECTION FOR SUCTION PIPE - CONNECTIONS ORIENTABLE THROUGH 360°
R = CONNECTION FOR PROTECTIVE RELAY (EXCHANGEABLE WITH CONNECTION Q) - CONNECTIONS ORIENTABLE THROUGH 360°
T = THERMOMETER BAG (OPTIONALLY)
SR = INSPECTION WINDOW, RIGHT
SL = INSPECTION WINDOW, LEFT

SCALE 1 : 2,5 (1 : 1)

(M) DRIVE SIDE OF THE TAP SELECTOR

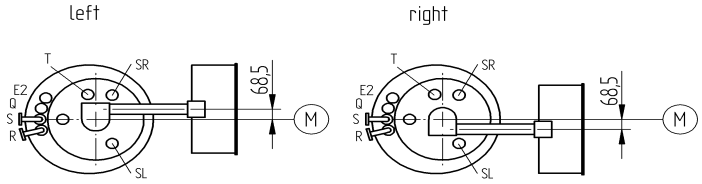
 © Maschinenfabrik Reinhausen 2010

3.1.24 VACUTAP® VM — variants of on-load tap-changer head (720026)

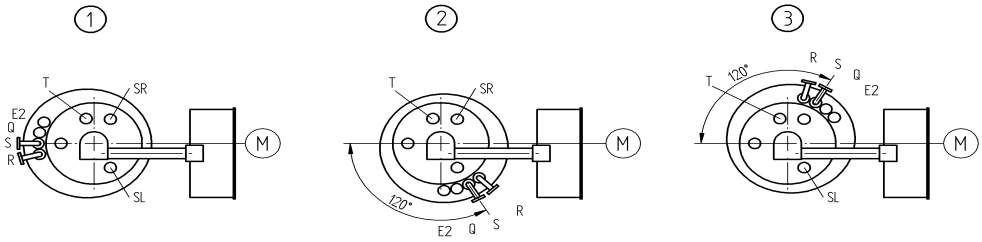


A = tap selector
K = change-over selector
B = diverter switch oil compartment
} the represented version Typ M
C = on-load tap-changer head
D = pipe connections (R, S, Q, E2)
DV = pressure relief valve
E = on-load tap-changer head cover
F = upper gear unit
G = drive shaft, horizontal
H = bevel gear
I = drive shaft, vertical
MA = motor drive unit
(M) = drive side of tap selector
SR = inspection window on the right
SL = inspection window on the left
T = temperature sensor

Position of drive shaft of gear unit



Head variants



Swivelling ranges

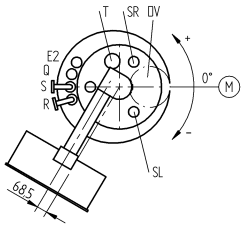
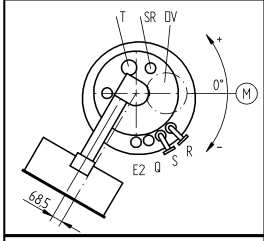
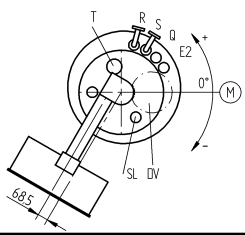
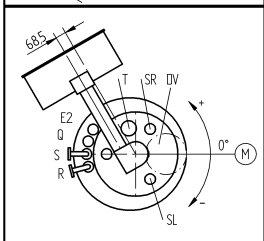
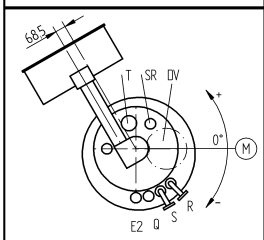
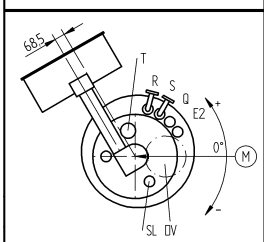
A considerabel number of variants of the on-load tap-changer head are available for adapting the horizontal part of the drive shaft to the transformer tank.

The mounting position of the tap selector A and diverter switch oil compartment B is determined by the drive side of tap selector (M).

The on-load tap-changer head C together with its pipe connections D may be turned through 120 degrees clockwise or anti-clockwise. This results in the variants 1, 2 and 3.

The upper gear unit F can be turned continuously on its own axis. Table 720027: lists the limitation of the swivelling range for the particular head variant. The angel specifications refer to the center of rotation of the gear unit. Pay particular attention to the offset of the drive shaft.

3.1.25 VACUTAP® VM — swivel range of gear unit (720027)

SKETCH	HEAD VERSION COMPONENTS USED	LIMITATION OF THE SWIVELLING RANGE
	DRIVE SHAFT RIGHT HEAD VERSION 1	-180° 0° 180°
	PIPE CONNECTION R	-168° -140°
	PIPE CONNECTION S	-155° 177°
	PIPE CONNECTION Q	-171° 162°
	PIPE CONNECTION E2	147° 174°
	PRESSURE RELIEF VALVE DV	-150° -35°
	TEMPERATURE SENSOR T	96° 175°
	INSPECTION WINDOW SL / SR	-64° SL -8° 56° SR 112°
	DRIVE SHAFT RIGHT HEAD VERSION 2	-180° 0° 180°
	PIPE CONNECTION R	-48° -21°
	PIPE CONNECTION S	-63° -36°
	PIPE CONNECTION Q	-78° -51°
	PIPE CONNECTION E2	-93° -66°
	PRESSURE RELIEF VALVE DV	-150° -35°
	TEMPERATURE SENSOR T	96° 175°
	INSPECTION WINDOW SR	56° SR 112°
	DRIVE SHAFT RIGHT HEAD VERSION 3	-180° 0° 180°
	PIPE CONNECTION R	72° 99°
	PIPE CONNECTION S	57° 84°
	PIPE CONNECTION Q	42° 69°
	PIPE CONNECTION E2	27° 54°
	PRESSURE RELIEF VALVE DV	-150° -35°
	TEMPERATURE SENSOR T	96° 175°
	INSPECTION WINDOW SL	-64° SL -8°
	DRIVE SHAFT LEFT HEAD VERSION 1	-180° 0° 180°
	PIPE CONNECTION R	-162° 171°
	PIPE CONNECTION S	-177° 156°
	PIPE CONNECTION Q	141° 168°
	PIPE CONNECTION E2	126° 153°
	PRESSURE RELIEF VALVE DV	35° 150°
	TEMPERATURE SENSOR T	34° 114°
	INSPECTION WINDOW SL / SR	-112° SL -56° 8° SR 64°
	DRIVE SHAFT LEFT HEAD VERSION 2	-180° 0° 180°
	PIPE CONNECTION R	-69° -42°
	PIPE CONNECTION S	-84° -57°
	PIPE CONNECTION Q	-99° -72°
	PIPE CONNECTION E2	-114° -87°
	PRESSURE RELIEF VALVE DV	35° 150°
	TEMPERATURE SENSOR T	34° 114°
	INSPECTION WINDOW SR	8° SR 64°
	DRIVE SHAFT LEFT HEAD VERSION 3	-180° 0° 180°
	PIPE CONNECTION R	50° 78°
	PIPE CONNECTION S	35° 62°
	PIPE CONNECTION Q	21° 48°
	PIPE CONNECTION E2	6° 33°
	PRESSURE RELIEF VALVE DV	35° 150°
	TEMPERATURE SENSOR T	34° 114°
	INSPECTION WINDOW SL	-112° SL -56°

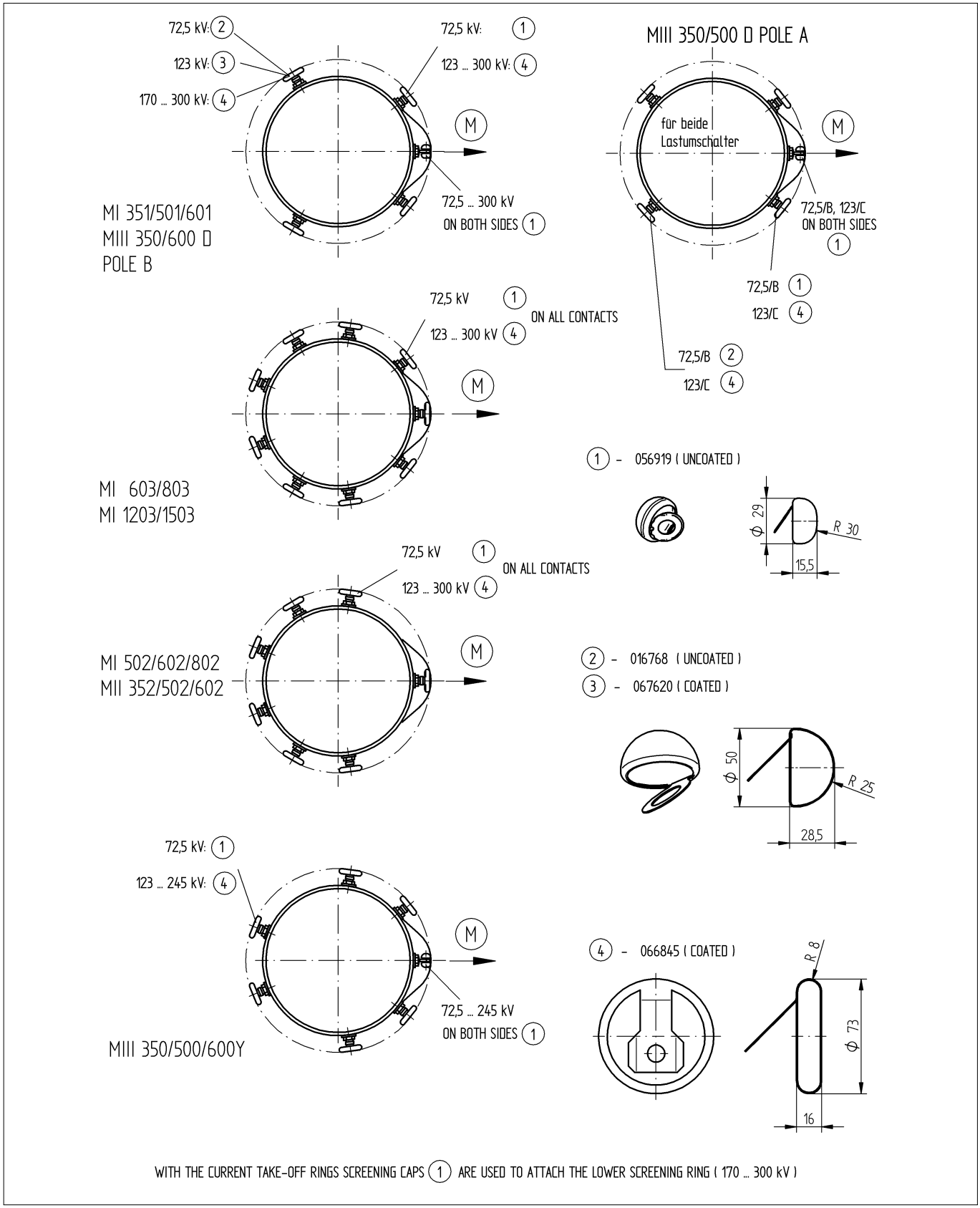
LIMITATION OF THE SWIVELLING RANGE TROUGH PIPE CONNECTIONS R AND S

LIMITATION OF THE SWIVELLING RANGE TROUGH OPTIONAL EXISTING PIPE CONNECTIONS Q , E2 AND PRESSURE RELIEF VALVE DV

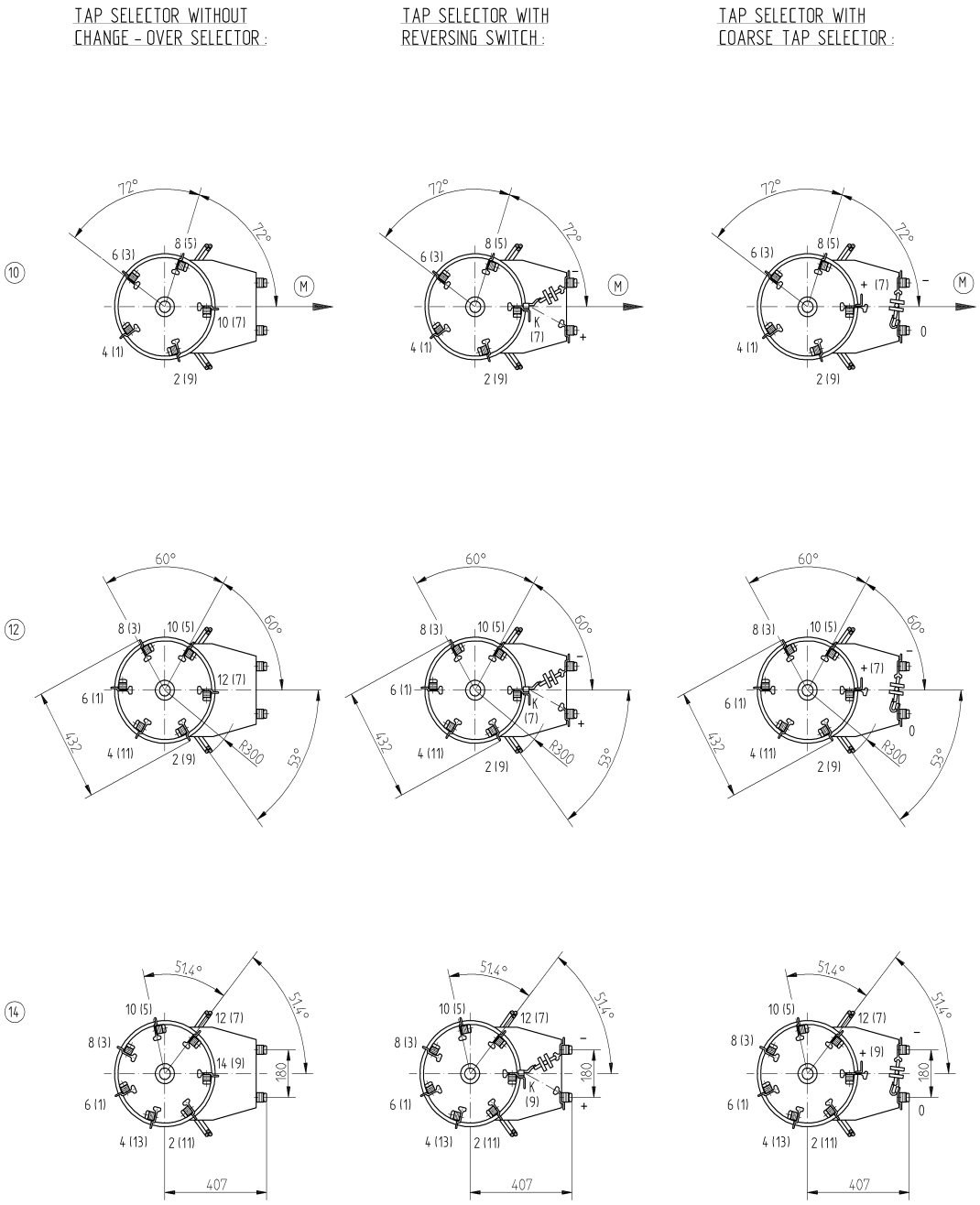
SWIRELLING RANGE POSIBLE, BUT THE TEMPERATURE SENSOR T AND THE INSPECTION WINDOW SL / SR ARE NOT VISIBLE

 © Maschinenfabrik Reinhausen 2010

3.1.26 VACUTAP® VM — screenings on oil compartment contacts (730336)



3.1.27 VACUTAP® VM 300 — tap selector cross-sections (898041)



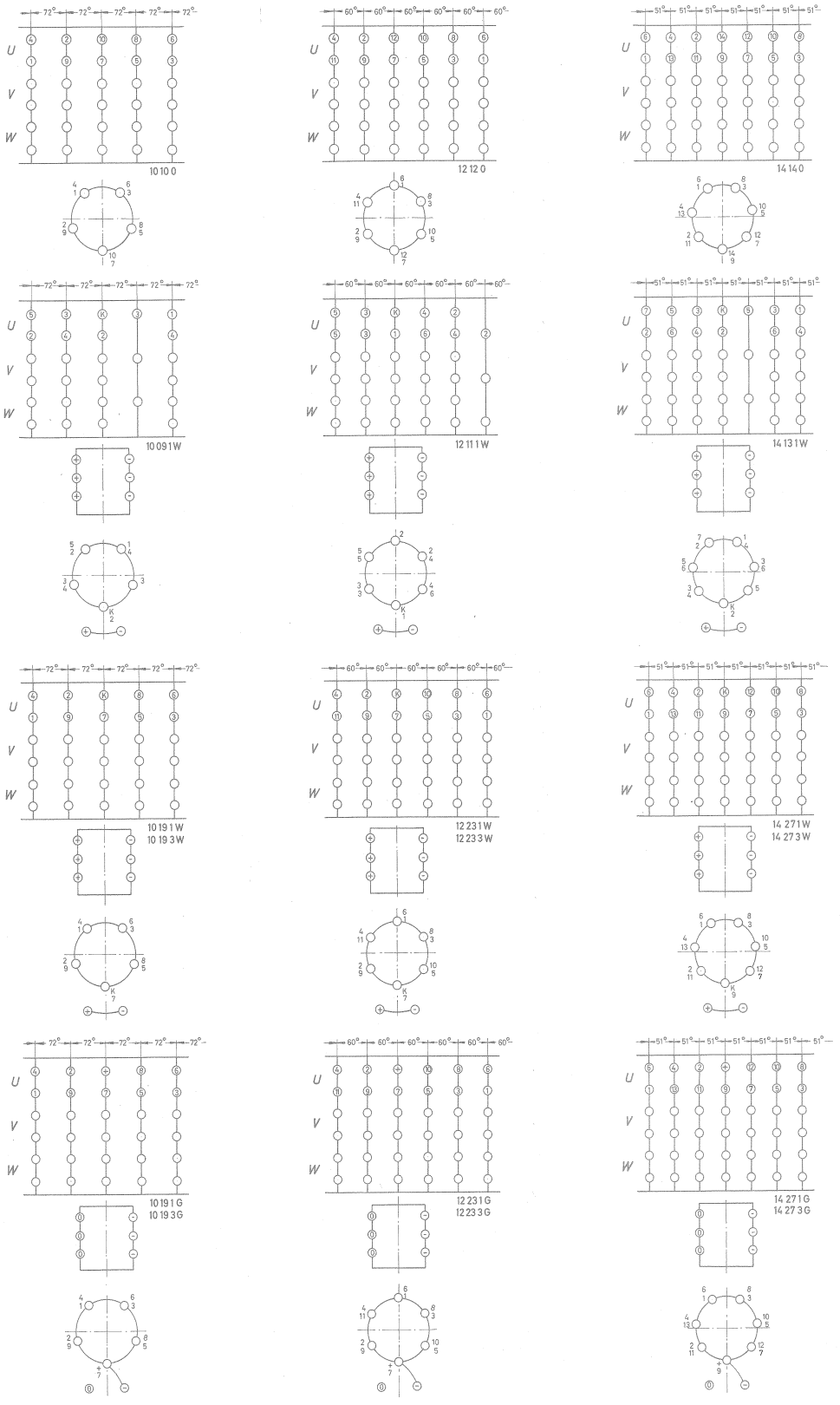
DESIGNATION OF TAP SELECTOR TERMINALS :
(PLAN VIEW)
E . G . : 4 UPPER CONTACT PLANE
(13) LOWER CONTACT PLANE

(M) DRIVE SIDE OF TAP SELECTOR
(10) (12) (14) TAP SELECTOR PITCH

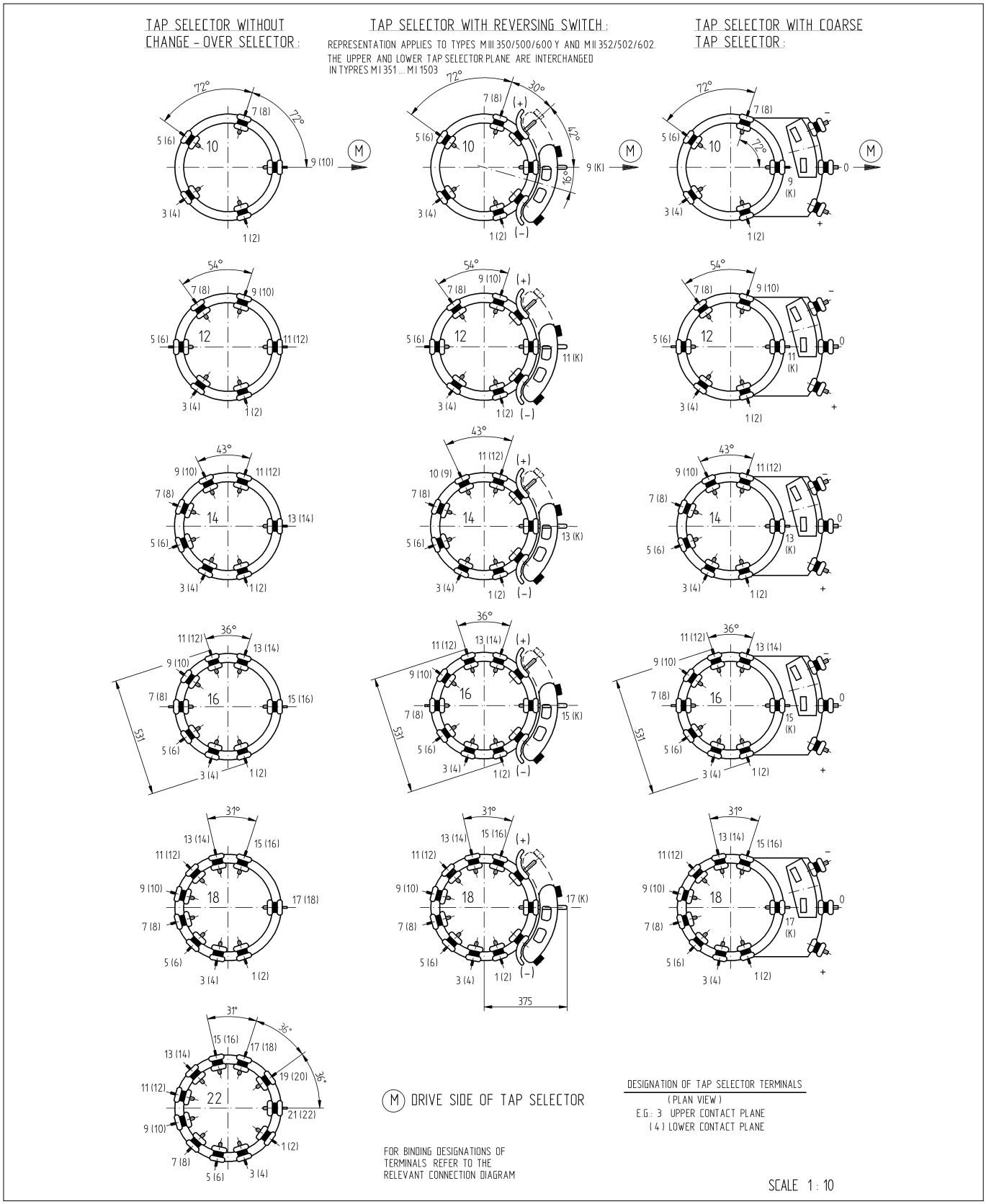
FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE RELEVANT CONNECTION DIAGRAM

 © Maschinenfabrik Reinhausen 2010

3.1.28 VACUTAP® VM 300 — arrangement of contacts on tap selector (blanks, 891114)

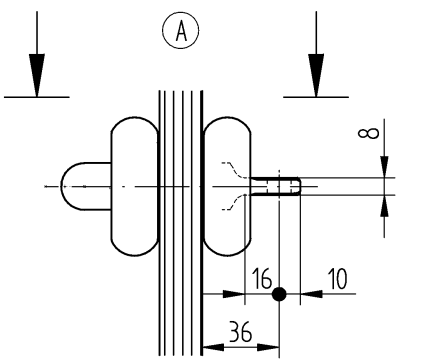


3.1.29 VACUTAP® VM — arrangement of contacts on tap selector, tap selector division 10...22 (898013)

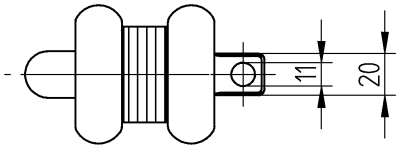


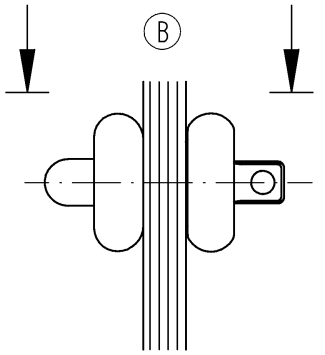
 © Maschinenfabrik Reinhausen 2010

3.1.30 VACUTAP® VM — installation position of tap selector connection contacts (890477)

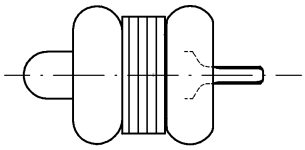


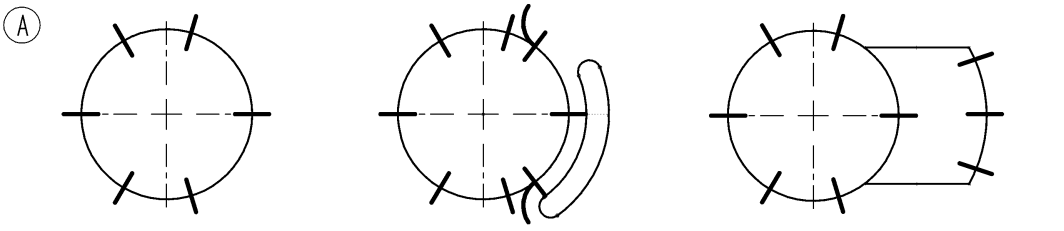
Through hole, vertical



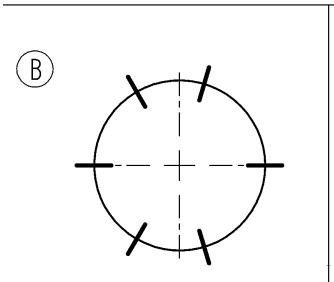


Through hole, horizontal

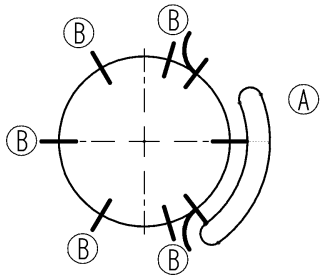




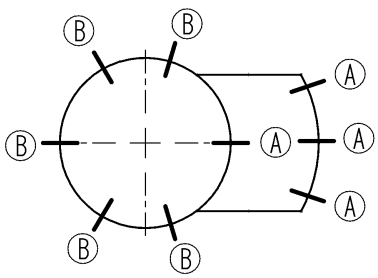
M III 350 / 500 / 600Y – 0 VM III 350 / 500 / 650Y – 0	M III 350 / 500 / 600Y – W VM III 350 / 500 / 650Y – W	M III 350 / 500 / 600Y – G VM III 350 / 500 / 650Y – G
M II 352 / 502 / 602 – 0 VM II 352 / 502 / 652 – 0	M II 352 / 502 / 602 – W VM II 352 / 502 / 652 – W	M II 352 / 502 / 602 – G VM II 352 / 502 / 652 – G
M I 351 / 501 / 601 – 0 VM I 351 / 501 / 651 – 0	M I 351 / 501 / 601 – W VM I 351 / 501 / 651 – W	M I 351 / 501 / 601 – G VM I 351 / 501 / 651 – G



M I 802 – 0
VM I 802 – 0
VM I 1002 – 0
M I 1203 / 1503 – 0
VM I 1203 / 1503 – 0



M I 802 – W
VM I 802 – W
VM I 1002 – W
M I 1203 / 1503 – W
VM I 1203 / 1503 – W



M I 802 – G
VM I 802 – G
VM I 1002 – G
M I 1203 / 1503 – G
VM I 1203 / 1503 – G

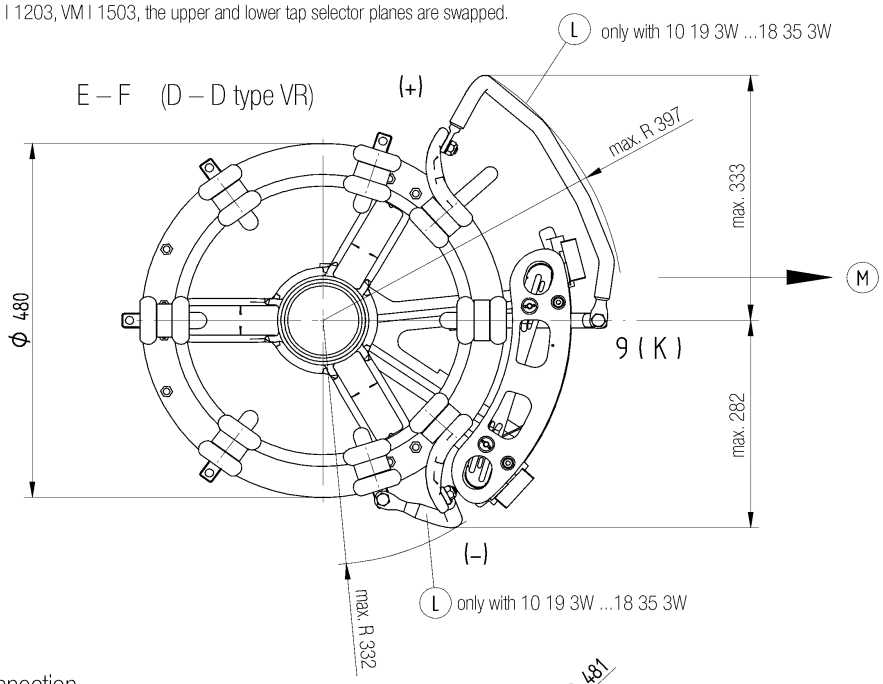
A + B

3.1.31 VACUTAP® VM — connecting leads 3W, 1G, 3G (723590)

Reversing change-over selector

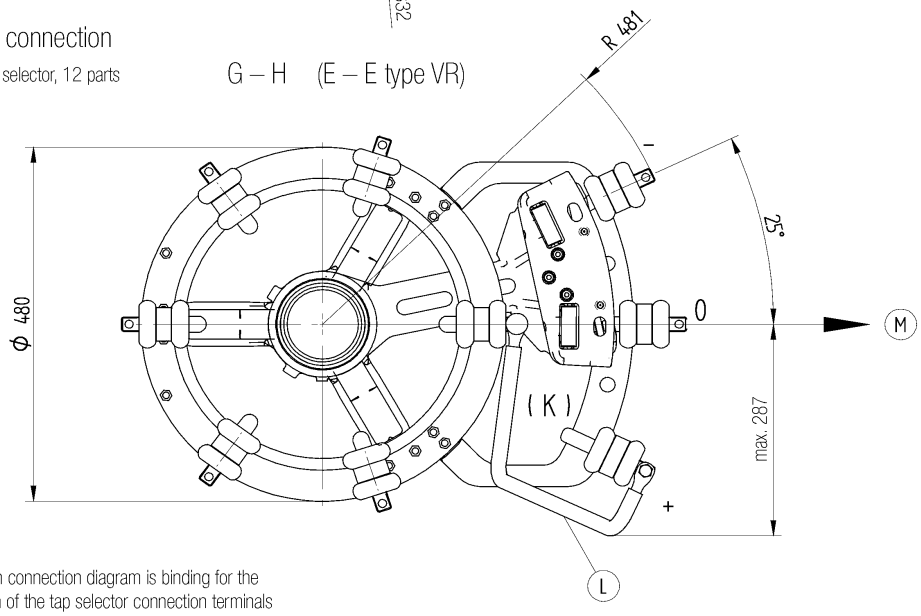
Diagram applies to M III 350/500/600Y and M II 352/502/602, 12 parts
With M I 351 ... M I 1203, M I 1503, the upper and lower tap selector planes are swapped.

and for VM III 350/500/650Y and VM II 352/502/652, 12 parts
With VM I 351 ... VM I 1203, VM I 1503, the upper and lower tap selector planes are swapped.



Coarse tap connection

Diagram for tap selector, 12 parts

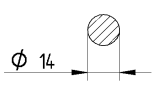


The version connection diagram is binding for the designation of the tap selector connection terminals

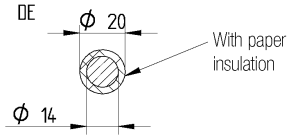
(M) Drive side of tap selector

(L) CD connecting leads

Tap selector size
B, C, D

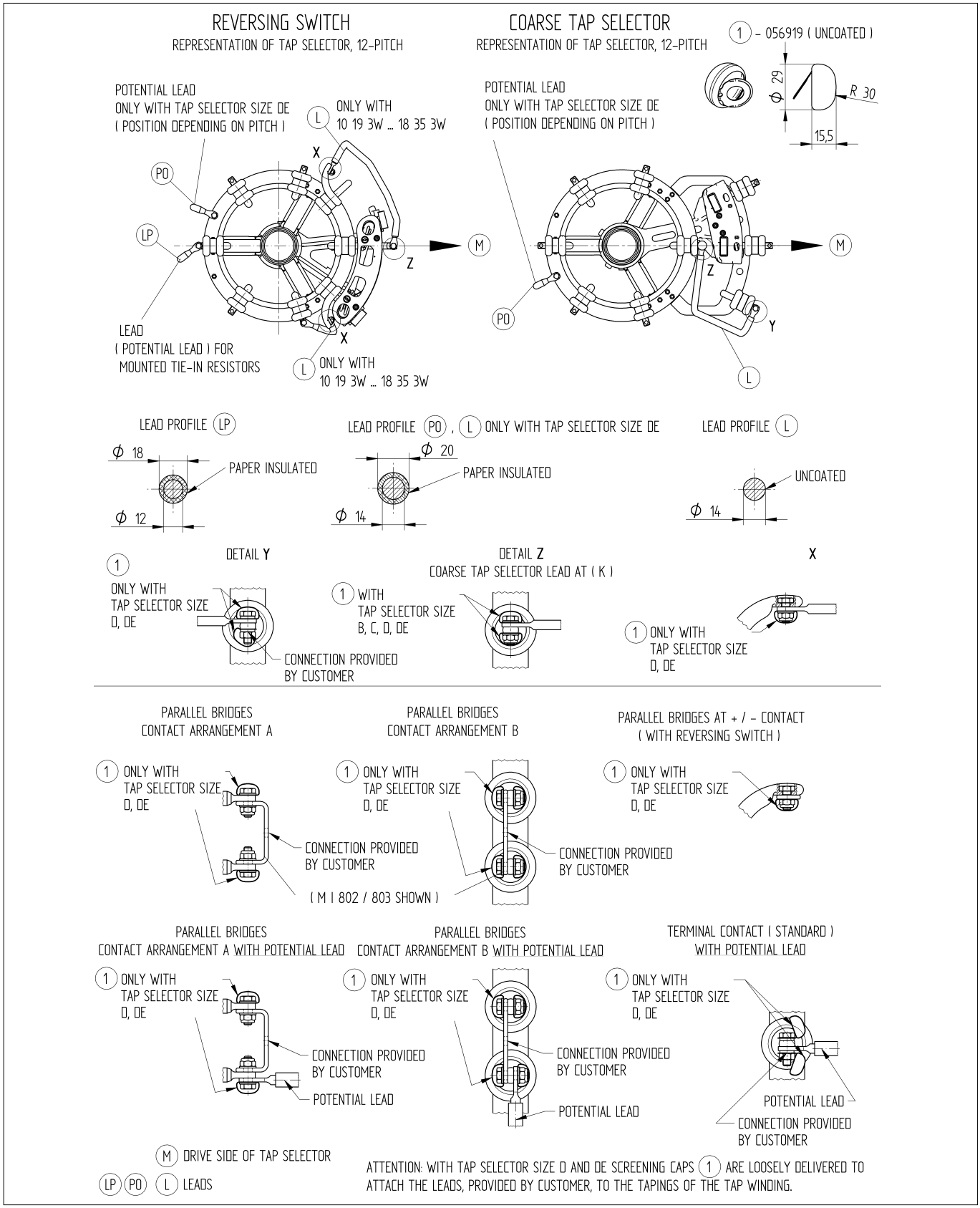


DE



 © Maschinenfabrik Reinhausen 2010

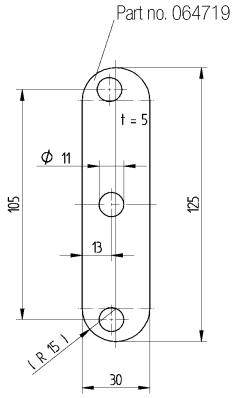
3.1.32 VACUTAP® VM — screenings on fine tap selector and change-over selector (730335)



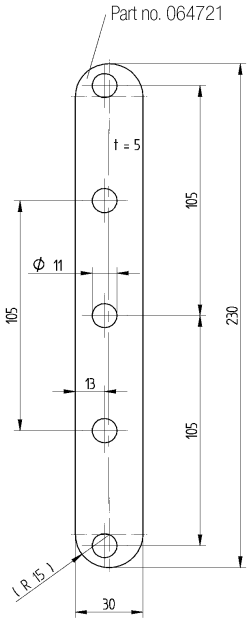
3.1.33 VACUTAP® VM 802/803/1203/1503 — bridges to parallel connection of tap selector connection contacts (899598)

Arrangement of contacts B

M I 802 / 803
VM I 802 / 803 / 1002

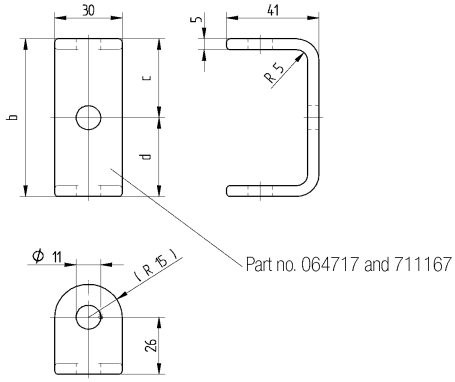


M I 1203 / 1503
VM I 1203 / 1503



Arrangement of contacts A

M I 802 / 803 / 1203 / 1503
VM I 802 / 803 / 1203 / 1503

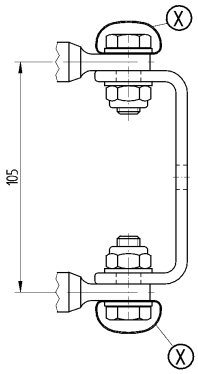


Arrangement of contacts (see 890477:)	Part no.	Dimension b	Dimension c	Dimension d
A without lead	064717	97	48,5	48,5
A with lead	711167	91	48,5	42,5

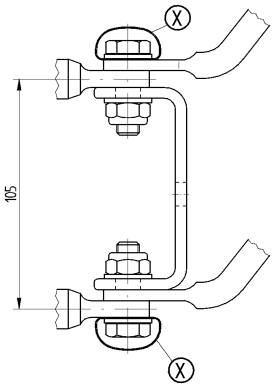
Installation of parallel bridges for contact arrangement A with and without connecting lead for 3W tap change operation

M I 802 / 803
VM I 802 / 803 / 1002

Without lead

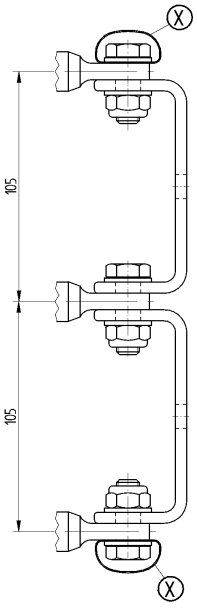


With lead

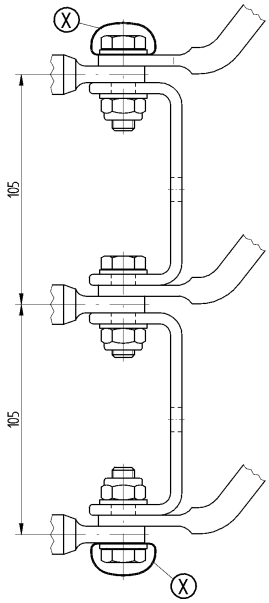


M I 1203 / 1503
VM I 1203 / 1503

Without lead



With lead



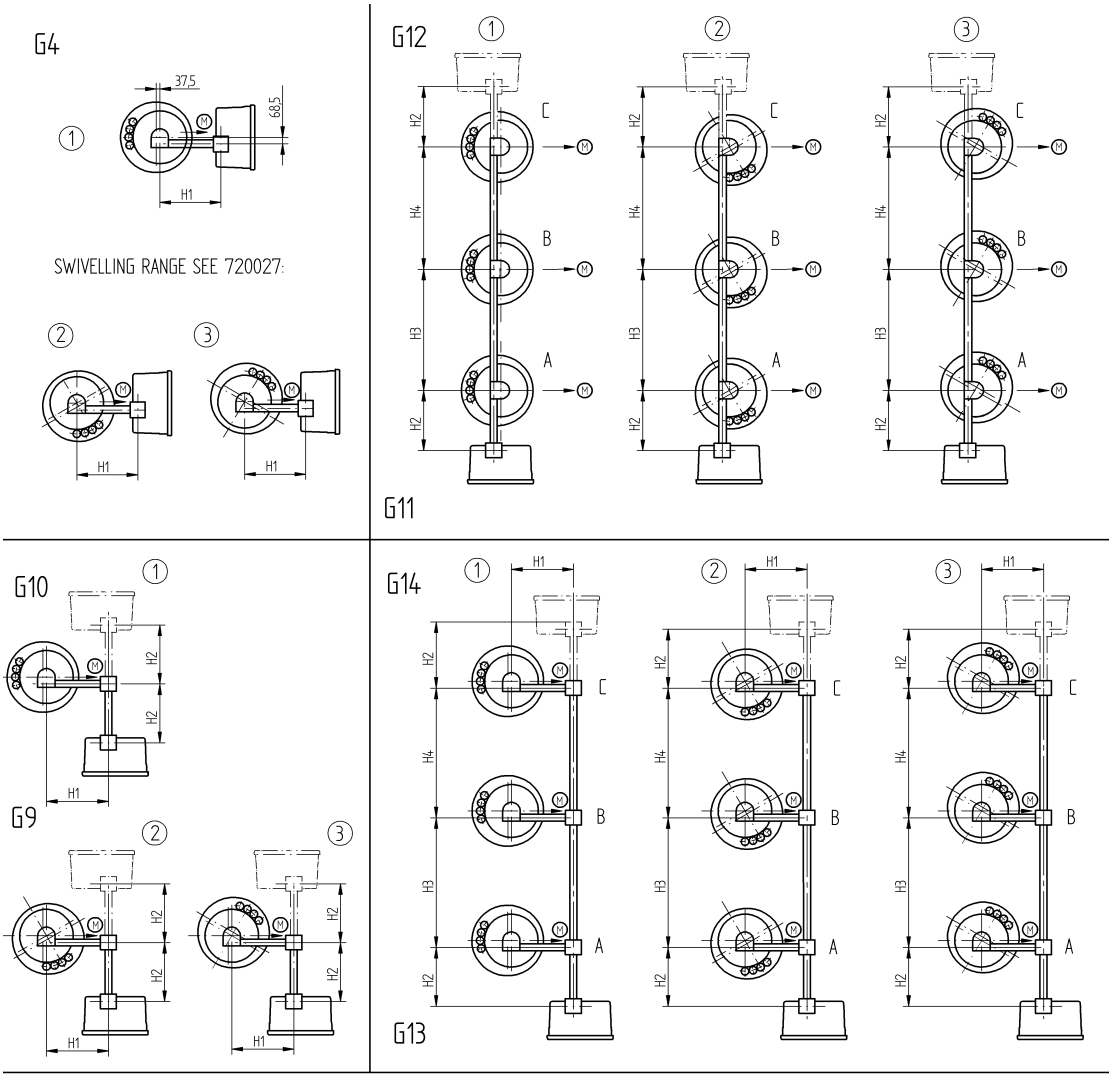
(X) only with tap selector size 0 and OE

Caution: Parallel bridges are not included in the standard scope of supply

Scale 1 : 1

 © Maschinenfabrik Reinhausen 2010

3.1.34 VACUTAP® VM — horizontal drive shaft, centrical drive (893896)

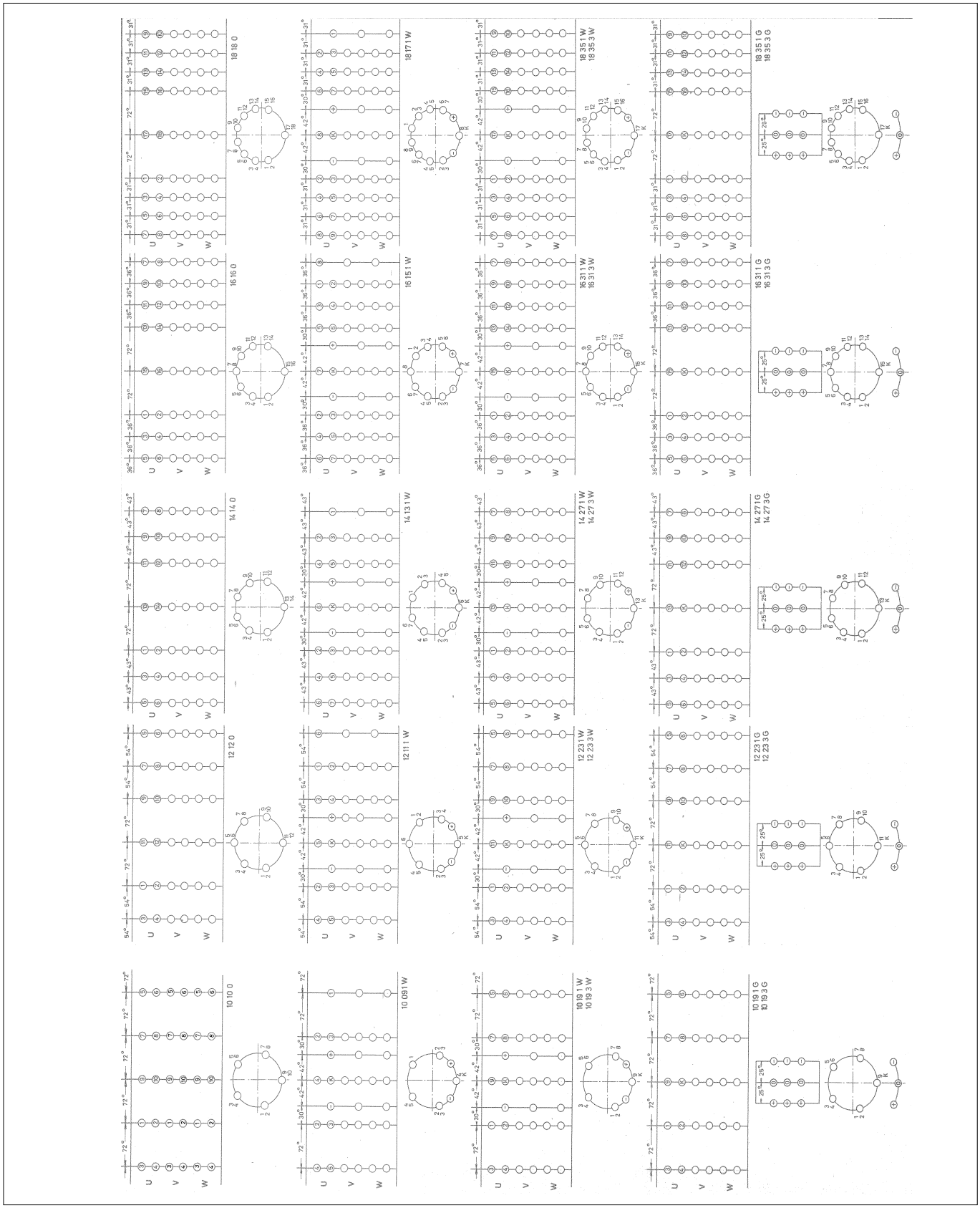


ARRANGEMENT		G4	G9, G10	G11, G12	G13, G14
STANDARD DESIGN		■		■	
SPECIAL DESIGN			■		■
MINIMUM DIMENSIONS[1] (DETERMINED FOR MECHANICAL REASONS: NECESSARY INSULATING DISTANCES NOT CONSIDERED !)	H1	535	545	-	545
	H2	-	318	515	318
	H3 [2]	-	-	840	840
	H4 [2]	-	-	840	840
INTERMEDIATE BEARING FOR	H1 >	2254	2309	-	2309
	H2 >	-	2259	2254	2259
	H3 >	-	-	2249	2259
	H4 >	-	-	2249	2259

NOTE:
1) FOR OLTCS WITH THE CHANGE-OVER SELECTOR ATTACHED LATERALLY, THE DIMENSIONS OF THE CHANGE-OVER SELECTOR AFTER INSTALLED IN POSITION HAVE TO BE TAKEN INTO ACCOUNT (SEE THE CORRESPONDING OLTC DIMENSION DRAWING)
2) IN GENERAL DETERMINED BY THE INSULATION DISTANCE BETWEEN POLES A, B, C .

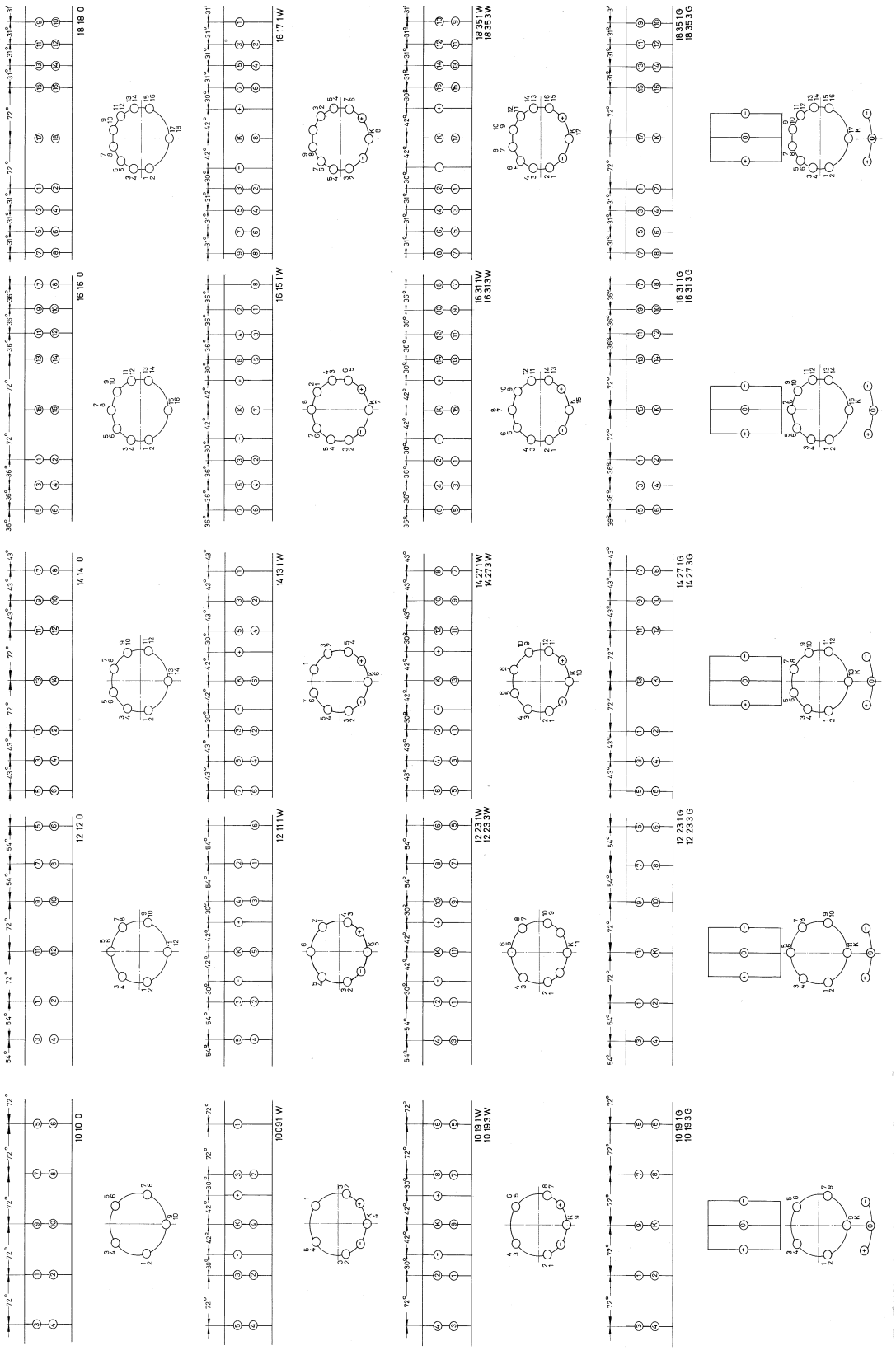
① ② ③ – HEAD VERSION
→ Ⓜ – DRIVE SIDE OF TAP SELECTOR/ SIDE OF CHANGE-OVER SELECTOR

3.1.35 VACUTAP® VM III 350/500/650 Y — arrangement of tap selectors (891107)

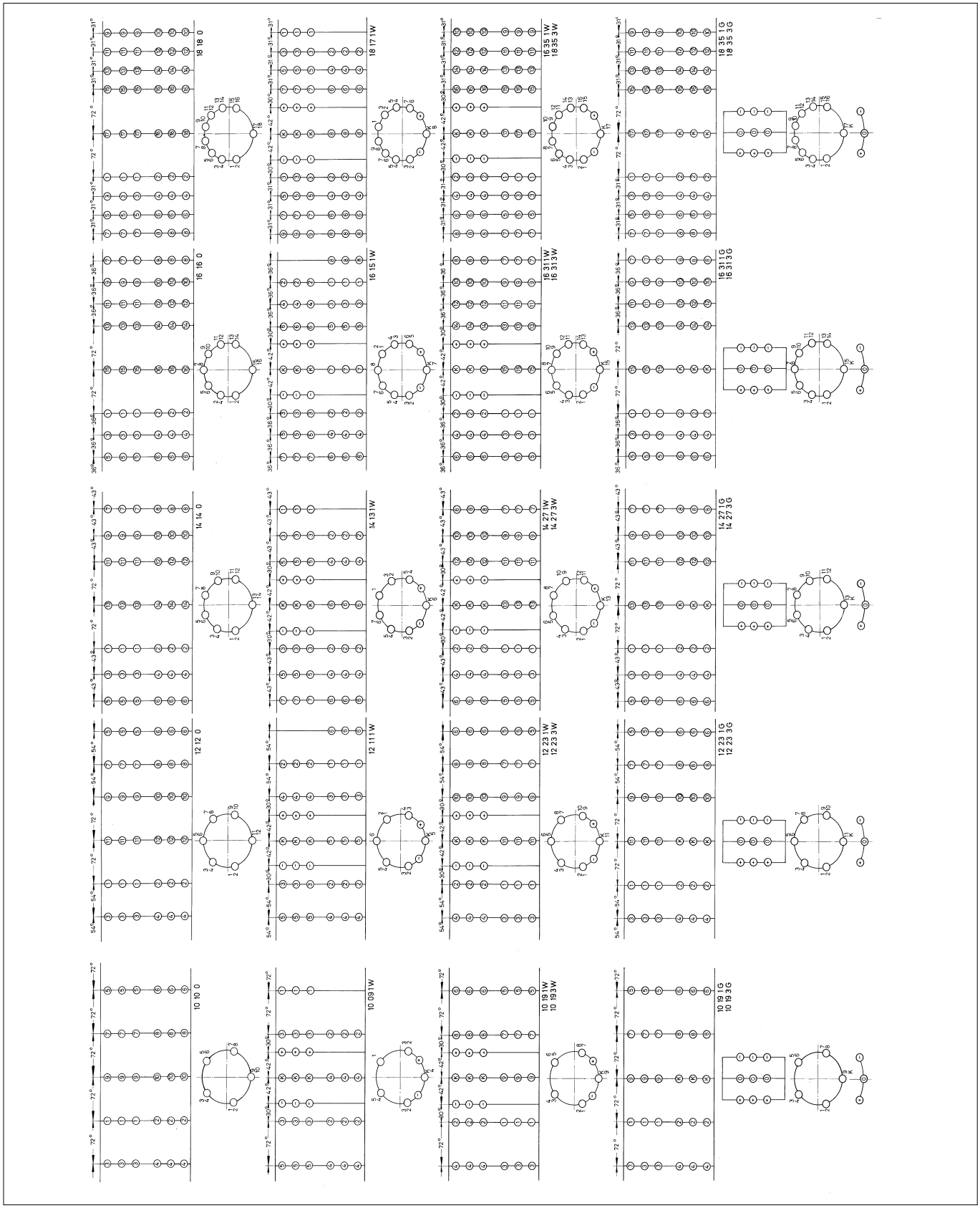


 © Maschinenfabrik Reinhausen 2010

3.1.36 VACUTAP® VM I 351/501/651 — arrangement of tap selectors (891108)

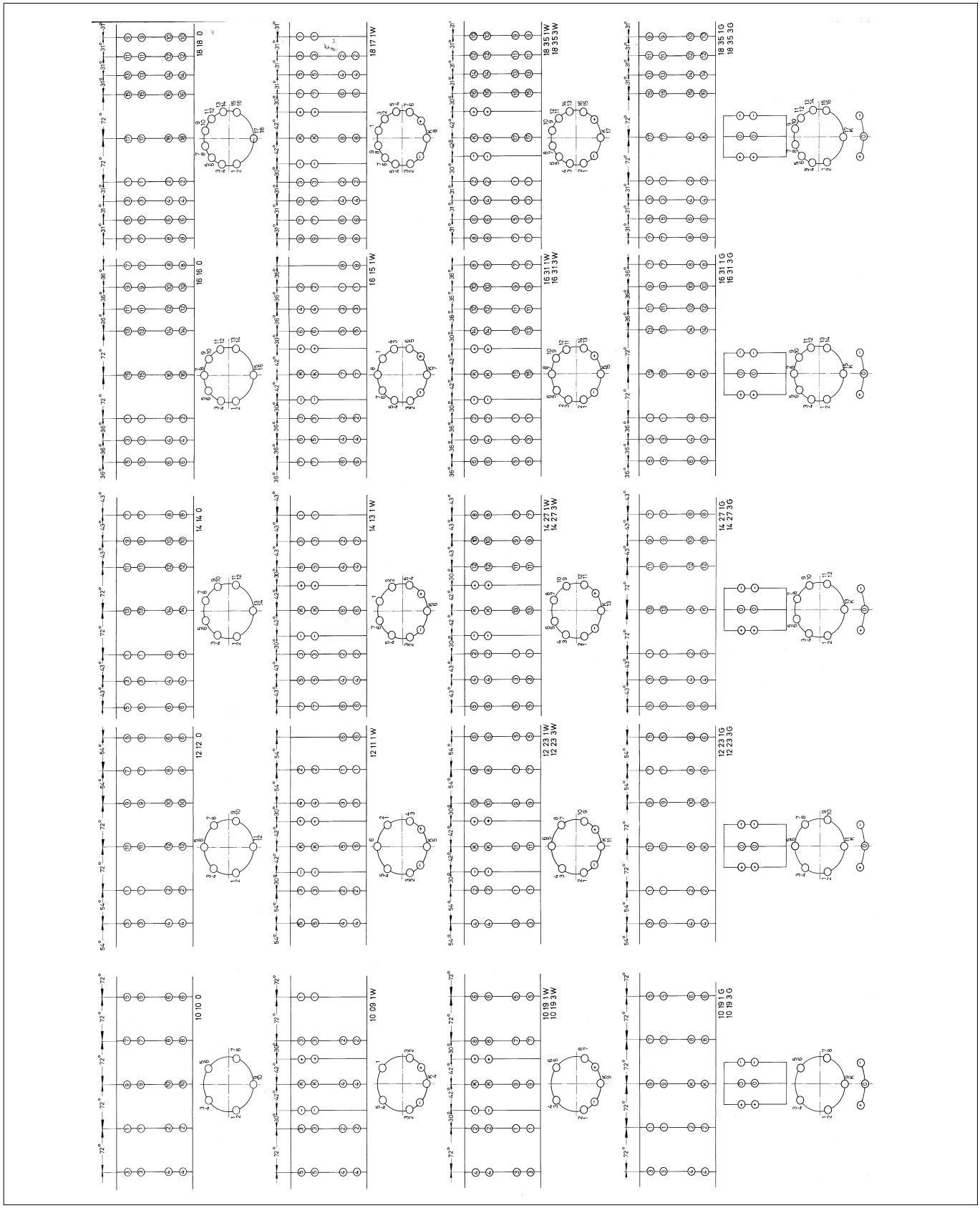


3.1.37 VACUTAP® VM I 1503 — arrangement of tap selectors (891109)

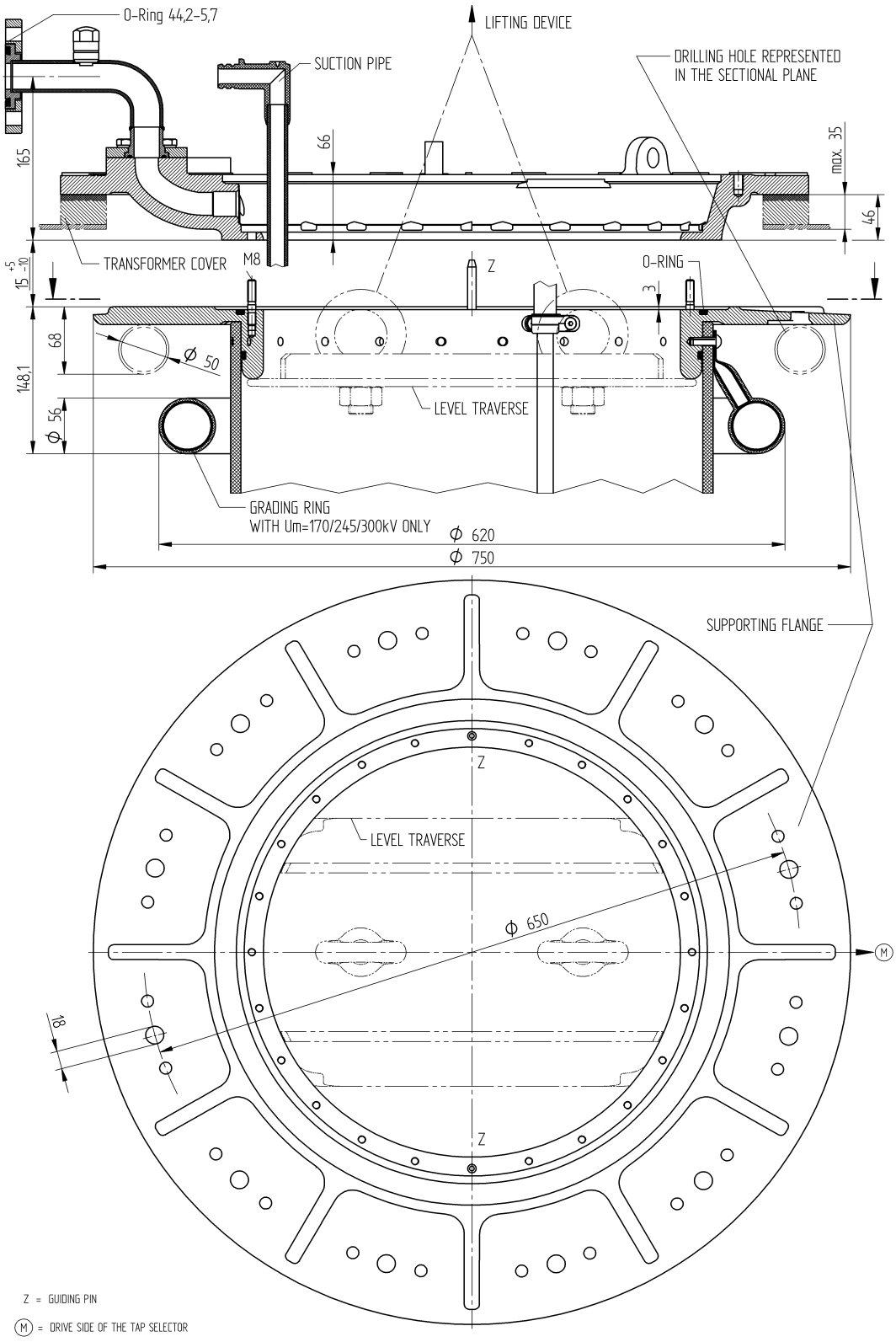


 © Maschinenfabrik Reinhausen 2010

3.1.38 VACUTAP® VM I 802/1002 — arrangement of tap selectors (891110)

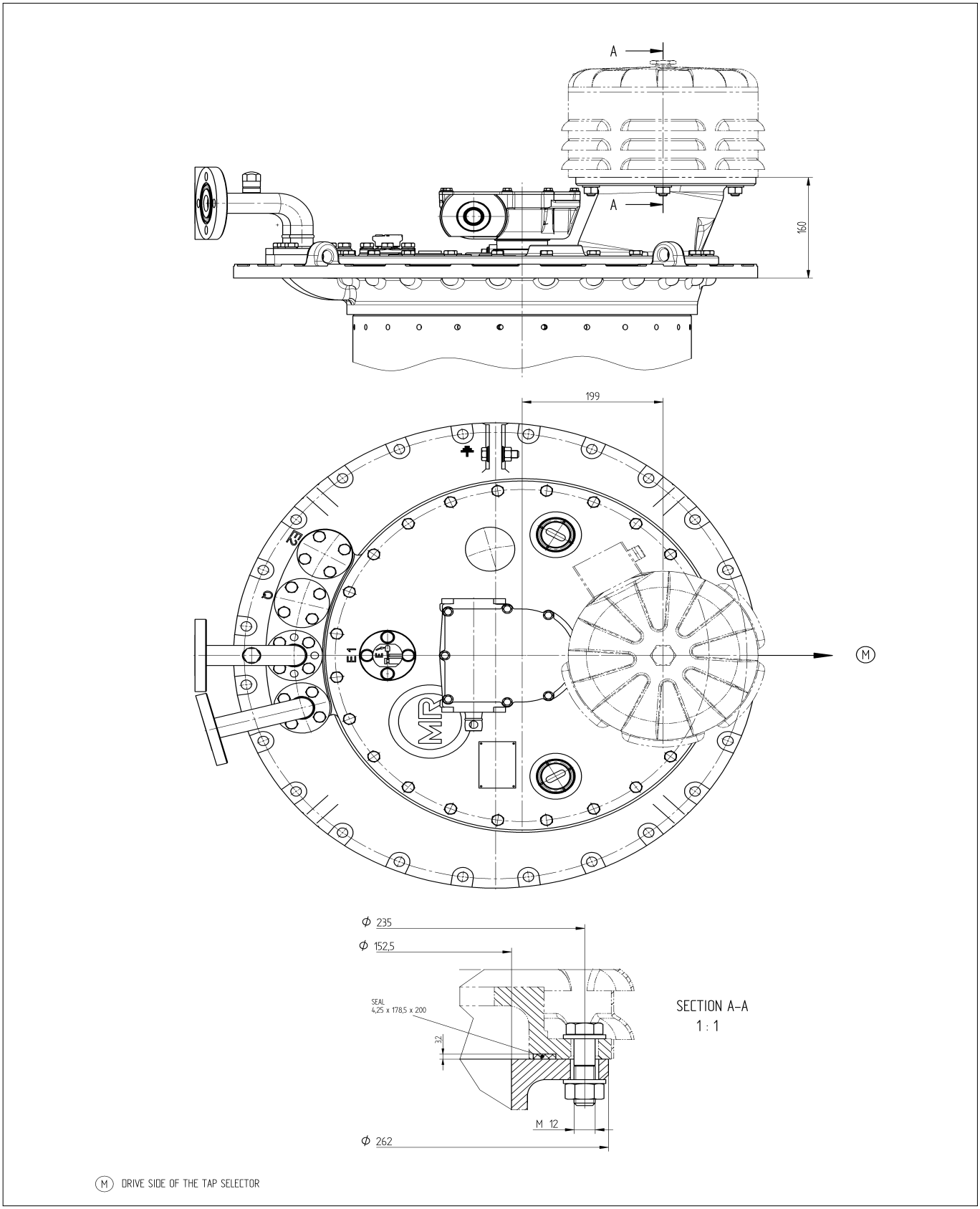


3.1.39 VACUTAP® VM, supporting flange, special design for bell-type tank installation for U_m up to 300 kV (896762)

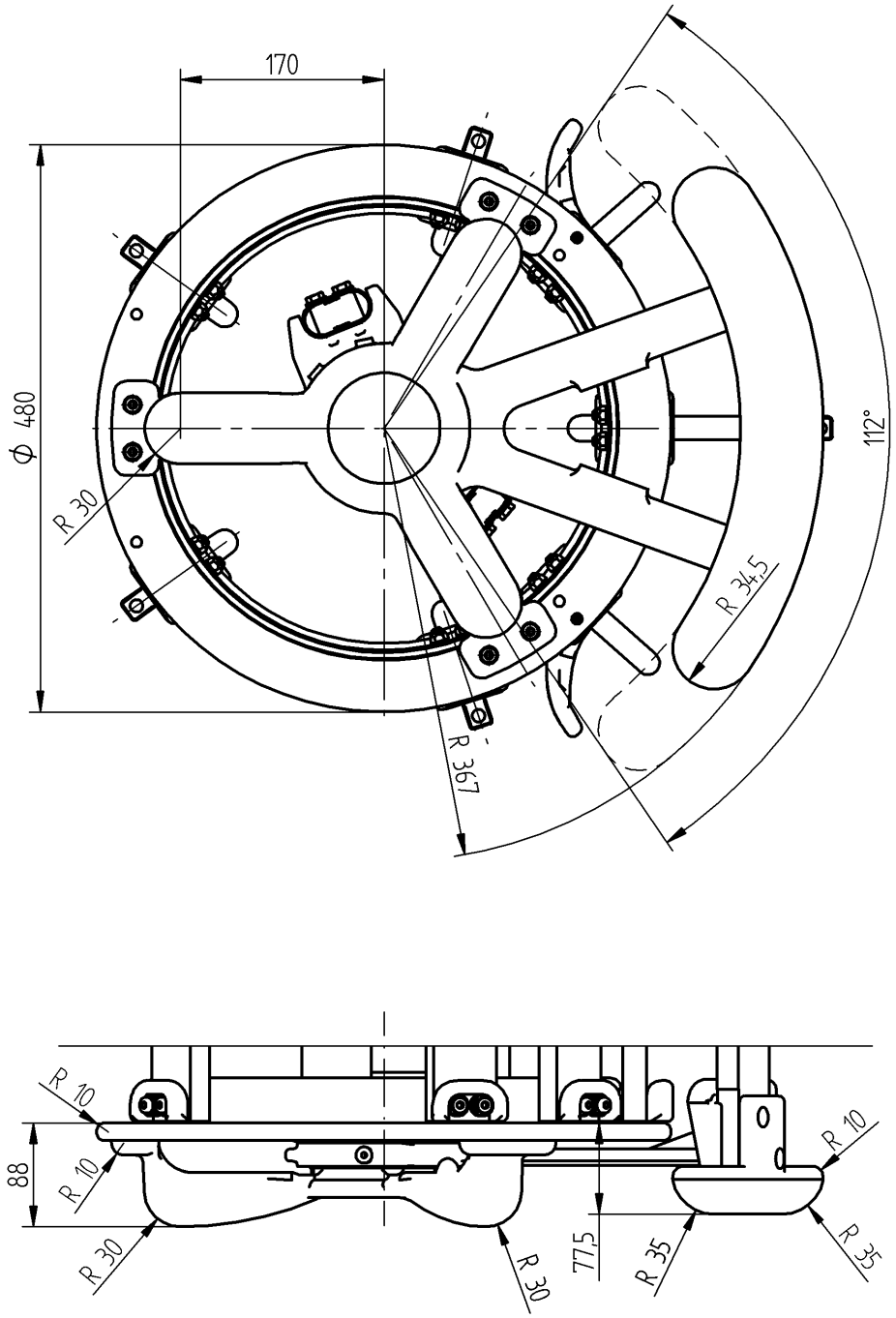


 © Maschinenfabrik Reinhausen 2010

3.1.40 VACUTAP® VM — flange for pressure relief valve (895168)

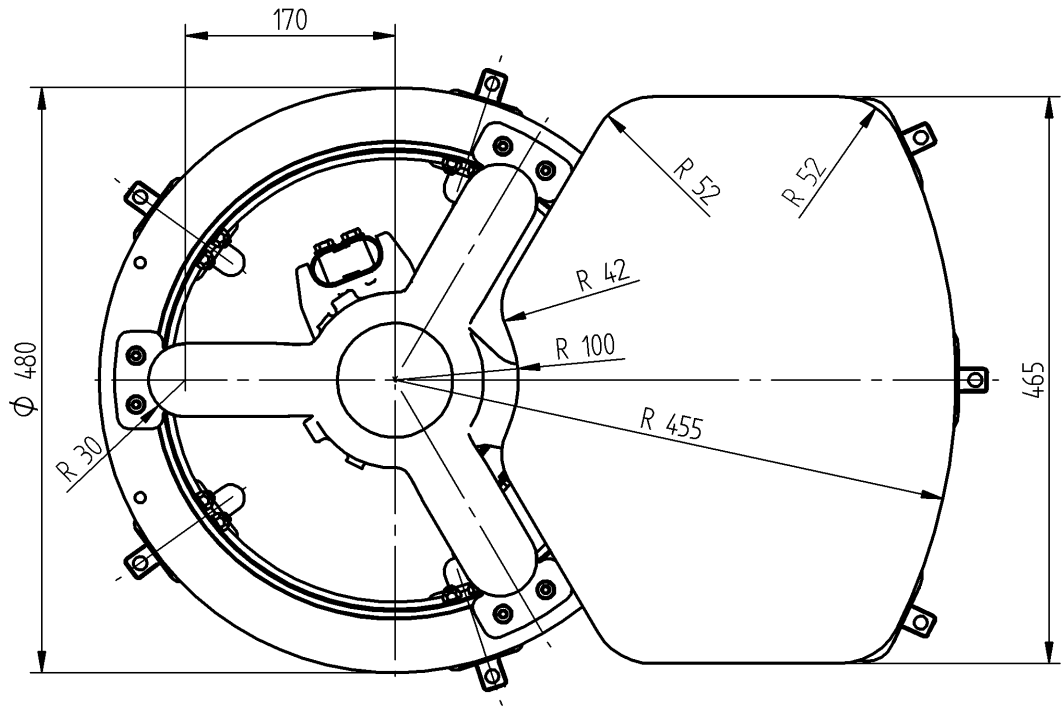


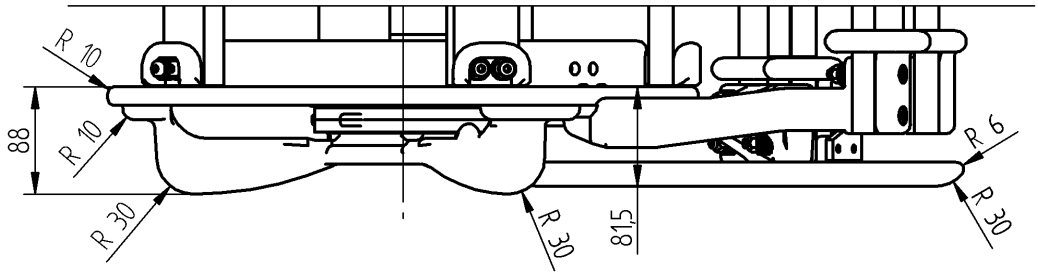
3.1.41 VACUTAP® VM — tap selector base with additional screening, reversing change-over selector design (893934)



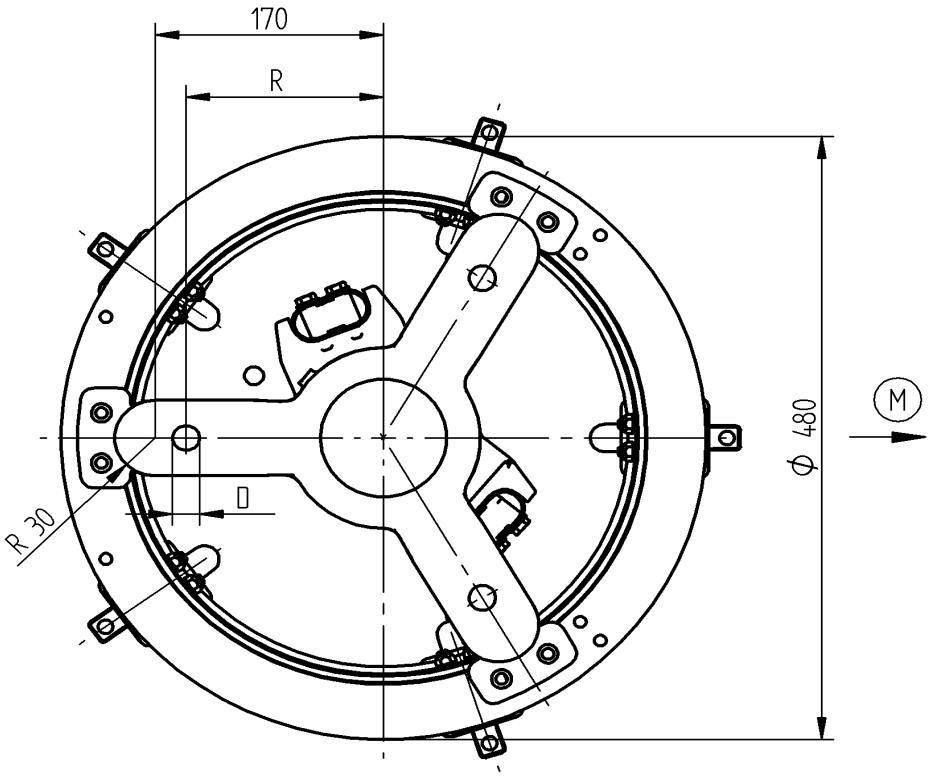
 © Maschinenfabrik Reinhausen 2010

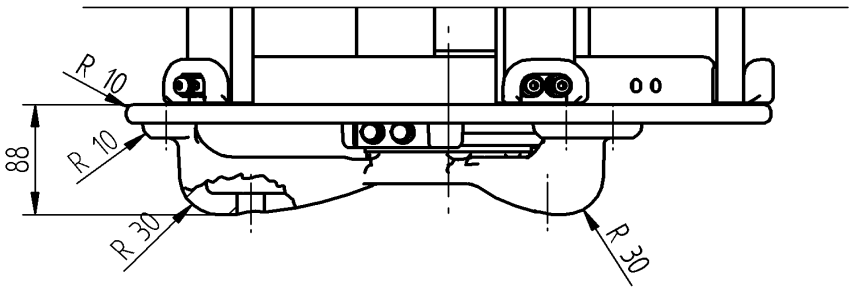
3.1.42 VACUTAP® VM — tap selector base with additional screening, coarse tap connection design (893935)





3.1.43 VACUTAP® VM — tap selector base with through hole D20 and D13 (725649)





R	D	BOTTOM OF TAP SELECTOR
147	20	097251
160	13	097252

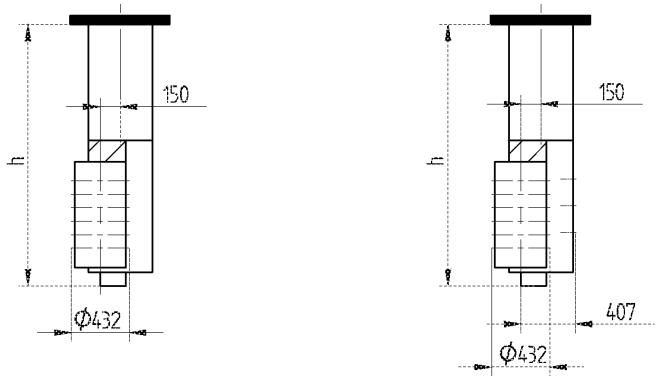
(M) DRIVE SIDE OF TAP SELECTOR

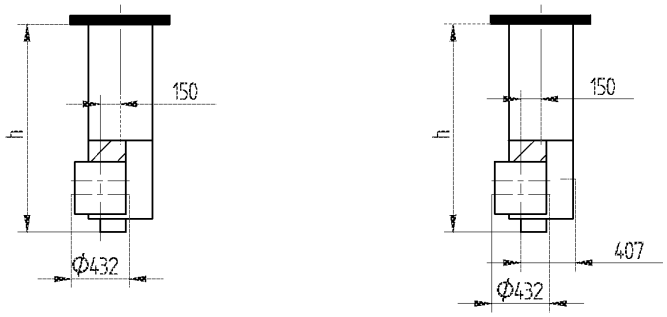
 © Maschinenfabrik Reinhausen 2010

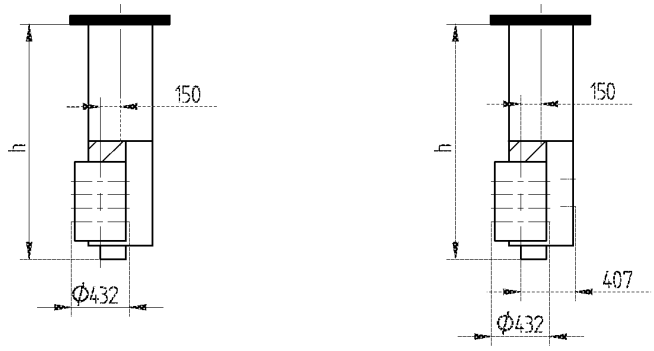
3.2 VACUTAP® VM 300 — overview of on-load tap-changer designs (765835)

without change-over selector

with change-over selector







Installation length h in mm

VM III 300 Y

U_m	Tap selector size B
72,5	1942
123	2072
170	2202
245	2302

VM I 301

U_m	Tap selector size B
72,5	1542
123	1672
170	1802
245	1902

VM II 302

U_m	Tap selector size B
72,5	1742
123	1872
170	2002
245	2102

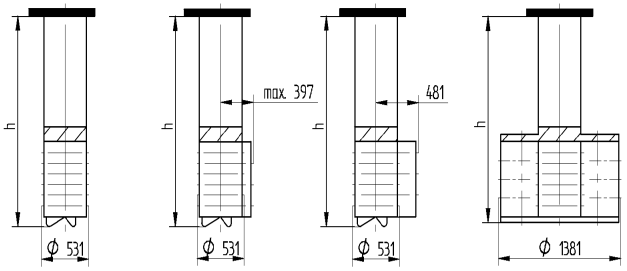
3.3 VACUTAP® VM — overview of on-load tap-changer designs (899740)

Without change-over selector | With reversing change-over selector | With coarse change-over selector | with multiple coarse change-over selector (MG) only size B, C and O

Installation length h in mm

M III 350 Y *)
M III 500 Y *)
M III 600 Y
*) not as MG

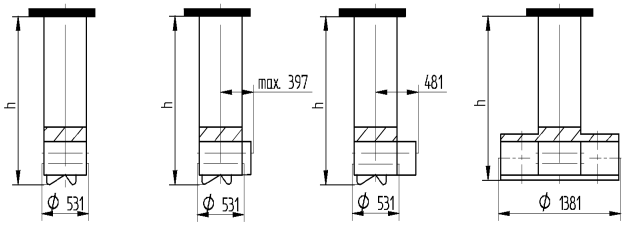
VM III 350 Y *)
VM III 500 Y *)
VM III 650 Y
*) not as MG



Tap selector size						
	B		C		D/DE	D
U_m	0/W/G	MG	0/W/G	MG	0/W/G	MG
72,5	1894	1856	2069	2031	2524	2486
123	2024	1986	2199	2161	2654	2616
170	2154	2116	2329	2291	2784	2746
245	2254	2216	2429	2391	2884	2846

M I 351 *)
M I 501 *)
M I 601
*) not as MG

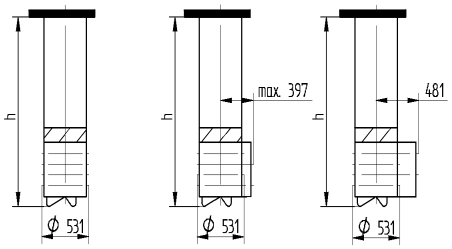
VM I 351 *)
VM I 501 *)
VM I 651
*) not as MG



Tap selector size						
	B		C		D/DE	D
U_m	0/W/G	MG	0/W/G	MG	0/W/G	MG
72,5	1514	1476	1589	1551	1784	1746
123	1644	1606	1719	1681	1914	1876
170	1774	1736	1849	1811	2044	2006
245	1874	1836	1949	1911	2144	2106
300	2026	1988	2101	2063	2296	2258

M II 352
M II 502
M II 602

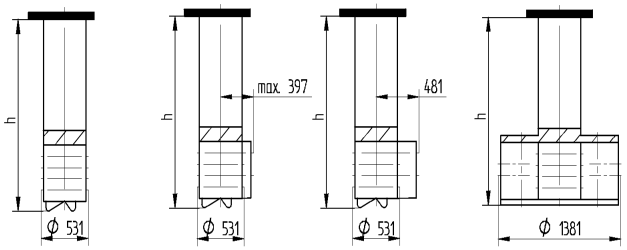
VM II 352
VM II 502
VM II 652



Tap selector size			
	B	C	D/DE
U_m	0/W/G	0/W/G	0/W/G
72,5	1704	1829	2154
123	1834	1959	2284
170	1964	2089	2414
245	2064	2189	2514
300	2216	2341	2666

M I 802

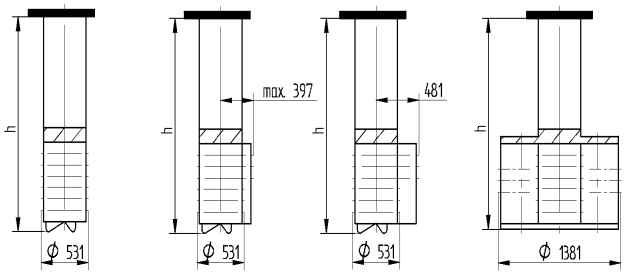
VM I 802
VM I 1002



Tap selector size						
	B		C		D/DE	D
U_m	0/W/G	MG	0/W/G	MG	0/W/G	MG
72,5	1724	1686	1799	1761	1994	1956
123	1854	1816	1929	1891	2124	2086
170	1984	1956	2059	2021	2254	2216
245	2084	2046	2159	2121	2354	2316
300	2236	2198	2311	2273	2506	2468

M I 1203
M I 1503

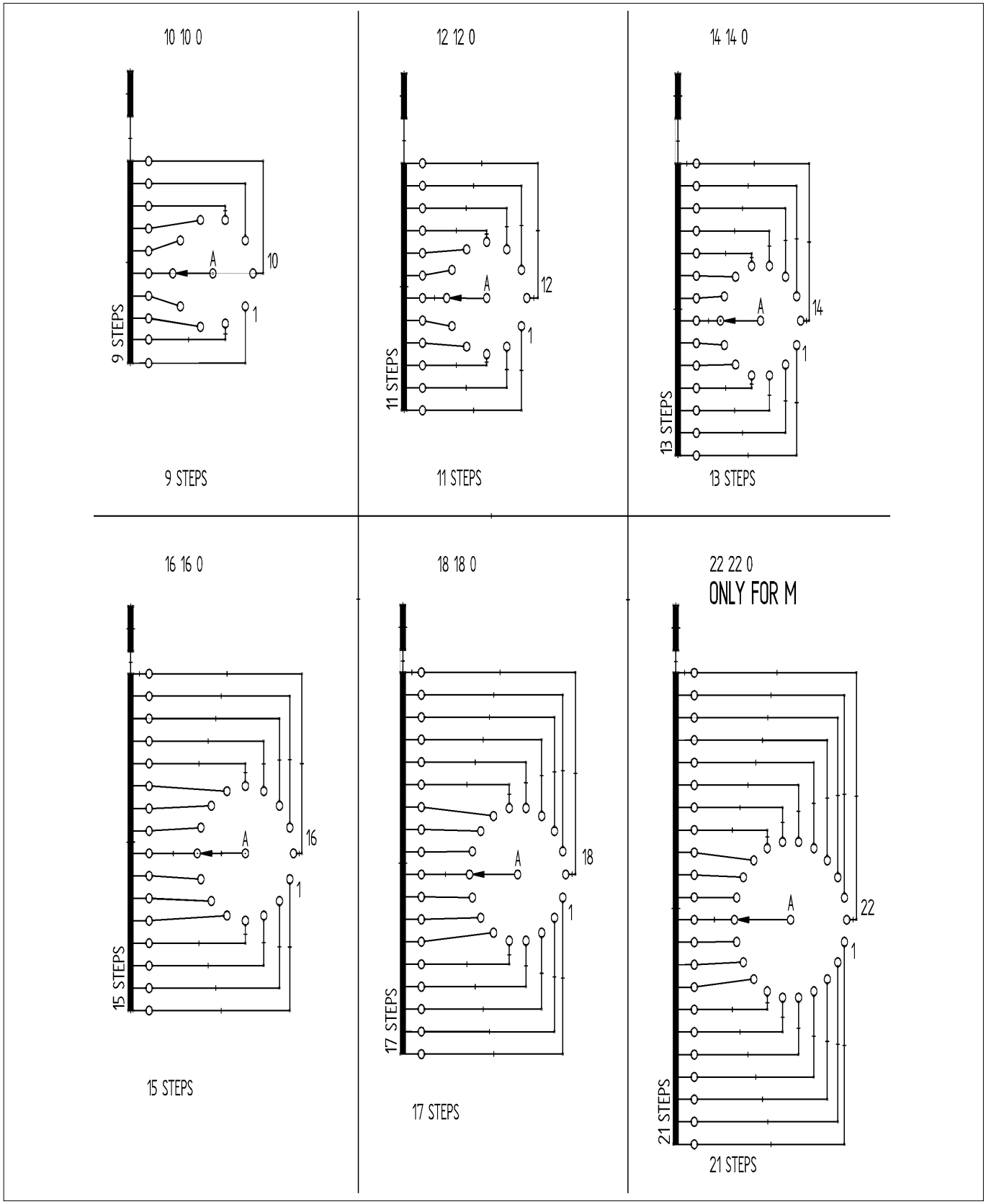
VM I 1203
VM I 1503



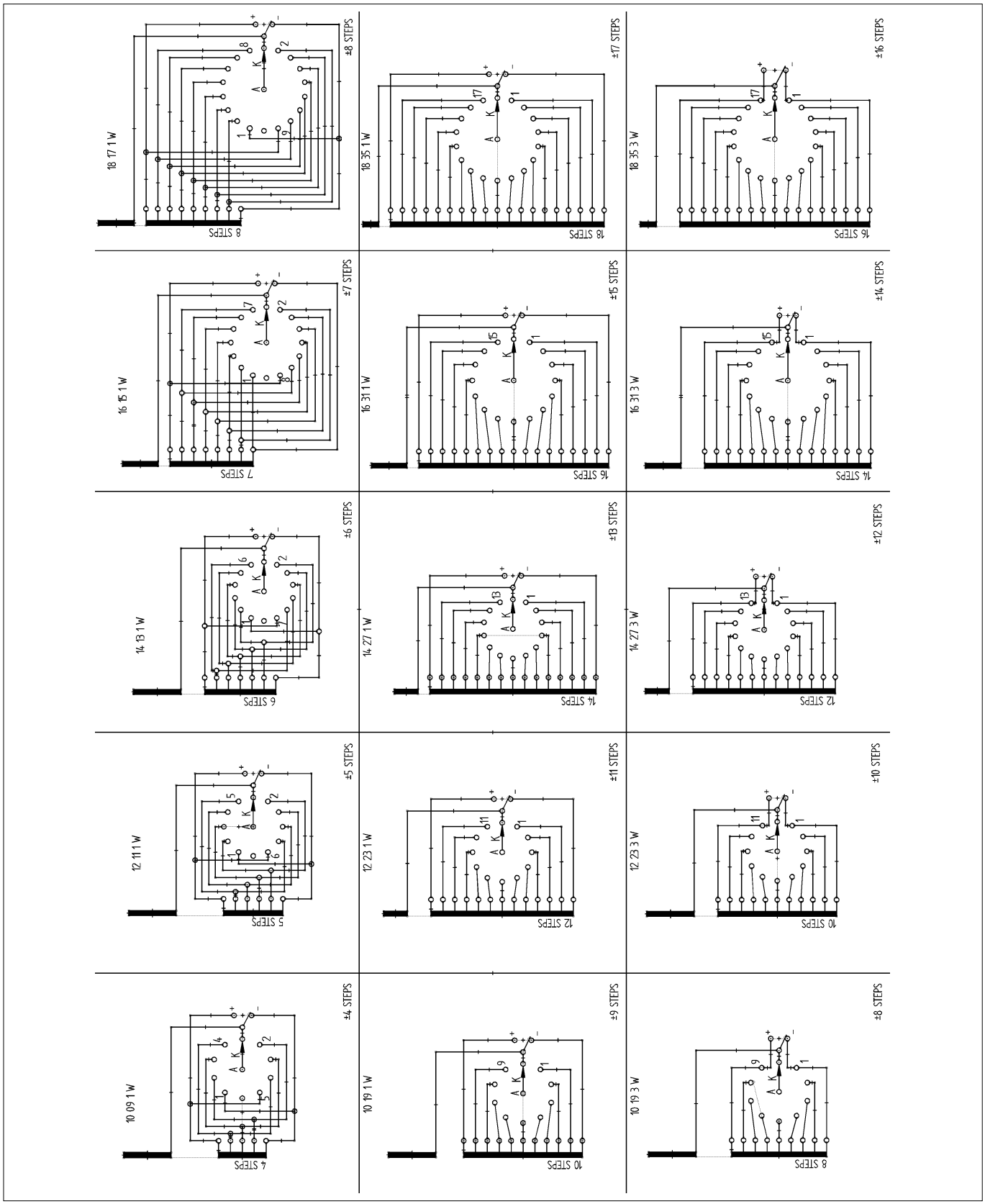
Tap selector size						
	B		C		D/DE	D
U_m	0/W/G	MG	0/W/G	MG	0/W/G	MG
72,5	1934	1896	2009	1971	2204	2166
123	2064	2026	2139	2101	2334	2296
170	2194	2156	2269	2231	2464	2426
245	2294	2256	2369	2331	2564	2526
300	2446	2408	2521	2483	2716	2678

 © Maschinenfabrik Reinhausen 2010

3.4 Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616)

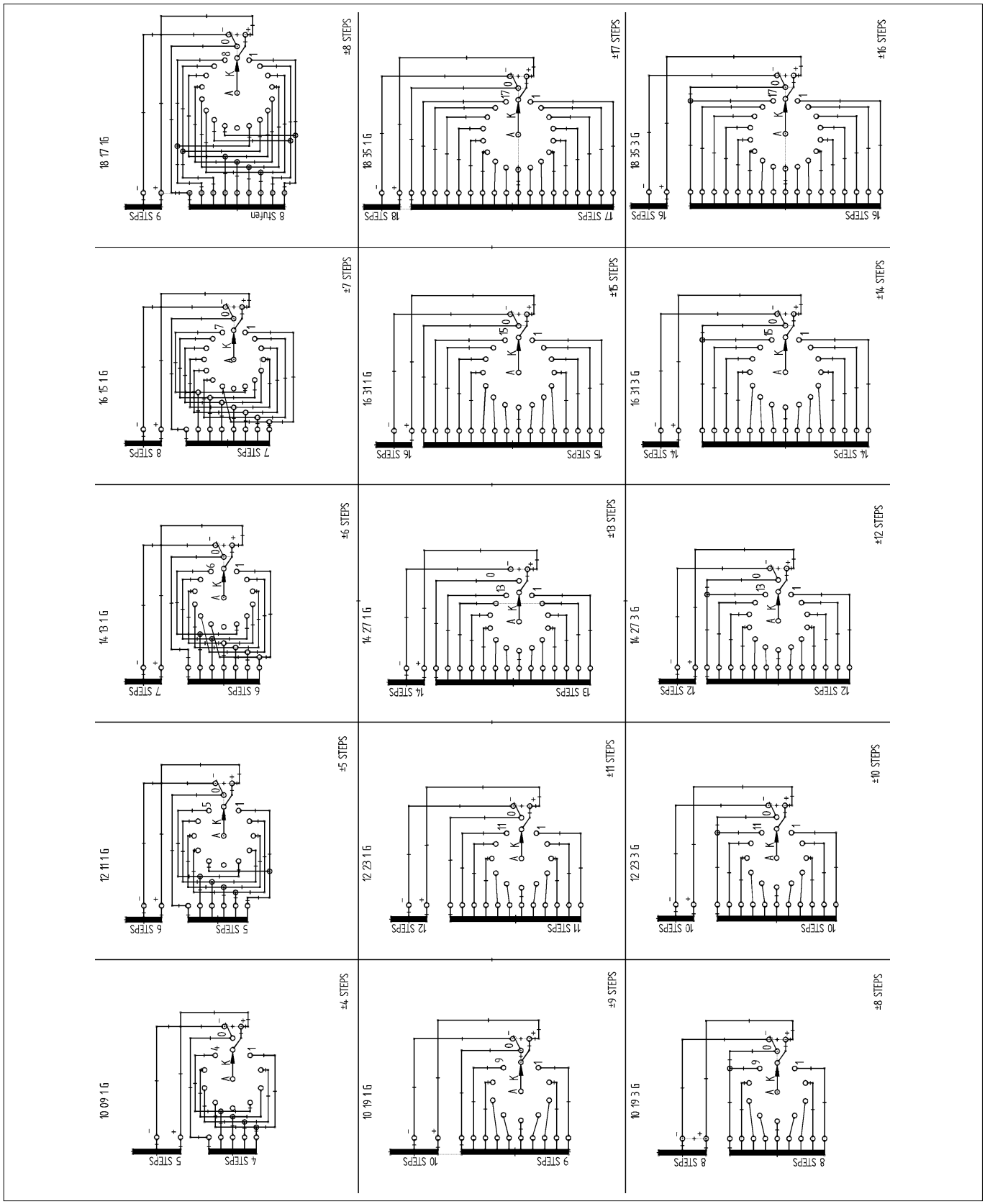


Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616)



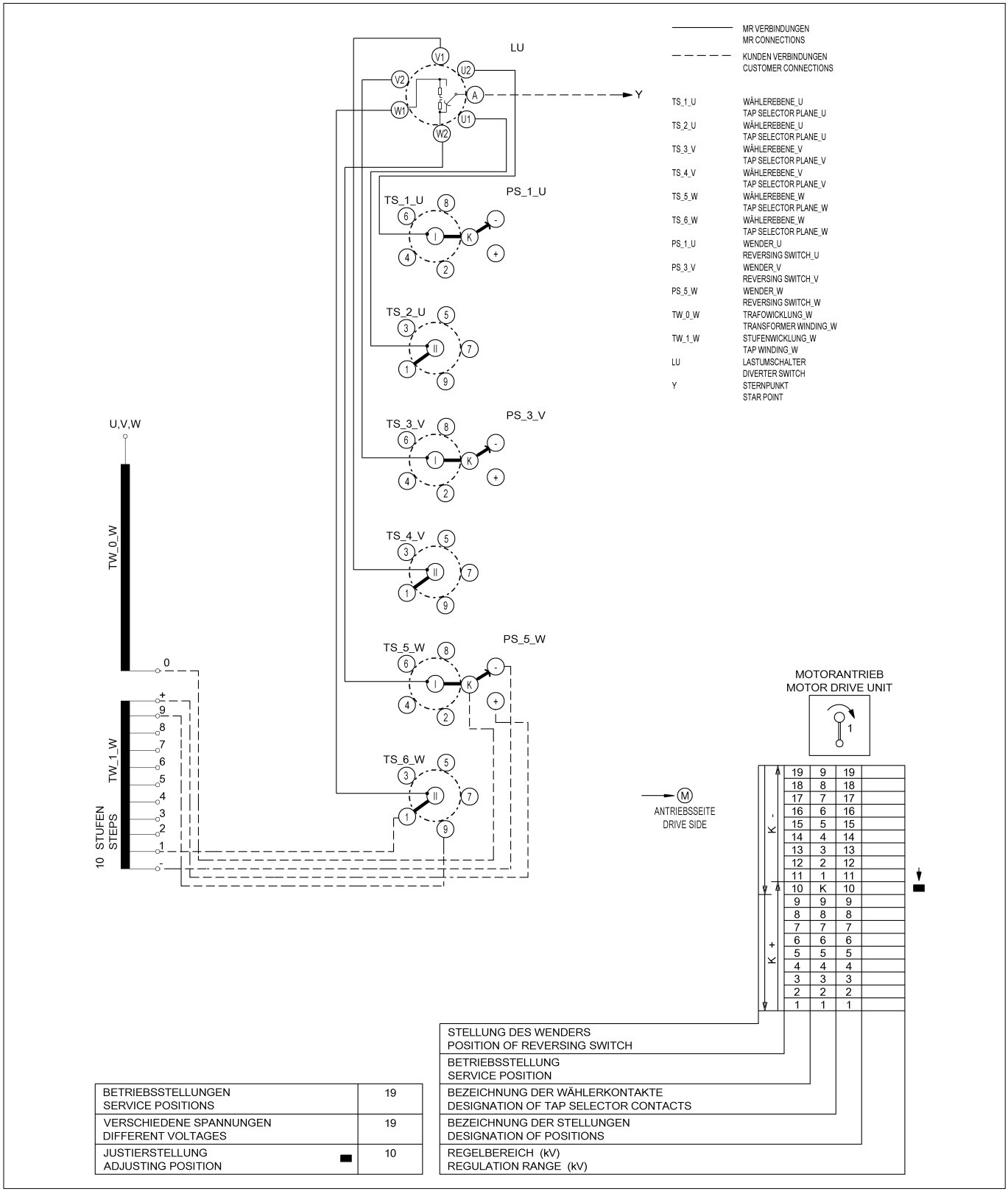
 © Maschinenfabrik Reinhausen 2010

Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616)



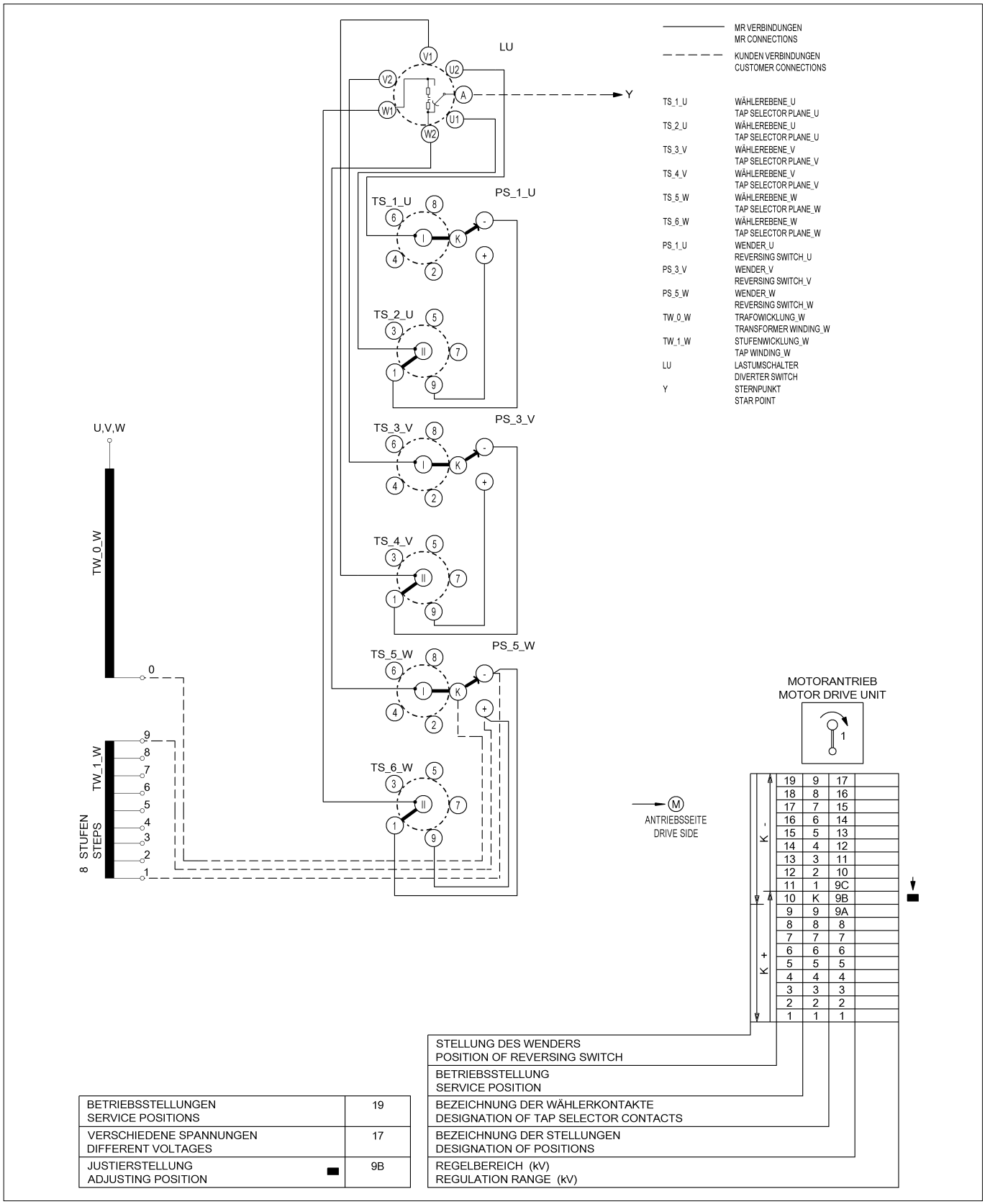
3.5 Connection diagram (contacts designated in accordance with MR standard)

3.5.1 VACUTAP® VM III 300 Y, basic connection diagram 10 19 1 W (2414642)

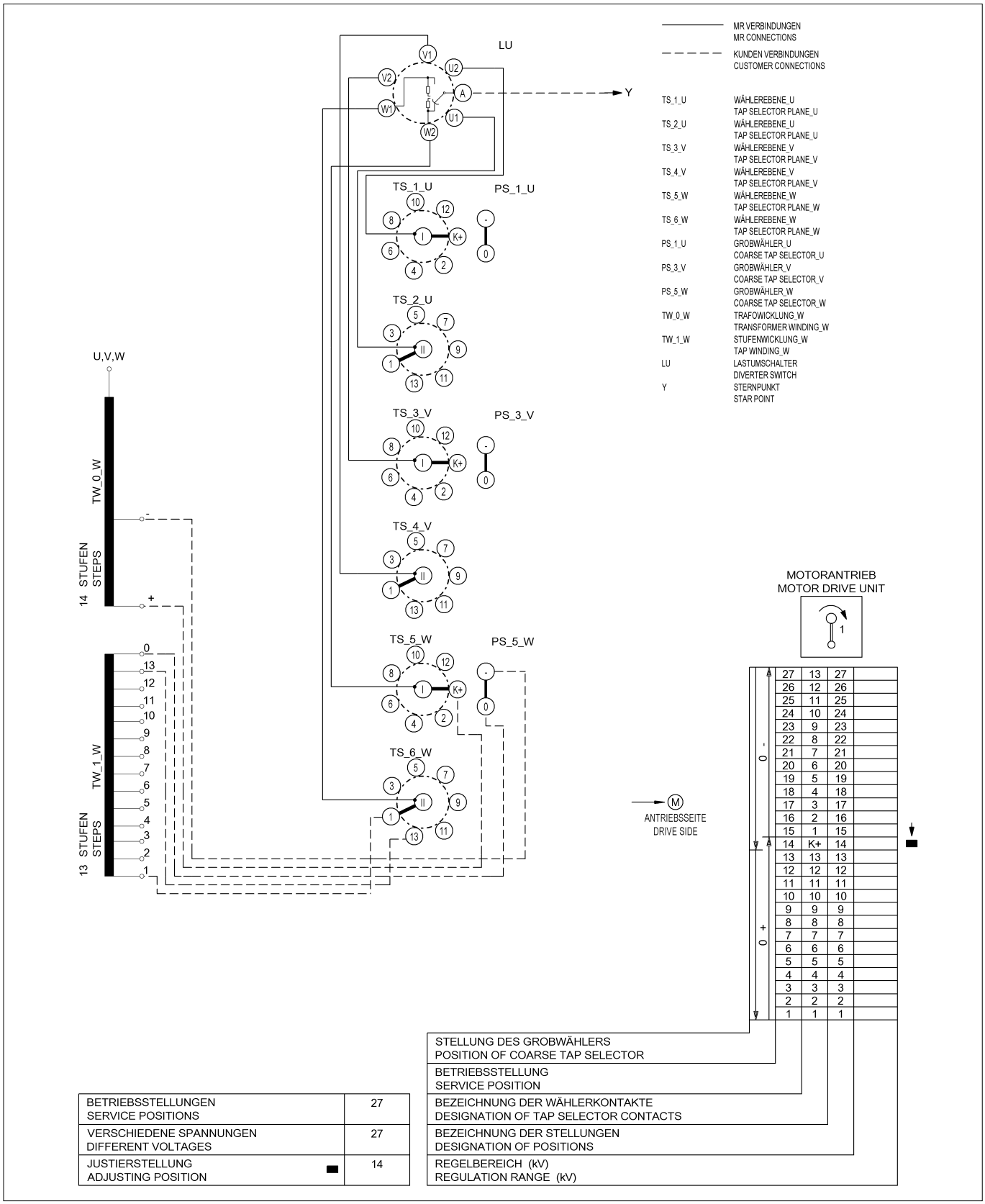


 © Maschinenfabrik Reinhausen 2010

3.5.2 VACUTAP® VM III 300 Y, basic connection diagram 10 19 3 W (2414644)

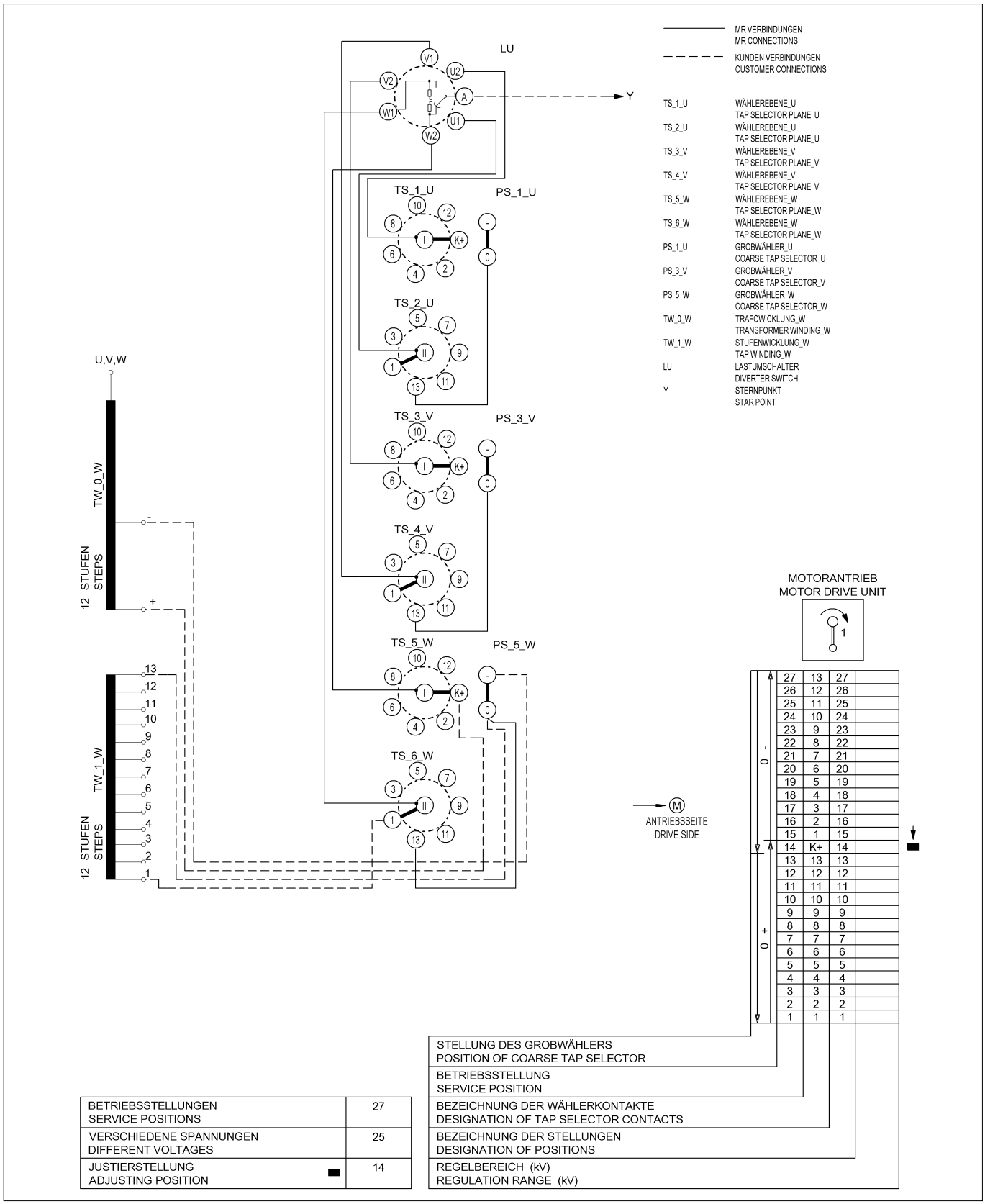


3.5.3 VACUTAP® VM III 300 Y, basic connection diagram 14 27 1 G (2414648)

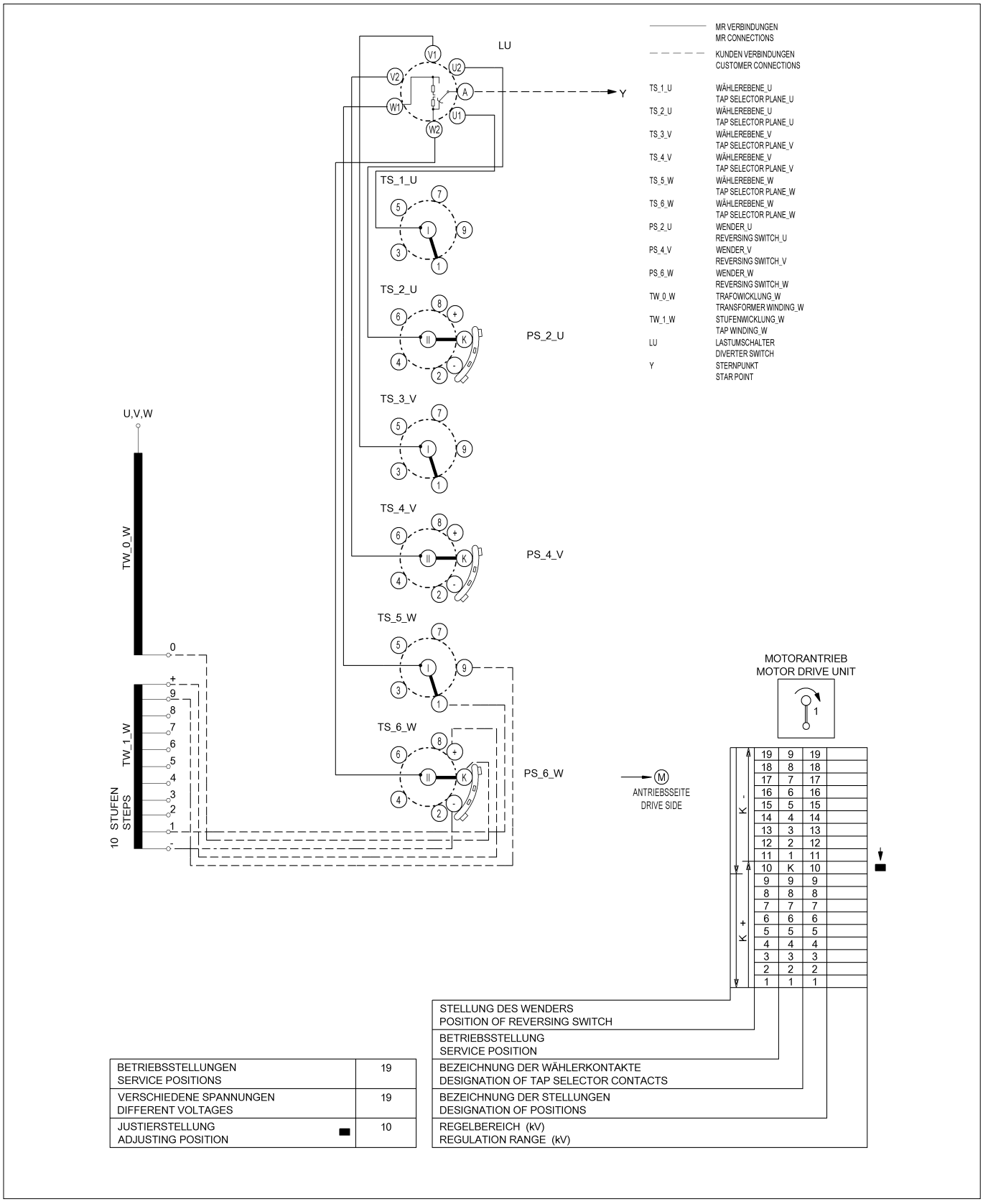


 © Maschinenfabrik Reinhausen 2010

3.5.4 VACUTAP® VM III 300 Y, basic connection diagram 14 27 3 G (2414649)

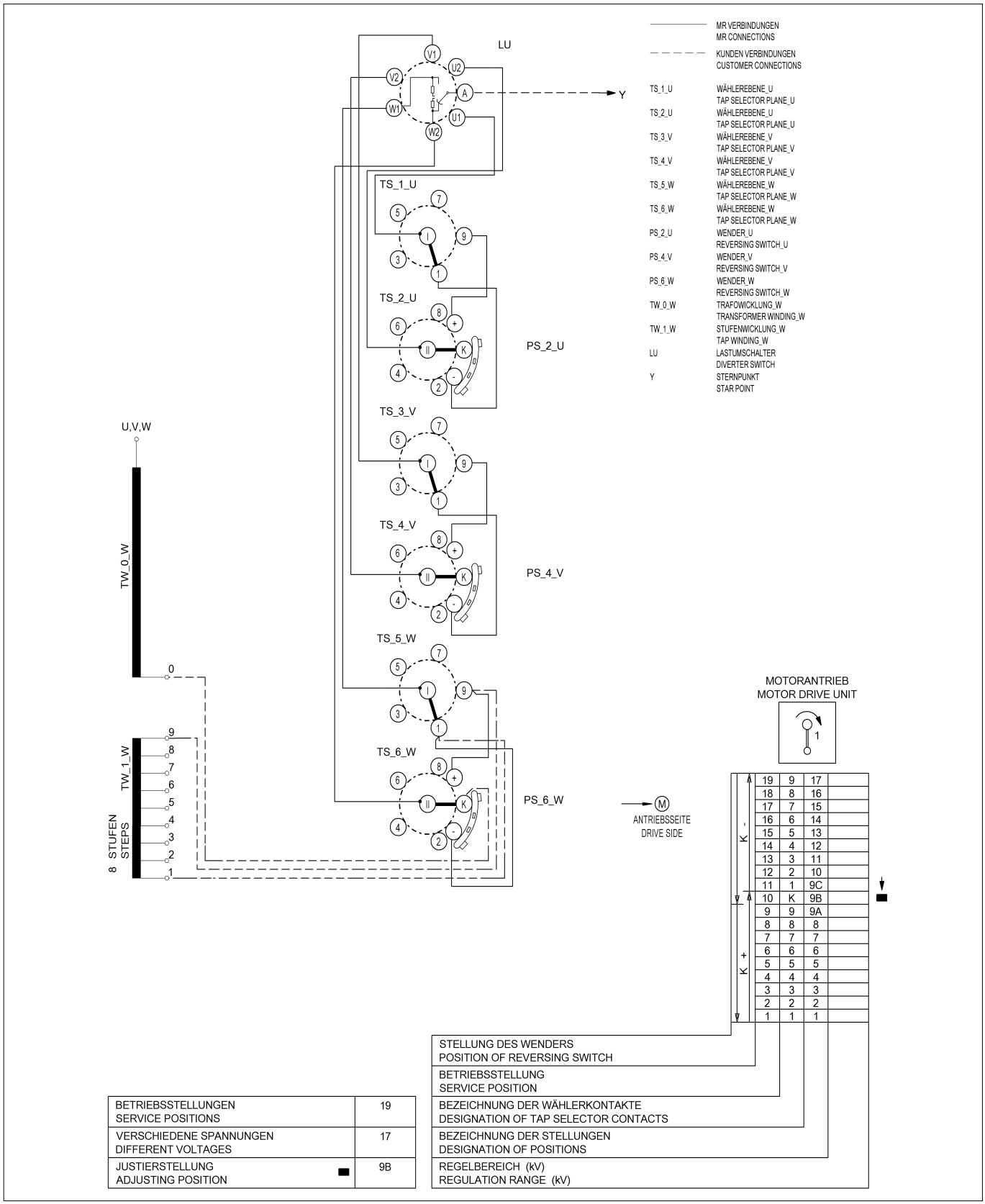


3.5.5 VACUTAP® VM III 350/500/650 Y, basic connection diagram 10 19 1 W (2414658)

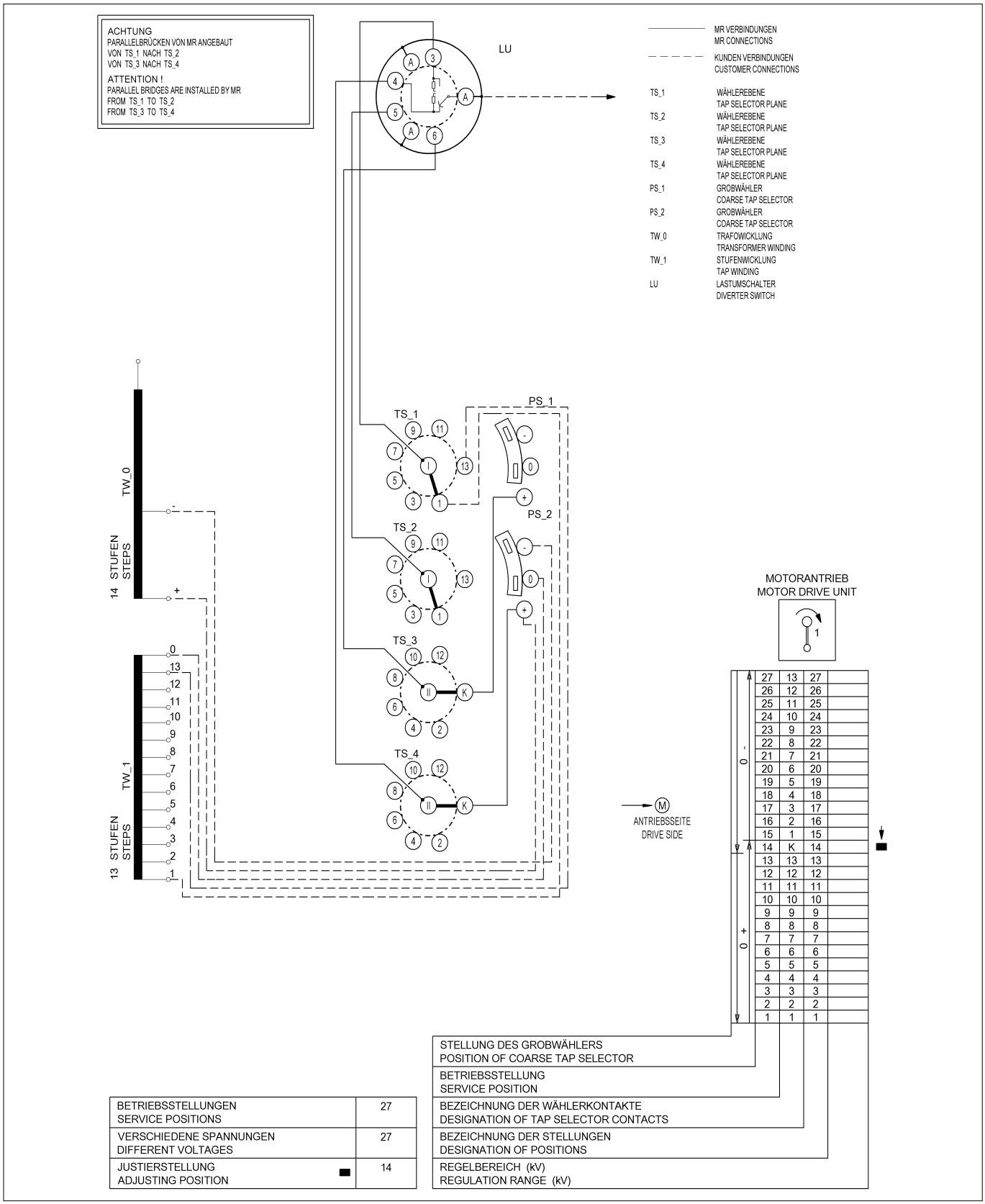


 © Maschinenfabrik Reinhausen 2010

3.5.6 VACUTAP® VM III 350/500/650 Y, basic connection diagram 10 19 3 W (2414670)

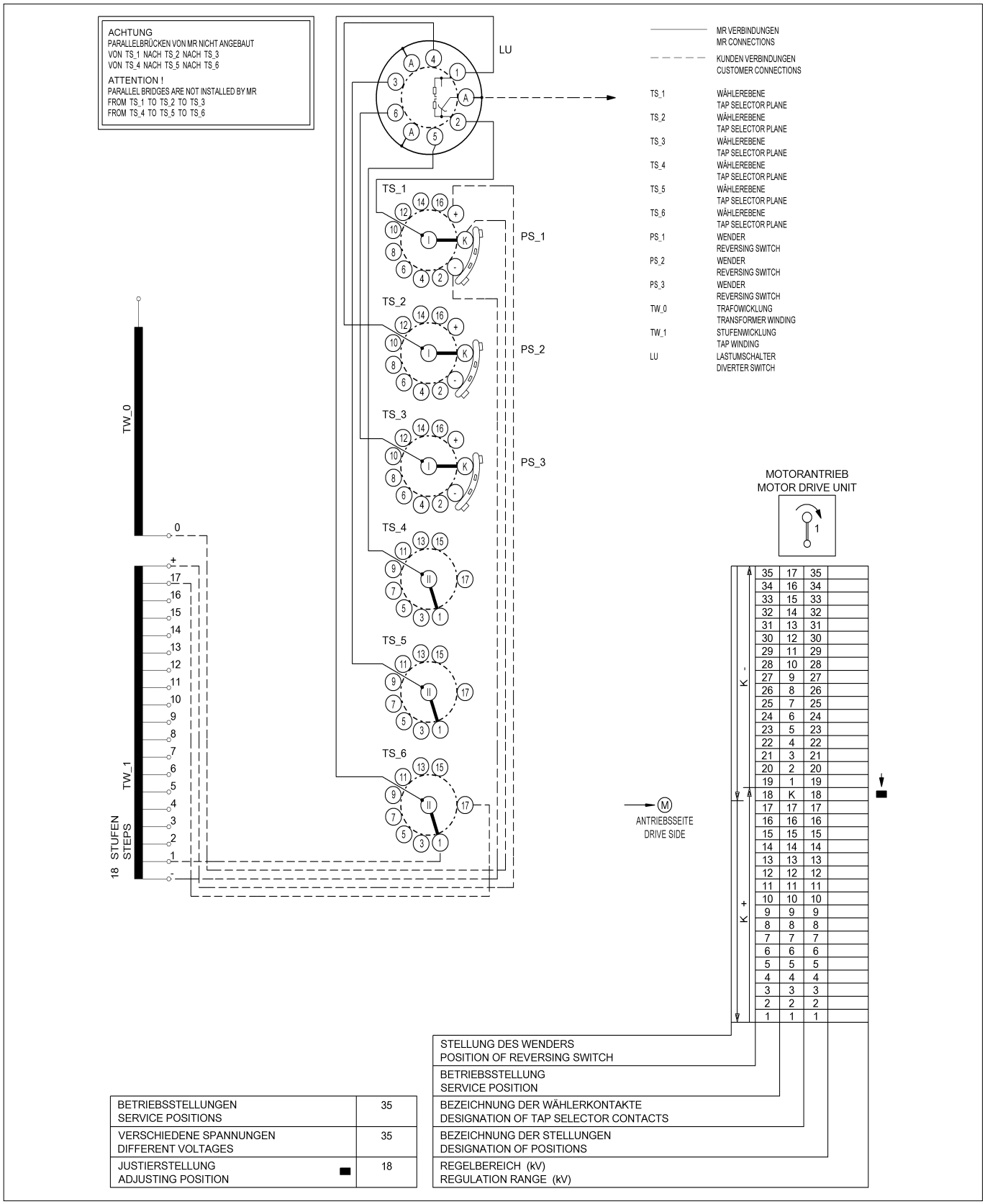


3.5.7 VACUTAP® VM I 802/1002, basic connection diagram 14 27 1 G (2414631)

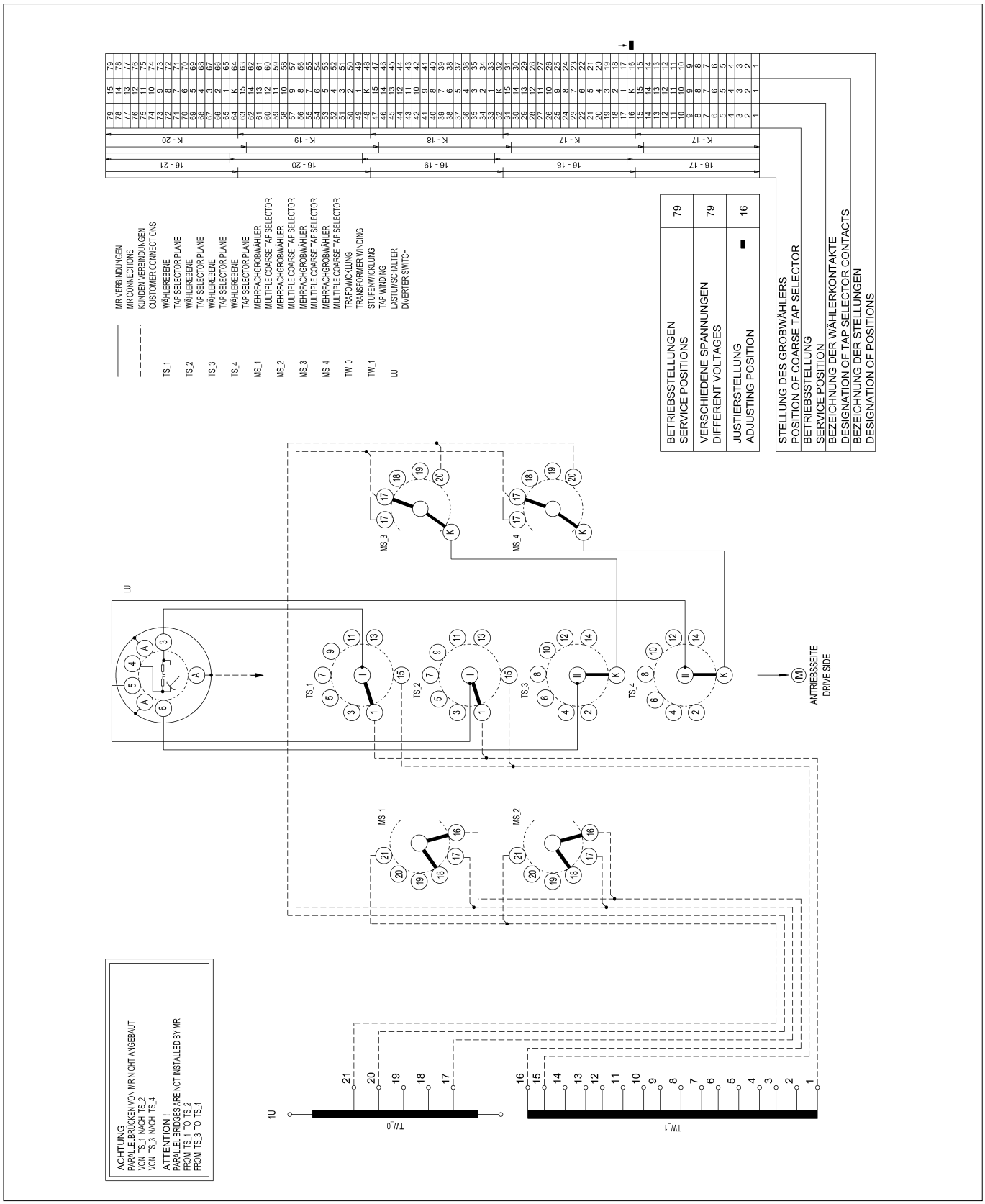


 © Maschinenfabrik Reinhausen 2010

3.5.8 VACUTAP® VM I 1203/1503, basic connection diagram 18 35 1 W (2414636)



3.5.9 VACUTAP® VM I 802/1002, basic connection diagram 16 79 1 G, multiple coarse change-over selector design (2407535)



 © Maschinenfabrik Reinhausen 2010

4 MR worldwide

Australia Reinhausen Australia Pty. Ltd. Ground Floor 6-10 Geeves Avenue Rockdale N. S. W. 2216 Phone: +61 2 9556 2133 Fax: +61 2 9597 1339 E-Mail: sales@au.reinhausen.com	**Italy** Reinhausen Italia S.r.l. Via Alserio, 16 20159 Milano Phone: +39 02 6943471 Fax: +39 02 69434766 E-Mail: sales@it.reinhausen.com	**Russian Federation** OOO MR Naberezhnaya Akademika Tupoleva 15, Bld. 2 ("Tupolev Plaza") 105005 Moscow Tel. +7 495 980 89 67 Fax. +7 495 980 89 67 E-Mail: mrr@reinhausen.ru
Brazil MR do Brasil Indústria Mecánica Ltda. Av. Elias Yazbek, 465 CEP: 06803-000 Embu - São Paulo Phone: +55 11 4785 2150 Fax: +55 11 4785 2185 E-Mail: vendas@reinhausen.com.br	**Japan** MR Japan Corporation German Industry Park 1-18-2 Hakusan, Midori-ku Yokohama 226-0006 Phone: +81 45 929 5728 Fax: +81 45 929 5741	**South Africa** Reinhausen South Africa (Pty) Ltd. No. 15, Third Street, Booysens Reserve Johannesburg Phone: +27 11 8352077 Fax: +27 11 8353806 E-Mail: support@reinhausen.co.za
Canada Reinhausen Canada Inc. 1010 Sherbrooke West, Suite 1800 Montréal, Québec H3A 2R7, Canada Phone: +1 514 286 1075 Fax: +1 514 286 0520 Mobile: +49 170 7807 696 E-Mail: m.foata@ca.reinhausen.com	**Luxembourg** Reinhausen Luxembourg S.A. 72, Rue de Prés L-7333 Steinsel Phone: +352 27 3347 1 Fax: +352 27 3347 99 E-Mail: sales@lu.reinhausen.com	**South Korea** Reinhausen Korea Ltd. Baek Sang Bldg. Room No. 1500 197-28, Kwanhun-Dong, Chongro-Ku Seoul 110-718, Korea Phone: +82 2 767 4909 Fax: +82 2 736 0049 E-Mail: you-mi.jang@kr.reinhausen.com
India Easun-MR Tap Changers Ltd. 612, CTH Road Tiruninravur, Chennai 602 024 Phone: +91 44 26300883 Fax: +91 44 26390881 E-Mail: easunmr@vsnl.com	**Malaysia** Reinhausen Asia-Pacific Sdn. Bhd Level 11 Chulan Tower No. 3 Jalan Conlay 50450 Kuala Lumpur Phone: +60 3 2142 6481 Fax: +60 3 2142 6422 E-Mail: mr_rap@my.reinhausen.com	**U.S.A.** Reinhausen Manufacturing Inc. 2549 North 9th Avenue Humboldt, TN 38343 Phone: +1 731 784 7681 Fax: +1 731 784 7682 E-Mail: sales@reinhausen.com
Iran Iran Transfo After Sales Services Co. Zanjan, Industrial Township No. 1 (Aliabad) Corner of Morad Str. Postal Code 4533144551 E-Mail: itass@iran-transfo.com	**P.R.C. (China)** MR China Ltd. (MRT) 开德贸易（上海）有限公司 中国上海浦东新区浦东南路360号 新上海国际大厦4楼E座 邮编： 200120 电话：＋86 21 61634588 传真：＋86 21 61634582 邮箱：mr-sales@cn.reinhausen.com mr-service@cn.reinhausen.com	**United Arab Emirates** Reinhausen Middle East FZE Dubai Airport Freezone Building Phase 6, 3rd floor, Office No. 6EB 341 Dubai Phone: +971 4 6091828 Fax: +971 4 6091829 E-Mail: service@ae.reinhausen.com

Maschinenfabrik Reinhausen GmbH
Falkensteinstraße 8
93059 Regensburg

Telefon: +49 (0)941 40 90-0
Telefax: +49 (0)941 40 90-7001
E-Mail: sales@reinhausen.com

www.reinhausen.com

2332907/00 EN • 11/10 • F0248500